



**Cochrane
Library**

Cochrane Database of Systematic Reviews

Disease-modifying antirheumatic drugs (DMARDs) for rheumatoid arthritis after failure of biologic or targeted synthetic therapy: a systematic review and network meta-analysis (Review)

Thomas J, Kamso MM, Whittle SL, Wells GA, Kelly SE, Pardo Pardo J, Tugwell P, Johnston RV, Glennon V, Ejaredar M, Thomas M, Lee C, Elliott J, Mokbel A, Avery JC, Grobler L, Cyril S, Deardon R, Akl EA, Maxwell LJ, Buchbinder R, Hazlewood GS, Supported by the Cochrane Musculoskeletal Health and Pain Thematic Group

Thomas J, Kamso MM, Whittle SL, Wells GA, Kelly SE, Pardo Pardo J, Tugwell P, Johnston RV, Glennon V, Ejaredar M, Thomas M, Lee C, Elliott J, Mokbel A, Avery JC, Grobler L, Cyril S, Deardon R, Akl EA, Maxwell LJ, Buchbinder R, Hazlewood GS, Supported by the Cochrane Musculoskeletal Health and Pain Thematic Group.

Disease-modifying antirheumatic drugs (DMARDs) for rheumatoid arthritis after failure of biologic or targeted synthetic therapy: a systematic review and network meta-analysis.

Cochrane Database of Systematic Reviews 2026, Issue 7. Art. No.: CD013562.

DOI: [10.1002/14651858.CD013562.pub2](https://doi.org/10.1002/14651858.CD013562.pub2).

www.cochranelibrary.com

TABLE OF CONTENTS

ABSTRACT	1
PLAIN LANGUAGE SUMMARY	3
SUMMARY OF FINDINGS	5
BACKGROUND	18
OBJECTIVES	19
METHODS	19
RESULTS	26
Figure 1.	27
Figure 2.	29
Figure 3.	30
Figure 4.	33
Figure 5.	34
Figure 6.	36
Figure 7.	37
Figure 8.	38
Figure 9.	39
DISCUSSION	40
AUTHORS' CONCLUSIONS	42
SUPPLEMENTARY MATERIALS	43
ADDITIONAL INFORMATION	43
REFERENCES	46
ADDITIONAL TABLES	50

[Intervention Review]

Disease-modifying antirheumatic drugs (DMARDs) for rheumatoid arthritis after failure of biologic or targeted synthetic therapy: a systematic review and network meta-analysis

Jocelyn Thomas^{1a}, Mohammed Mujaab Kamso^{2a}, Samuel L Whittle^{3,4b}, George A Wells^{5c}, Shannon E Kelly^{6d}, Jordi Pardo Pardo^{7,8e}, Peter Tugwell^{9f}, Renea V Johnston^{4g}, Vanessa Glennon^{4h}, Maede Ejaredar¹⁰, Megan Thomas^{11,12i}, Chloe Lee¹³, Jesse Elliott¹⁴, Abir Mokbel¹⁵, Jodie C Avery^{16j}, Liesl Grobler^{4k}, Sheila Cyril⁴, Rob Deardon^{17l}, Elie A Akl^{18,19m}, Lara J Maxwell^{20,21}, Rachel Buchbinder⁴ⁿ, Glen S Hazlewood^{22o}, Supported by the Cochrane Musculoskeletal Health and Pain Thematic Group

¹Department of Community Health Sciences - Epidemiology, University of Calgary, Calgary, Canada. ²Department of Community Health Sciences- Biostatistics, University of Calgary, Calgary, Canada. ³Rheumatology Unit, Queen Elizabeth Hospital, Woodville South, Australia. ⁴School of Public Health and Preventive Medicine, Monash University, Melbourne, Australia. ⁵School of Epidemiology and Public Health, University of Ottawa, Ottawa, Canada. ⁶Cardiovascular Research Methods Centre, University of Ottawa Heart Institute, Ottawa, Canada. ⁷Department of Medicine, University of Ottawa, Ottawa, Canada. ⁸Centre for Practice Changing Research, Ottawa Hospital Research Institute, Ottawa, Canada. ⁹Clinical Epidemiology Program, Ottawa Hospital Research Institute, Ottawa, Canada. ¹⁰Department of Medicine, Cumming School of Medicine, University of Calgary, Calgary, Canada. ¹¹Department of Community Health Sciences, University of Calgary, Calgary, Canada. ¹²University of British Columbia, Vancouver, Canada. ¹³McCaig Institute for Bone and Joint Health, University of Calgary, Calgary, Canada. ¹⁴Canadian Agency for Drugs and Technologies in Health, Ottawa, Canada. ¹⁵Rheumatology, Immunology, Cairo University, Cairo, Egypt. ¹⁶Robinson Research Institute, Adelaide University, Adelaide, Australia. ¹⁷Department of Mathematics & Statistics, Faculty of Science, University of Calgary, Calgary, Canada. ¹⁸Department of Internal Medicine, American University of Beirut Medical Center, Beirut, Lebanon. ¹⁹American University of Beirut, Beirut, Lebanon. ²⁰Faculty of Medicine, University of Ottawa, Ottawa, Canada. ²¹Bruyère Health Research Institute, Ottawa, Canada. ²²Department of Medicine and Department of Community Health Sciences, University of Calgary, Calgary, Canada

^aThese authors should be considered joint first author ORCID iD: <https://orcid.org/0000-0002-5851-3998>; ORCID iD: <https://orcid.org/0000-0003-0436-1850>. ^bORCID iD: <https://orcid.org/0000-0001-7417-7691>. ^cORCID iD: <https://orcid.org/0000-0002-2289-9139>. ^dORCID iD: <https://orcid.org/0000-0002-5520-6095>. ^eORCID iD: <https://orcid.org/0000-0002-3985-5335>. ^fORCID iD: <https://orcid.org/0000-0001-5062-0556>. ^gORCID iD: <https://orcid.org/0000-0003-3901-1164>. ^hORCID iD: <https://orcid.org/0000-0003-2792-7706>. ⁱORCID iD: <https://orcid.org/0000-0002-3210-2533>. ^jORCID iD: <https://orcid.org/0000-0002-8857-9162>. ^kORCID iD: <https://orcid.org/0000-0002-0177-9672>. ^lORCID iD: <https://orcid.org/0000-0003-0151-6028>. ^mORCID iD: <https://orcid.org/0000-0002-3444-8618>. ⁿORCID iD: <https://orcid.org/0000-0002-0597-0933>. ^oORCID iD: <https://orcid.org/0000-0001-7709-3741>

Contact: Glen S Hazlewood, glenhazlewood@gmail.com.

Editorial group: Cochrane Central Editorial Service.

Publication status and date: New, published in Issue 7, 2026.

Citation: Thomas J, Kamso MM, Whittle SL, Wells GA, Kelly SE, Pardo Pardo J, Tugwell P, Johnston RV, Glennon V, Ejaredar M, Thomas M, Lee C, Elliott J, Mokbel A, Avery JC, Grobler L, Cyril S, Deardon R, Akl EA, Maxwell LJ, Buchbinder R, Hazlewood GS, Supported by the Cochrane Musculoskeletal Health and Pain Thematic Group. Disease-modifying antirheumatic drugs (DMARDs) for rheumatoid arthritis after failure of biologic or targeted synthetic therapy: a systematic review and network meta-analysis. *Cochrane Database of Systematic Reviews* 2026, Issue 7. Art. No.: CD013562. DOI: [10.1002/14651858.CD013562.pub2](https://doi.org/10.1002/14651858.CD013562.pub2).

Copyright © 2026 The Cochrane Collaboration. Published by John Wiley & Sons, Ltd.

ABSTRACT

Rationale

After inadequate response to first-line biologic or targeted synthetic (b/ts) disease-modifying antirheumatic drug (DMARD) therapy in adults with rheumatoid arthritis, there are numerous alternative DMARD options, and current understanding of their comparative benefits and harms is limited.

Disease-modifying antirheumatic drugs (DMARDs) for rheumatoid arthritis after failure of biologic or targeted synthetic therapy: a systematic review and network meta-analysis (Review)

Copyright © 2026 The Cochrane Collaboration. Published by John Wiley & Sons, Ltd.

Objectives

The aim of this living systematic review and network meta-analysis was to compare the benefits and harms of DMARDs after failure of biologic or targeted synthetic DMARDs in adults with rheumatoid arthritis.

Search methods

We searched CENTRAL, MEDLINE, Embase, and two trial registries (ClinicalTrials.gov and the WHO ICTRP) from inception until 28 November 2025, with no restrictions on language or date of publication.

Eligibility criteria

We included randomised controlled trials (RCTs) of adults aged 18 years or older diagnosed with rheumatoid arthritis according to 1958, 1987, or 2010 classification criteria who previously demonstrated inadequate response to a b/ts DMARD. Eligible interventions included conventional synthetic DMARDs (methotrexate, antimalarials, sulfasalazine, leflunomide, ciclosporin, and azathioprine), biologic DMARDs (adalimumab, certolizumab, etanercept, golimumab, infliximab, abatacept, rituximab, tocilizumab, sarilumab, and anakinra), and targeted synthetic DMARDs (tofacitinib, baricitinib, and upadacitinib).

Outcomes

Our critical outcomes were American College of Rheumatology 50% (ACR50) response, withdrawals due to adverse events, radiographic progression, Disease Activity Score 28 (DAS28) remission, pain as measured by visual analogue scale, function as measured by the Health Assessment Questionnaire (HAQ), and serious adverse events. Important outcomes included ACR20, ACR70, serious infections, fatigue, and quality of life.

Risk of bias

We used Cochrane's RoB 1 tool to assess risk of bias in the included studies.

Synthesis methods

We first screened the records using an approach that combined machine learning and crowdsourcing to identify probable RCTs. We then reviewed the records identified as RCTs for eligibility and simultaneously classified them to the appropriate Population, Intervention, Comparator, and Outcome (PICO) question(s). Two review authors then extracted relevant data from the included studies in duplicate and independently, with any disagreements resolved by a third review author. A Bayesian random-effects network meta-analysis was conducted using a semi-informative prior probability distribution. We assessed the certainty of evidence for each outcome using the GRADE approach.

Included studies

We included 19 unique studies (4779 participants) in the review, all of which were parallel-design RCTs. Eleven trials were placebo controlled; two trials had an inactive comparator arm; and six trials had an active comparator arm. The trials were performed in a well-established rheumatoid arthritis population, with the median baseline disease duration ranging from 6.4 to 14 years, median age of participants ranging from 49 to 58 years, and median baseline disease activity (DAS28) ranging from 4.87 to 6.79.

Synthesis of results

ACR50 response

We found moderate-/high-certainty evidence that using a tumour necrosis factor (TNF) inhibitor not previously tried, interleukin-6 (IL-6) inhibitors, abatacept, rituximab, and Janus kinase (JAK) inhibitors was more effective than placebo: TNF inhibitor not previously tried (odds ratio (OR) 6.04, 95% credible interval (CrI) 2.49 to 16.3; high-certainty evidence), sarilumab (OR 3.11, 95% CrI 1.25 to 7.76; high-certainty evidence), tocilizumab 4 mg/kg intravenous (OR 5.31, 95% CrI 2.09 to 12.09; high-certainty evidence), tocilizumab 8 mg/kg intravenous (OR 10.03, 95% CrI 3.65 to 31.27; high-certainty evidence), subcutaneous abatacept (OR 4.31, 95% CrI 0.97 to 18.28; moderate-certainty evidence), intravenous abatacept (OR 4.57, 95% CrI 2.21 to 10.18; high-certainty evidence), rituximab (OR 5.50, 95% CrI 2.31 to 13.12; high-certainty evidence), upadacitinib (OR 3.93, 95% CrI 1.53 to 10.32; high-certainty evidence), tofacitinib (OR 3.93, 95% CrI 1.53 to 10.32; high-certainty evidence), baricitinib 2 mg (OR 2.00, 95% CrI 0.77 to 5.23; moderate-certainty evidence), baricitinib 4 mg (OR 2.79, 95% CrI 1.10 to 7.22; high-certainty evidence). With an assumed risk for placebo of 78 out of 1000 patients, the expected effects for the active drugs ranged from 143 (baricitinib 2 mg) to 455 (tocilizumab 8 mg/kg).

Withdrawals due to adverse events

For most interventions, there were sparse data with low-certainty evidence, except for TNF inhibitor not previously tried (risk ratio (RR) 0.32, 95% CrI 0.07 to 1.1; moderate-certainty evidence), which is probably less harmful than placebo, and sarilumab (RR 1.98, 95% CrI 0.57 to 7.19; moderate-certainty evidence), which is probably more harmful than placebo. Low-certainty evidence suggests that intravenous abatacept (RR 0.92, 95% CrI 0.34 to 2.79), upadacitinib (RR 0.41, 95% CrI 0.08 to 1.83), and baricitinib 2 mg (RR 0.93, 95% CrI 0.22 to 4.01) may be less harmful than placebo. Low-certainty evidence suggests that tocilizumab 4 mg/kg (RR 1.39, 95% CrI 0.39 to 5.28), tocilizumab 8

mg/kg (RR 1.49, 95% CrI 0.51 to 4.81), subcutaneous abatacept (RR 3.11, 95% CrI 0.05 to 249.6), rituximab (RR 2.38, 95% CrI 0.44 to 23.31), tofacitinib (RR 1.37, 95% CrI 0.35 to 5.58), and baricitinib 4 mg (RR 1.43, 95% CrI 0.38 to 5.81) may be more harmful than placebo.

Data were insufficient to perform a network meta-analysis for radiographic progression.

For DAS28 and the HAQ, there was mostly moderate-/high-certainty evidence of a benefit, with some exceptions for comparisons with indirect evidence only that was of low or very low certainty.

For the other efficacy outcomes, data were sparse with wide credible intervals, and the certainty of evidence was typically low.

Authors' conclusions

We found high-certainty evidence that nine therapies and moderate-certainty evidence that two therapies provide a clinically important benefit in improving disease activity compared to placebo for people with rheumatoid arthritis after failure of b/ts DMARD therapy. There was significant uncertainty surrounding treatment-related harms, with the evidence having been downgraded for serious or extremely serious imprecision. Pair-wise comparisons showed no significant differences among therapies, although the certainty of evidence was low. The lack of clarity regarding safety and comparative efficacy suggests that treatment decisions should be guided by individual patient characteristics and preferences.

Funding

This work was supported by grants from the Canadian Institutes for Health Research (CIHR) [Funding Reference Numbers (FRN) 178375 and 180324] and the National Health and Medical Research Council (NHMRC) Cochrane Collaboration. This research was supported by Arthritis Society Canada (Doctoral Studentship TGP-23-0211).

Registration

This study was outlined in a Cochrane protocol (CD013562; DOI 10.1002/14651858.CD013562).

PLAIN LANGUAGE SUMMARY

In adults with rheumatoid arthritis, which disease-modifying antirheumatic medicines (DMARDs) (a type of medication that stops or slows the disease process) work after a biologic or targeted synthetic DMARD has failed?

Key messages

- Several medications can help rheumatoid arthritis symptoms after failure of a biologic or targeted synthetic (b/ts) disease-modifying antirheumatic medicine (DMARD) (a type of medication that stops or slows the disease process).
- We are unsure how the medications compare with placebo (dummy treatment) or with each other in terms of unwanted effects.
- We need more studies that directly compare drugs to get a better idea of which is best. Treatment decisions should consider individual health factors and preferences.

What is rheumatoid arthritis?

Rheumatoid arthritis is a disease where the body's immune system, which normally fights infections, mistakenly attacks the joints. This causes joint pain, swelling, stiffness, and disability. Rheumatoid arthritis is a long-term condition, but treatments can help manage symptoms and slow the damage.

How is rheumatoid arthritis treated?

Rheumatoid arthritis is typically treated with disease-modifying antirheumatic drugs (DMARDs), which work to control the inflammation responsible for symptoms and also prevent joint damage. There are various kinds of DMARDs. Targeted synthetic (ts) DMARDs work by targeting specific cells or proteins. Biologic (b) DMARDs also target specific cells or proteins, but are made from material that comes from living organisms. Lastly, conventional synthetic DMARDs affect the whole immune system rather than targeting specific parts of it.

What did we want to find out?

We wanted to explore the effectiveness of DMARDs in adults with rheumatoid arthritis for whom treatment with a b/ts DMARD didn't work. This is a 'living' systematic review, meaning that it will be updated regularly. As this field of medicine is always growing and changing, we want to have the most up-to-date evidence reflected in this review.

What did we do?

We searched for studies that tested the effectiveness of DMARDs in adults with rheumatoid arthritis for whom treatment with one b/ts DMARD didn't work.

We looked at how well the available DMARDs reduced the symptoms of rheumatoid arthritis, including pain, joint swelling, and issues with joint movement. We measured this using the American College of Rheumatology 50% response criteria, or ACR50 response. An ACR50 response means the rheumatoid arthritis medication is effective and symptoms have improved by at least half. We also looked at any unwanted effect that might cause a person to stop using the medication.

We compared and summarised the results of the studies and rated our confidence in the evidence based on factors such as study methods and sizes.

What did we find?

We found 19 studies that involved 4779 people with rheumatoid arthritis who had tried a tumour necrosis factor inhibitor (a type of bDMARD) in the past, but it did not improve their symptoms. Most people in these studies were women, with an average age of 49 to 58 years, who had been living with rheumatoid arthritis for an average of 6 to 14 years. The studies were conducted worldwide, including in North America, Europe, and Asia. Nine studies were funded by pharmaceutical companies, while the other 10 were funded by national research grants.

Several medications reduced the symptoms of rheumatoid arthritis in patients who have failed a b/ts DMARD:

1. a different tumour necrosis factor inhibitor (2 studies; 343 people);
2. sarilumab (1 study, 365 people);
3. tocilizumab 4 mg/kg (1 study, 319 people);
4. tocilizumab 8 mg/kg (1 study, 328 people);
5. intravenous (into a vein) abatacept (2 studies, 665 people);
6. rituximab (1 study, 499 people);
7. upadacitinib (1 study, 333 people);
8. tofacitinib (1 study, 263 people);
9. a higher dose of baricitinib (1 study, 353 people).

We are unsure how the medications compare with placebo (dummy treatment) or with each other in terms of unwanted effects, as the evidence for this outcome was generally of low quality.

What are the limitations of the evidence?

Our confidence in the evidence varied from high to very low. In cases where our confidence was limited, this was because there were not enough studies to be certain about the results. When we searched for studies to include in the review, we found some gaps in the available evidence. Specifically, we found that there were fewer studies that directly compared medications with each other and more studies that compared medications to a placebo. More studies that compare drugs directly would better help us understand which drug is best.

How up to date is this evidence?

The evidence is current to 28 November 2025. As this is a living review, we will be updating this research yearly to ensure our results reflect new evidence.

SUMMARY OF FINDINGS

Summary of findings 1. Summary of findings for ACR50

ACR50						
Total studies: 12	Relative effect	Anticipated absolute effect			Certainty of evidence	Interpretation of results
Total participants: 3671	OR (95% CrI)	Events per 1000 persons (95% CrI)				
		Without intervention	With intervention	Difference		

Patient population: individuals with rheumatoid arthritis who have previously failed a TNFi

Intervention: TNFi not previously tried, IL-6 inhibitors (sarilumab 200 mg SC every 2 weeks, tocilizumab 4 mg/kg IV every 4 weeks, and tocilizumab 8 mg/kg IV every 4 weeks), T-cell co-stimulation inhibitors (abatacept 125 mg SC weekly, and abatacept 10 mg/kg (rounded) IV every 4 weeks (+/- IV load at 2 weeks)), B-cell depletor (rituximab 1000 mg IV week 0 and 2), and JAK inhibitors (upadacitinib 15 mg oral once daily, tofacitinib 5 mg oral twice a day, baricitinib 2 mg oral once daily, baricitinib 4 mg oral once daily)

Comparator: placebo

Outcome: 50% improvement in rheumatoid arthritis symptoms from baseline (ACR50)

Setting: outpatient

TNF inhibitor

TNFi not previously tried (2 RCTs; 343 participants)	6.04 (2.49, 16.3)	77	334 (172 to 576)	258 more (95 more to 499 more)	⊕⊕⊕⊕ ^a High	More effective than placebo
---	-------------------	----	------------------	--------------------------------	---------------------------	-----------------------------

IL-6 inhibitor

Sarilumab 200 mg SC every 2 weeks (1 RCT; 365 participants)	3.11 (1.25, 7.76)	77	206 (94 to 392)	129 more (17 more to 316 more)	⊕⊕⊕⊕ ^b High	More effective than placebo
Tocilizumab 4 mg/kg IV every 4 weeks (1 RCT; 319 participants)	5.31 (2.09, 12.09)	77	313 (124 to 632)	235 more (45 more to 554 more)	⊕⊕⊕⊕ ^c High	More effective than placebo
Tocilizumab 8 mg/kg IV every 4 weeks (1 RCT; 328 participants)	10.03 (3.65, 31.27)	77	455 (233 to 722)	378 more (156 more to 646 more)	⊕⊕⊕⊕ ^d High	More effective than placebo

T-cell co-stimulation inhibitor

Abatacept 125 mg SC weekly (indirect evidence only)	4.31 (0.97, 18.28)	77	264 (75 to 603)	187 more (2 fewer to 527 more)	⊕⊕⊕○ ^e Moderate	Probably more effective than placebo, but the credible interval crosses the null effect
Abatacept 10 mg/kg (rounded) IV every 4 weeks (+/- IV load at 2 weeks) (2 RCTs; 665 participants)	4.57 (2.21, 10.18)	77	275 (155 to 459)	199 more (79 more to 382 more)	⊕⊕⊕⊕ ^f High	More effective than placebo

B-cell depletor

Rituximab 1000 mg IV week 0 and 2 (1 RCT; 499 participants)	5.50 (2.31, 13.12)	77	314 (161 to 522)	237 more (84 more to 445 more)	⊕⊕⊕⊕ ^g High	More effective than placebo
--	--------------------	----	------------------	--------------------------------	---------------------------	-----------------------------

JAK inhibitor

Upadacitinib 15 mg oral once daily (1 RCT; 333 participants)	3.93 (1.53, 10.32)	77	246 (113 to 462)	170 more (36 more to 385 more)	⊕⊕⊕⊕ ^h High	More effective than placebo
Tofacitinib 5 mg oral twice a day (1 RCT; 263 participants)	4.09 (1.43, 12.13)	77	254 (106 to 502)	177 more (30 more to 425 more)	⊕⊕⊕⊕ ⁱ High	More effective than placebo
Baricitinib 2 mg oral once daily (1 RCT; 350 participants)	2.00 (0.77, 5.23)	77	143 (60 to 303)	66 more (17 fewer to 226 more)	⊕⊕⊕○ ^j Moderate	Probably more effective than placebo, but the credible interval crosses the null effect
Baricitinib 4 mg oral once daily (1 RCT; 353 participants)	2.79 (1.10, 7.22)	77	188 (84 to 375)	112 more (7 more to 298 more)	⊕⊕⊕⊕ ^k High	More effective than placebo

Abbreviations: ACR50, American College of Rheumatology 50% improvement; CrI, credible interval; JAK, Janus kinase; IL-6, interleukin-6; IV, intravenous; OR, odds ratio; RCT, randomised controlled trial; SC, subcutaneous; TNFi, tumour necrosis factor inhibitor

^aNetwork evidence starts rating high (direct evidence: some concerns for risk of bias).

^bNetwork evidence starts rating high (direct evidence: no concerns).

^cNetwork evidence starts rating high (direct evidence: some concerns for risk of bias).

^dNetwork evidence starts rating high (direct evidence: some concerns for risk of bias).

^eNetwork evidence starts rating high (indirect evidence: no concerns), downgraded -1 for serious imprecision.

- ^f Network evidence starts rating high (direct evidence: no concerns).
^g Network evidence starts rating high (direct evidence: some concerns for risk of bias).
^h Network evidence starts rating high (direct evidence: no concerns).
ⁱ Network evidence starts rating high (direct evidence: no concerns).
^j Network evidence starts rating high (direct evidence: no concerns), downgraded –1 for serious imprecision.
^k Network evidence starts rating high (direct evidence: no concerns).

Summary of findings 2. Summary of findings for withdrawals due to adverse events

Withdrawals due to adverse events						
Total studies: 10	Relative effect	Anticipated absolute effect			Certainty of evidence	Ranking
Total participants: 2863	RR (95% CrI)	Events per 1000 person-years (95% CrI)				
		Without intervention	With intervention	Difference		

Patient population: individuals with rheumatoid arthritis who have previously failed a TNFi

Intervention: TNFi not previously tried, IL-6 inhibitor (tocilizumab 4 mg/kg IV every 4 weeks, tocilizumab 8 mg/kg IV every 4 weeks), T-cell co-stimulation inhibitor (abatacept 10 mg/kg (rounded) IV every 4 weeks (+/- IV load at 2 weeks)), B-cell depletor (rituximab 1000 mg IV week 0 and 2), and JAK inhibitor (upadacitinib 15 mg oral once daily, tofacitinib 5 mg oral twice a day, baricitinib 2 mg oral once daily, baricitinib 4 mg oral once daily)

Comparator: placebo

Outcome: withdrawals due to adverse events

Setting: outpatient

TNF inhibitor

TNFi not previously tried (2 RCTs; 343 participants)	0.32 (0.07, 1.1)	107	34 (7 to 111)	73 fewer (99 fewer to 4 more)	 ^a Moderate	Probably less harmful than placebo, but the credible interval crosses the null effect
---	------------------	-----	---------------	-------------------------------	--	---

IL-6 inhibitor

Sarilumab 200 mg SC every 2 weeks (1 RCT; 365 participants)	1.98 (0.57, 7.19)	107	190 (59 to 536)	84 more (48 fewer to 429 more)	 ^b Moderate	Probably more harmful than placebo, but the credible interval crosses the null effect
--	-------------------	-----	-----------------	--------------------------------	--	---

Tocilizumab 4 mg/kg IV every 4 weeks (1 RCT; 319 participants)	1.39 (0.39, 5.28)	107	138 (41 to 431)	31 more (66 fewer to 324 more)	 ^c Low	May be more harmful than placebo, but the credible interval crosses the null effect
Tocilizumab 8 mg/kg IV every 4 weeks (1 RCT; 328 participants)	1.49 (0.51, 4.81)	107	147 (53 to 401)	40 more (54 fewer to 295 more)	 ^d Low	May be more harmful than placebo, but the credible interval crosses the null effect
T-cell co-stimulation inhibitor						
Abatacept 10 mg/kg (rounded) IV every 4 weeks (+/- IV load at 2 weeks) (2 RCTs; 665 participants)	0.92 (0.34, 2.79)	107	93 (36 to 257)	13 fewer (71 fewer to 151 more)	 ^e Low	May be less harmful than placebo, but the credible interval crosses the null effect
Abatacept 125 mg SC weekly (indirect evidence only)	3.11 (0.05, 249.6)	107	282 (5 to 1000)	176 more (101 fewer to 893 more)	 ^f Low	May be more harmful than placebo, but the credible interval crosses the null effect
B-cell depletor						
Rituximab 1000 mg IV week 0 and 2 (1 RCT; 499 participants)	2.38 (0.44, 23.31)	107	224 (46 to 917)	118 more (61 fewer to 810 more)	 ^g Low	May be more harmful than placebo, but the credible interval crosses the null effect
JAK inhibitor						
Upadacitinib 15 mg oral once daily (1 RCT; 333 participants)	0.41 (0.08, 1.83)	107	43 (8 to 177)	64 fewer (98 fewer to 71 more)	 ^h Low	May be less harmful than placebo, but the credible interval crosses the null effect
Tofacitinib 5 mg oral twice a day (1 RCT; 263 participants)	1.37 (0.35, 5.58)	107	136 (37 to 449)	29 more (70 fewer to 449 more)	 ⁱ Low	May be more harmful than placebo, but the credible interval crosses the null effect
Baricitinib 2 mg oral once daily (1 RCT; 350 participants)	0.93 (0.22, 4.01)	107	94 (23 to 348)	12 fewer (83 fewer to 241 more)	 ^j Low	May be less harmful than placebo, but the credible interval crosses the null effect

Baricitinib 4 mg oral once daily (1 RCT; 353 participants)	1.43 (0.38, 5.81)	107	141 (40 to 462)	35 more (67 fewer to 355 more)	   ^k Low	May be more harmful than placebo, but the credible interval crosses the null effect
---	-------------------	-----	-----------------	--------------------------------	---	---

Abbreviations: CrI, credible interval; JAK, Janus kinase; IL-6, interleukin-6; IV, intravenous; RCT, randomised controlled trial; RR, risk ratio; SC, subcutaneous; TNFi, tumour necrosis factor inhibitor

- ^aNetwork evidence starts rating high (direct evidence: some concerns for risk of bias), downgraded –1 for serious imprecision.
^bNetwork evidence starts rating high (direct evidence: some concerns for risk of bias), downgraded –1 for serious imprecision.
^cNetwork evidence starts rating high (direct evidence: some concerns for risk of bias), downgraded –2 for very serious imprecision.
^dNetwork evidence starts rating high (direct evidence: some concerns for risk of bias), downgraded –2 for very serious imprecision.
^eNetwork evidence starts rating high (direct evidence: no concerns), downgraded –2 for very serious imprecision.
^fNetwork evidence starts rating high (indirect evidence: no concerns), downgraded –2 for very serious imprecision.
^gNetwork evidence starts rating high (direct evidence: some concerns for risk of bias), downgraded –2 for very serious imprecision.
^hNetwork evidence starts rating high (direct evidence: no concerns), downgraded –2 for very serious imprecision.
ⁱNetwork evidence starts rating high (direct evidence: no concerns), downgraded –2 for very serious imprecision.
^jNetwork evidence starts rating high (direct evidence: no concerns), downgraded –2 for very serious imprecision.
^kNetwork evidence starts rating high (direct evidence: no concerns), downgraded –2 for very serious imprecision.

Summary of findings 3. Summary of findings for the Disease Activity Score 28 (DAS28) remission (defined as DAS28 < 2.6)

Disease Activity Score 28 (DAS28) remission						
Total studies: 11 Total participants: 2933	Relative effect	Anticipated absolute effect			Certainty of evidence	Ranking
	OR (95% CrI)	Events per 1000 patients (95% CrI)				
		Without intervention	With intervention	Difference		

Patient population: individuals with rheumatoid arthritis who have previously failed a TNFi

Intervention: TNFi not previously tried, IL-6 inhibitors (sarilumab 200 mg SC every 2 weeks, tocilizumab 4 mg/kg IV every 4 weeks, and tocilizumab 8 mg/kg IV every 4 weeks), T-cell co-stimulation inhibitors (abatacept 10 mg/kg (rounded) IV every 4 weeks (+/– IV load at 2 weeks) and abatacept 10 mg/kg (rounded) IV every 4 weeks (+/– IV load at 2 weeks)), B-cell depletor (rituximab 1000 mg IV week 0 and 2), and JAK inhibitors (tofacitinib 5 mg oral twice a day)

Comparator: placebo

Outcome: remission according to Disease Activity Score 28 (DAS28) (defined as DAS28 < 2.6)

Setting: outpatient

TNF inhibitors



TNFi not previously tried (2 RCTs; 343 participants)	8.14 (2.65, 34.54)	15	112 (39 to 348)	96 more (24 more to 333 more)	⊕⊕⊕⊕ ^a High	More effective than placebo
Non-TNF biologic						
Non-TNF biologic (indirect evidence only)	19.33 (4.19, 124.31)	15	230 (61 to 657)	215 more (46 more to 642 more)	⊕⊕⊕⊕ ^b High	More effective than placebo
IL-6 inhibitors						
Sarilumab 200 mg SC every 2 weeks (1 RCT; 365 participants)	5.33 (1.77, 16.69)	15	76 (27 to 205)	61 more (11 more to 190 more)	⊕⊕⊕⊕ ^c High	More effective than placebo
Tocilizumab 4 mg/kg IV every 4 weeks (1 RCT; 319 participants)	4.60 (1.21, 13.79)	15	66 (18 to 175)	51 more (3 more to 160 more)	⊕⊕⊕○ ^d Moderate	Probably more effective than placebo
Tocilizumab 8 mg/kg IV every 4 weeks (1 RCT; 328 participants)	22.8 (8.29, 78.5)	15	260 (113 to 548)	245 more (98 more to 533 more)	⊕⊕⊕⊕ ^e High	More effective than placebo
T-cell co-stimulation inhibitor						
Abatacept 125 mg SC weekly (indirect evidence only)	3.39 (0.31, 29.04)	15	50 (5 to 310)	35 more (10 fewer to 294 more)	⊕⊕⊕○ ^f Moderate	Probably more effective than placebo, but the credible interval crosses the null effect
Abatacept 10 mg/kg (rounded) IV every 4 weeks (+/- IV load at 2 weeks) (2 RCTs; 665 participants)	21.99 (2.99, 927.69)	15	253 (44 to 935)	238 more (29 more to 920 more)	⊕⊕⊕⊕ ^g High	More effective than placebo
B-cell depletor						
Rituximab 1000 mg IV week 0 and 2 (1 RCT; 499 participants)	5.51 (1.76, 21.35)	15	78 (26 to 248)	63 more (11 more to 233 more)	⊕⊕⊕⊕ ^h High	More effective than placebo
JAK inhibitors						
Tofacitinib 5 mg oral twice a day	4.95 (0.87, 44.54)	15	71 (13 to 407)	56 more (2 fewer to 392 more)	⊕⊕⊕○ ⁱ Moderate	Probably more effective than placebo, but the

(1 RCT; 263 participants)

credible interval crosses
the null effect

Abbreviations: CrI, credible interval; JAK, Janus kinase; IL-6, interleukin-6; IV, intravenous; OR, odds ratio; RCT, randomised controlled trial; SC, subcutaneous; TNFi, tumour necrosis factor inhibitor

- ^aNetwork evidence starts rating high (direct evidence: some concerns for risk of bias).
^bNetwork evidence starts rating high (indirect evidence: no concerns).
^cNetwork evidence starts rating high (direct evidence: no concerns).
^dNetwork evidence starts rating high (direct evidence: some concerns for risk of bias), downgraded –1 for serious imprecision.
^eNetwork evidence starts rating high (direct evidence: some concerns for risk of bias).
^fNetwork evidence starts rating high (indirect evidence: no concerns), downgraded –1 for serious imprecision.
^gNetwork evidence starts rating high (direct evidence: no concerns).
^hNetwork evidence starts rating high (direct evidence: some concerns for risk of bias).
ⁱNetwork evidence starts rating high (direct evidence: no concerns), downgraded –1 for serious imprecision.

Summary of findings 4. Summary of findings for pain as measured by visual analogue scale (VAS)

Visual analogue scale (VAS), 0 to 100 points, 0 is no pain

Total studies: 3	Relative effect	Anticipated absolute effect			Certainty of evidence	Ranking
Total participants: 1168	MD (95% CrI)	Mean (95% CrI)				
		Without intervention	With intervention	Difference		

Patient population: individuals with rheumatoid arthritis who have previously failed a TNFi

Intervention: JAK inhibitors (upadacitinib 15 mg oral once daily, baricitinib 2 mg oral once daily, baricitinib 4 mg oral once daily)

Comparator: placebo

Outcome: pain as measured by visual analogue scale (VAS), 0-to-100-point scale, lower scores indicate less pain

Setting: outpatient

JAK inhibitors

Upadacitinib 15 mg oral once daily (1 RCT; 333 participants)	–16.25 (–20.86, –11.77)	61.5	45.25 (40.64 to 49.73)	16.25 lower (20.86 lower to 11.77 lower)	 ^a High	More effective than placebo
---	-------------------------	------	------------------------	--	--	-----------------------------

Baricitinib 2 mg oral once daily (1 RCT; 350 participants)	-7.5 (-11.7, -3.4)	61.5	54 (49.8 to 58.1)	7.5 lower (11.7 lower to 3.4 lower)	⊕⊕⊕⊕ ^b High	More effective than placebo
Baricitinib 4 mg oral once daily (1 RCT; 353 participants)	-4.02 (-8.39, 0.52)	61.5	57.48 (53.11 to 62.02)	4.02 lower (8.39 lower to 0.52 higher)	⊕ ○ ○ ○ ^c Very low	The evidence is very uncertain about the effect.

Abbreviations: CrI, credible interval; JAK, Janus kinase; MD, mean difference; RCT, randomised controlled trial; TNFi, tumour necrosis factor inhibitor

^aNetwork evidence starts rating high (direct evidence: no concerns).

^bNetwork evidence starts rating high (direct evidence: no concerns).

^cNetwork evidence starts rating high (direct evidence: no concerns), downgraded -3 due to extremely serious imprecision.

Summary of findings 5. Summary of findings for the Health Assessment Questionnaire (HAQ)

Health Assessment Questionnaire (HAQ), 0-to-3 scale, lower scores indicate better functioning					
Total studies: 15 Total participants: 4289	Relative effect MD (95% CrI)	Anticipated absolute effect			Certainty of evidence
		Mean (95% CrI)			
		Without intervention	With intervention	Difference	Ranking

Patient population: individuals with rheumatoid arthritis who have previously failed a TNFi

Intervention: TNFi not previously tried, non-TNF biologic, IL-6 inhibitors (sarilumab 200 mg SC every 2 weeks, tocilizumab 4 mg/kg IV every 4 weeks, and tocilizumab 8 mg/kg IV every 4 weeks), T-cell co-stimulation inhibitors (abatacept 10 mg/kg (rounded) IV every 4 weeks (+/- IV load at 2 weeks) and abatacept 10 mg/kg (rounded) IV every 4 weeks (+/- IV load at 2 weeks)), B-cell depletor (rituximab 1000 mg IV week 0 and 2), and JAK inhibitors (upadacitinib 15 mg oral once daily, tofacitinib 5 mg oral twice a day, baricitinib 2 mg oral once daily, baricitinib 4 mg oral once daily)

Comparator: placebo

Outcome: Health Assessment Questionnaire (HAQ), which assesses pain and disability relating to arthritis, 0-to-3 scale, lower scores indicate better functioning

Setting: outpatient

TNF inhibitors

TNFi not previously tried (2 RCTs; 343 participants)	-0.19 (-0.5, 0.07)	1.56	1.37 (1.06 to 1.63)	0.19 lower (0.5 lower to 0.07 higher)	⊕⊕⊕○ ^a Moderate	Probably not different than placebo
---	--------------------	------	---------------------	---------------------------------------	-------------------------------	-------------------------------------

Non-TNF biologic						
Non-TNF biologic (indirect evidence only)	-0.16 (-0.62, 0.22)	1.56	1.4 (0.94 to 1.78)	0.16 lower (0.62 lower to 0.22 higher)	⊕⊕○○ ^b Low	May not be different than placebo
IL-6 inhibitors						
Sarilumab 200 mg SC every 2 weeks (1 RCT; 365 participants)	-0.24 (-0.55, 0.07)	1.56	1.32 (1.01 to 1.63)	0.24 lower (0.55 lower to 0.07 higher)	⊕⊕⊕○ ^c Moderate	Probably more effective than placebo, but the credible interval crosses the null effect
Tocilizumab 4 mg/kg IV every 4 weeks (1 RCT; 319 participants)	-0.26 (-0.39, -0.13)	1.56	1.3 (1.17 to 1.43)	0.26 lower (0.39 lower to 0.13 lower)	⊕⊕⊕⊕ ^d High	More effective than placebo
Tocilizumab 8 mg/kg IV every 4 weeks (1 RCT; 328 participants)	-0.25 (-0.44, -0.05)	1.56	1.31 (1.12 to 1.51)	0.25 lower (0.44 lower to 0.05 lower)	⊕⊕⊕⊕ ^e High	More effective than placebo
T-cell co-stimulation inhibitor						
Abatacept 125 mg SC weekly (indirect evidence only)	-0.3 (-0.97, 0.35)	1.56	1.36 (0.59 to 1.91)	0.3 lower (0.97 lower to 0.35 higher)	⊕⊕○○ ^f Low	May be more effective than placebo, but the credible interval crosses the null effect
Abatacept 10 mg/kg (rounded) IV every 4 weeks (+/- IV load at 2 weeks) (2 RCTs; 665 participants)	-0.33 (-0.5, -0.1)	1.56	1.23 (1.06 to 1.46)	0.33 lower (0.55 lower to 0.1 lower)	⊕⊕⊕⊕ ^g High	More effective than placebo
B-cell depletor						
Rituximab 1000 mg IV week 0 and 2 (1 RCT; 499 participants)	-0.27 (-0.48, -0.04)	1.56	1.29 (1.08 to 1.52)	0.27 lower (0.48 lower to 0.04 lower)	⊕⊕⊕⊕ ^h High	More effective than placebo
JAK inhibitors						
Upadacitinib 15 mg oral once daily	-0.12 (-0.43, 0.19)	1.56	1.44 (1.13 to 1.75)	0.12 lower (0.43 lower to 0.19 higher)	⊕⊕○○ ⁱ Low	May not be different than placebo

(1 RCT; 333 participants)						
Tofacitinib 5 mg oral twice a day (1 RCT; 263 participants)	−0.24 (−0.57, 0.1)	1.56	1.32 (0.99 to 1.66)	0.24 lower (0.57 lower to 0.1 higher)	⊕⊕⊕O ^j Moderate	Probably more effective than placebo, but the credible interval crosses the null effect
Baricitinib 2 mg oral once daily (1 RCT; 350 participants)	−0.28 (−0.58, 0.03)	1.56	1.28 (0.98 to 1.59)	0.28 lower (0.58 lower to 0.03 higher)	⊕⊕⊕O ^k Moderate	Probably more effective than placebo, but the credible interval crosses the null effect
Baricitinib 4 mg oral once daily (1 RCT; 353 participants)	−0.3 (−0.61, 0.01)	1.56	1.26 (0.95 to 1.57)	0.3 lower (0.61 lower to 0.01 higher)	⊕⊕⊕O ^l Moderate	Probably more effective than placebo, but the credible interval crosses the null effect
Abbreviations: CrI, credible interval; JAK, Janus kinase; IL-6, interleukin-6; IV, intravenous; MD, mean difference; RCT, randomised controlled trial; SC, subcutaneous; TNFi, tumour necrosis factor inhibitor						

^aNetwork evidence starts rating high (direct evidence: some concerns for risk of bias), downgraded −1 for serious imprecision.

^bNetwork evidence starts rating high (indirect evidence: no concerns), downgraded −2 for very serious imprecision.

^cNetwork evidence starts rating high (direct evidence: no concerns), downgraded −1 for serious imprecision.

^dNetwork evidence starts rating high (direct evidence: some concerns for risk of bias).

^eNetwork evidence starts rating high (direct evidence: some concerns for risk of bias).

^fNetwork evidence starts rating high (indirect evidence: no concerns), downgraded −2 due to extremely serious imprecision.

^gNetwork evidence starts rating high (direct evidence: no concerns).

^hNetwork evidence starts rating high (direct evidence: no concerns).

ⁱNetwork evidence starts rating high (direct evidence: no concerns), downgraded −2 for very serious imprecision.

^jNetwork evidence starts rating high (direct evidence: no concerns), downgraded −1 for serious imprecision.

^kNetwork evidence starts rating high (direct evidence: no concerns), downgraded −1 for serious imprecision.

^lNetwork evidence starts rating high (direct evidence: no concerns), downgraded −1 for serious imprecision.

Summary of findings 6. Summary of findings for serious adverse events

Serious adverse events				
Total studies: (13)	Relative effect	Anticipated absolute effect		
Total participants: (3821)	RR (95% CrI)	Events per 1000 patient years (95% CrI)		
		Without inter- vention	With interven- tion	Difference

Patient population: individuals with rheumatoid arthritis who have previously failed a TNFi

Intervention: TNFi not previously tried, IL-6 inhibitors (sarilumab 200 mg SC every 2 weeks, tocilizumab 4 mg/kg IV every 4 weeks, and tocilizumab 8 mg/kg IV every 4 weeks), T-cell co-stimulation inhibitors (abatacept 10 mg/kg (rounded) IV every 4 weeks (+/- IV load at 2 weeks)), B-cell depletor (rituximab 1000 mg IV week 0 and 2), and JAK inhibitors (upadacitinib 6 mg oral twice a day, upadacitinib 15 mg oral once daily, tofacitinib 5 mg oral twice a day, baricitinib 2 mg oral once daily, baricitinib 4 mg oral once daily)

Comparator: placebo

Outcome: serious adverse events

Setting: outpatient

TNF inhibitors

TNFi not previously tried (2 RCTs; 343 participants)	0.43 (0.14, 1.24)	187	77 (26 to 207)	110 fewer (161 fewer to 20 more)	⊕ ○ ○ ○ ^a Very low	The evidence is very uncertain about the effect.
---	-------------------	-----	----------------	----------------------------------	----------------------------------	--

Non-TNF biologic

Non-TNF biologic (indirect evidence only)	0.89 (0.17, 4.4)	187	153 (31 to 561)	34 fewer (156 fewer to 374 more)	⊕ ○ ○ ○ ^b Very low	The evidence is very uncertain about the effect.
--	------------------	-----	-----------------	----------------------------------	----------------------------------	--

IL-6 inhibitors

Sarilumab 200 mg SC every 2 weeks (1 RCT; 365 participants)	1.01 (0.35, 2.9)	187	172 (63 to 419)	15 fewer (124 fewer to 232 more)	⊕ ○ ○ ○ ^c Very low	The evidence is very uncertain about the effect.
--	------------------	-----	-----------------	----------------------------------	----------------------------------	--

Tocilizumab 4 mg/kg IV every 4 weeks (1 RCT; 319 participants)	0.81 (0.33, 2.13)	187	141 (60 to 328)	46 fewer (127 fewer to 142 more)	⊕ ○ ○ ○ ^d Very low	The evidence is very uncertain about the effect.
---	-------------------	-----	-----------------	----------------------------------	----------------------------------	--

Tocilizumab 8 mg/kg IV every 4 weeks (2 RCT; 489 participants)	0.76 (0.33, 1.8)	187	132 (60 to 286)	54 fewer (127 fewer to 99 more)	⊕ ○ ○ ○ ^e Very low	The evidence is very uncertain about the effect.
---	------------------	-----	-----------------	---------------------------------	----------------------------------	--

T-cell co-stimulation inhibitor

Abatacept 125 mg SC weekly (indirect evidence only)	0.87 (0.15, 4.94)	187	150 (28 to 603)	37 fewer (159 fewer to 416 more)	⊕ ○ ○ ○ ^f Very low	The evidence is very uncertain about the effect.
--	-------------------	-----	-----------------	----------------------------------	----------------------------------	--

Abatacept 10 mg/kg (rounded) IV every 4 weeks (+/- IV load at 2 weeks) (2 RCTs; 665 participants)	0.61 (0.28, 1.26)	187	108 (51 to 210)	79 fewer (136 fewer to 23 more)	⊕⊕⊕○ ^g Moderate	Probably less harmful than placebo, but the credible inter- val crosses the null effect
B-cell depletor						
Rituximab 1000 mg IV week 0 and 2 (1 RCT; 499 participants)	0.6 (0.36, 1.72)	187	106 (47 to 275)	81 fewer (140 fewer to 88 more)	⊕⊕⊕○ ^h Moderate	Probably less harmful than placebo, but the credible inter- val crosses the null effect
JAK inhibitors						
Upadacitinib 15 mg oral once daily (1 RCT; 333 participants)	NE [‡]	NE [‡]	NE [‡]	NE [‡]	NE ^{i‡}	NE [‡]
Tofacitinib 5 mg oral twice a day (1 RCT; 263 participants)	0.29 (0.03, 1.6)	187	53 (6 to 259)	134 fewer (181 few- er to 72 more)	⊕⊕ ○ ○ ^j Low	May be less harmful than place- bo, but the credible interval crosses the null effect
Baricitinib 2 mg oral once daily (1 RCT; 350 participants)	0.49 (0.13, 1.63)	187	84 (24 to 262)	103 fewer (163 few- er to 76 more)	⊕⊕ ○ ○ ^k Low	May be less harmful than place- bo, but the credible interval crosses the null effect
Baricitinib 4 mg oral once daily (1 RCT; 353 participants)	1.25 (0.41, 3.9)	187	208 (74 to 517)	21 more (113 fewer to 331 more)	⊕ ○ ○ ○ ^l Very low	The evidence is very uncertain about the effect.
Abbreviations: CrI, credible interval; JAK, Janus kinase; IL-6, interleukin-6; IV, intravenous; NE, not estimable; RCT, randomised controlled trial; RR, risk ratio; SC, subcuta- neous; TNFi, tumour necrosis factor inhibitor						

‡ Insufficient data

^aNetwork evidence starts rating high (direct: some concerns for risk of bias), downgraded -3 for extremely serious imprecision.

^bNetwork evidence starts rating high (direct: some concerns for risk of bias), downgraded -3 for extremely serious imprecision.

^cNetwork evidence starts rating high (direct), downgraded -3 for extremely serious imprecision.

^dNetwork evidence starts rating high (direct: some concerns for risk of bias), downgraded -3 for extremely serious imprecision.

^eNetwork evidence starts rating high (direct: some concerns for risk of bias), downgraded -3 for extremely serious imprecision.

^fNetwork evidence starts rating high (indirect), downgraded -3 for extremely serious imprecision.

^gNetwork evidence starts rating high (direct), downgraded -1 for serious imprecision.

^hNetwork evidence starts rating high (direct: some concerns for risk of bias), downgraded -1 for serious imprecision.

ⁱNetwork estimate not estimable due to insufficient data - certainty rating not applicable.

^jNetwork evidence starts rating high (direct), downgraded -2 for very serious imprecision.

^kNetwork evidence starts rating high (direct), downgraded –2 for very serious imprecision.
^lNetwork evidence starts rating high (direct), downgraded –3 for extremely serious imprecision.

BACKGROUND

Description of the condition

Rheumatoid arthritis is a chronic systemic autoimmune disease estimated to affect 0.5% to 1% of the worldwide general population [1]. The synovial lining of joints is the most common target of the dysregulated immune response, although the disease can affect many organs including the skin, heart, lungs, and nervous system [2]. People with rheumatoid arthritis experience joint swelling, pain, and resultant disability. The pathophysiology is complex, involving an interplay between genetic and environmental factors [2]. While there is currently no cure for rheumatoid arthritis, many treatment options that control the activity of the disease are now available.

Description of the intervention and how it might work

The cornerstone of pharmacologic treatment for rheumatoid arthritis is disease-modifying antirheumatic drugs (DMARDs), which work to control the inflammation responsible for symptoms and also prevent joint damage. DMARDs can be categorised according to their structure and mechanisms of action. Conventional synthetic DMARDs, such as methotrexate, were not designed for a specific molecular target [3]. Conversely, targeted synthetic DMARDs, such as tofacitinib, are a newer class of medications, designed with a specific molecular target in mind. Biologic DMARDs are derived biologically, and similar to targeted synthetic DMARDs, were designed to target specific cells or proteins involved in the inflammatory response present in rheumatoid arthritis [3]. Given these are biologically derived proteins, it is impossible to exactly reproduce their complex structure. As such, biologic DMARDs can be further classified into bio-originator and biosimilar DMARDs, to specify whether they were made by the first or subsequent manufacturers.

All DMARDs target immune pathways to treat the immune dysregulation. The exact mechanism of action varies. Conventional synthetic DMARDs work through non-specific effects, typically targeting multiple immune pathways [3]. In contrast, biologic DMARDs and targeted synthetic DMARDs are designed to target specific immune processes [3]. The specific targets include cytokines that are proteins involved in cell signalling, such as tumour necrosis factor (TNF; targeted by anti-TNF agents, of which there are five currently available: adalimumab, certolizumab, etanercept, golimumab, infliximab), interleukin-6 (IL-6; targeted by tocilizumab and sarilumab), and interleukin-1 (IL-1; anakinra, although this is now seldom used). Other targets include T-cell co-stimulation molecules (targeted by abatacept) and markers of B-cells (targeted by anti-CD20, rituximab). To date, approved targeted synthetic DMARDs for rheumatoid arthritis (tofacitinib, baricitinib, upadacitinib) target the Janus kinase (JAK) signalling pathway, which are important intracellular molecules involved in signal transduction [4]. While biologic DMARDs share similarities, they work through different mechanisms of action and therefore may have different effects and side effect profiles.

Why it is important to do this review

Treatment options for rheumatoid arthritis have expanded considerably over the past 20 years, and new treatments continue to emerge. This has led to greater options for patients and clinicians and has contributed to improved outcomes for patients. Making evidence-informed choices between these treatment

options requires timely, high-quality synthesis of the best available evidence. Accomplishing this is challenging for a few reasons.

One challenge in developing meaningful evidence summaries is that trials are often designed with non-active (e.g. placebo) comparators [5]. While such comparisons may initially be necessary to prove the efficacy of new treatments, later on decision-makers need comparisons of different active treatments for their benefits and harms. To this end, network meta-analyses (NMAs) have emerged as a popular tool in comparative effectiveness research [6, 7]. NMAs use all the available data and combine both direct evidence (from head-to-head trials) and indirect evidence (where two treatments have been compared with the same third treatment) to provide pair-wise comparisons between all treatments.

Another challenge is keeping evidence summaries up to date. A potential solution is the 'living' systematic review approach, which is designed to incorporate new evidence as it becomes available over time [8]. A living systematic review approach is well-suited to rheumatoid arthritis drug therapy for several reasons. First, drug treatment for rheumatoid arthritis is consistently a high-priority topic for clinical societies and health policy decision-makers, given the availability of multiple effective, but costly, treatments. Second, while treatment outcomes relative to placebo are precise for many treatments, comparisons between treatments are not. In recent American College of Rheumatology (ACR) guidelines, the quality of evidence for most recommendations was low, reflecting the fact that there is considerable uncertainty when comparing active treatments [9]. There are, however, a growing number of comparative effectiveness trials that may help narrow the imprecision of these comparisons. Finally, as new treatments continue to emerge, it is necessary to understand how these fit in with existing therapeutic alternatives.

The aim of this review was to establish a sustainable living NMA that provides timely, high-quality, and reproducible evidence summaries for international guideline groups. We have established collaborations with the Canadian Rheumatology Association (CRA), Australia and New Zealand Musculoskeletal (ANZMUSC) Clinical Trials Network, and the Australian Rheumatology Association (ARA). These groups are using the results of this living systematic review to inform the development of trustworthy clinical practice guidelines in their respective regions [10, 11]. To ensure the feasibility and sustainability of the project, we have incorporated efficiencies throughout the review, including machine-learning and crowdsourcing approaches to identify and classify clinical trials [12]; use of Covidence [13], an online software platform for data extraction and workflow management; and automation of NMA analyses. To ensure transparency and reproducibility of our work, the model codes and readable datasets are hosted on an Open Science Framework website associated with this project [14].

This review compares the benefits and harms of DMARD therapy in people with rheumatoid arthritis after an inadequate response to advanced therapy (biologic DMARDs or targeted synthetic DMARDs). This is one of a series of three reviews. The other two reviews compare DMARD therapy at other decision points: 1) as initial therapy; and 2) after an inadequate response to conventional synthetic DMARDs. A common protocol for all three reviews has been published [15]. These reviews were conducted according to the guidelines recommended by the Cochrane Musculoskeletal Editorial Board and Cochrane guidance for living systematic

reviews [16, 17]. All changes made to the protocol are summarised in the 'Differences between protocol and review' section in the [Methods](#).

OBJECTIVES

The aim of this living systematic review and network meta-analysis was to compare the benefits and harms of disease-modifying antirheumatic drugs (DMARDs) after failure of biologic or targeted synthetic DMARDs in adults with rheumatoid arthritis.

METHODS

We followed the Methodological Expectations of Cochrane Intervention Reviews (MECIR) when conducting the review and PRISMA 2020 for the reporting [18, 19].

Differences between protocol and review

We conducted the review according to the published protocol. Any deviations from the protocol are outlined below.

Changes to selection of studies

Three questions embedded in the Population, Intervention, Comparator, and Outcome (PICO) annotation tool were removed. These questions were intended to facilitate the classification of identified randomised controlled trials (RCTs), but further evaluation found them to be of limited help [12].

Changes to types of interventions

While not applicable to this review, we made changes to eligible interventions across the three planned reviews, including the removal of subcutaneous methotrexate (scMTX) as a node. Given the bioavailable dose of MTX depends on both the route and the dose, and MTX doses vary across trials, having a separate node for scMTX was considered inappropriate. Prior research has not found MTX dose to be an effect modifier [20]. Intramuscular (IM) gold was excluded, as the production of IM gold was discontinued globally in 2019 [21]. Changes applicable to this review include the removal of intervention ranking and the inclusion of a baricitinib dose of 4 mg, which is approved in some countries (e.g. Australia) [22, 23]. We decided to remove intervention ranking because of the limitations in the interpretation and potential for misleading inferences, especially when differences between ranks are minimal [24].

Changes to synthesis method

Informative prior probability distributions were used for the variance of random effects across all analyses, instead of uninformative priors. Many networks had too sparse data to estimate this variance precisely; using an informative prior borrows information from the literature regarding plausible variation in random effects across studies [25, 26]. The results were compared in sensitivity analyses to an uninformative prior and a Bayesian fixed-effect model. In general, the estimates in the informative prior aligned with the informative prior in the Bayesian model. For continuous outcomes, final values were used with planned imputation methods.

Criteria for considering studies for this review

Types of studies

We included RCTs published in peer-reviewed journals, including cross-over RCTs, but excluding cluster-randomised trials. Finally, we identified conference abstracts but did not include them in the review, primarily because reporting of results in abstracts can often differ from the full-text reports.

Types of participants

We included studies that recruited adults aged 18 years or older, diagnosed with rheumatoid arthritis according to 1958, 1987, or 2010 classification criteria [27, 28, 29]. We included studies with a mixed population (e.g. undifferentiated inflammatory arthritis), provided data were reported separately for the subgroup of participants with rheumatoid arthritis, according to defined classification criteria, or if at least 75% of the overall population consisted of people with rheumatoid arthritis, even if stratified data were not available. Participants in the trial had to have active disease, as defined by the study. Studies that evaluated a switch to a different medication in participants with well-controlled disease were excluded. We also excluded studies where all participants were required to have another diagnosis in addition to rheumatoid arthritis (e.g. amyloidosis, diabetes).

We conducted separate NMAs according to the prior therapy received, to align with the decision points in rheumatoid arthritis therapy. Prior therapy is also an important effect modifier for rheumatoid arthritis treatments, as previously demonstrated [20, 30]. This supported our decision to create separate networks based on the prior biologic DMARD or targeted synthetic DMARD, thereby ensuring comparability of participants and the plausibility of the transitivity assumption. For each NMA, all participants in the trial were required to have failed:

1. TNF inhibitor;
2. two TNF inhibitors;
3. IL-6 inhibitor;
4. rituximab;
5. abatacept;
6. JAK inhibitor.

The definition of a treatment failure was at the discretion of the study investigators. We excluded trials where less than 80% of participants were required to have failed treatment, unless subgroup data were available. Trials were excluded if the prior treatment exposure was not reported in sufficient detail to assign the trial to one of the networks. These trials are included in the 'Characteristics of excluded studies' table.

Living systematic review considerations

As new evidence emerges, we may add new networks. For example, future trials may be conducted in people who have failed two IL-6 inhibitors or after a certain sequence of two or more biologics.

Types of interventions

The interventions listed below were eligible.

1. Conventional synthetic DMARDs: methotrexate, antimalarials (hydroxychloroquine/chloroquine), sulfasalazine, leflunomide, cyclosporin, and azathioprine, of any dosing strategy.

2. Biologic DMARDs: TNF inhibitors (TNFi) (adalimumab, certolizumab, etanercept, golimumab, infliximab), T-cell co-stimulation inhibitors (abatacept), B-cell depletors (rituximab), IL-6 inhibitors (tocilizumab, sarilumab), and IL-1 inhibitors (anakinra). For this group, the eligible dosing strategies reflect the approved doses, and are as follows:
 - a. anakinra 100 mg subcutaneously, once daily;
 - b. adalimumab 40 mg subcutaneously, every two weeks;
 - c. certolizumab 200 mg subcutaneously, every two weeks; or 400 mg subcutaneously, every four weeks with or without initial loading doses of 400 mg subcutaneously at baseline, two, and four weeks;
 - d. etanercept 50 mg subcutaneously, every week; or 25 mg subcutaneously, twice-weekly;
 - e. golimumab 50 mg subcutaneously, every four weeks;
 - f. golimumab 2 mg/kg intravenously, every eight weeks after initial dosing at baseline and four weeks;
 - g. infliximab 3 mg/kg intravenously, every eight weeks after initial dosing at baseline, two, and six weeks;
 - h. abatacept intravenously, every four weeks at ~10 mg/kg (500 mg in participants < 60 kg, 750 mg in participants 60 kg to 100 kg, and 1000 mg in participants > 100 kg), after the initial dosing regimen at baseline, two, and four weeks;
 - i. abatacept 125 mg subcutaneously, every week, with or without an intravenous loading dose of ~10 mg/kg;
 - j. rituximab, two 1000 mg intravenous doses two weeks apart;
 - k. tocilizumab intravenously, every four weeks at 4 mg/kg or 8 mg/kg (each dose was considered separately);
 - l. tocilizumab 162 mg subcutaneously, every week;
 - m. sarilumab 200 mg subcutaneously, every two weeks.
3. Targeted synthetic DMARDs. The eligible dosing strategies reflect the approved doses, and are as follows:
 - a. tofacitinib 5 mg orally, twice daily; or 11 mg orally, once daily (considered as the same dose);
 - b. baricitinib 2 mg, once daily or baricitinib 4 mg, once daily (each dose would be considered separately). The approved dose varies by country;
 - c. upadacitinib 15 mg, once daily.

The above interventions comprise the complete set of interventions across all three reviews (DMARD-naïve, DMARD-inadequate response, biologic-inadequate response), although it was expected for this review (biologic-inadequate response) that only trials of other biologic DMARDs or targeted synthetic DMARDs would be identified. All the eligible treatments in this review are options available to patients after failure of biologic/targeted synthetic DMARDs and can be jointly randomised. The list of conventional synthetic DMARDs excludes older drugs that are no longer in use (e.g. oral gold, IM gold, penicillamine). We excluded trials that evaluated the effect of glucocorticoids alone as an intervention (e.g. prednisone versus placebo), but included trials where glucocorticoids were part of an intervention being compared with other DMARDs. We excluded trials that administered other co-interventions (e.g. vaccination) as part of the intervention for all participants.

We also excluded trials evaluating the effect of biosimilars, which are biological pharmaceuticals that are almost identical to an original product manufactured by a different company (e.g. CT-

P13 is an infliximab biosimilar). It was judged that to adequately address the safety and efficacy of biosimilars, a separate review should be conducted that includes the complete evidence base for these therapies. Many biosimilar trials compare a switch to a biosimilar to continuation of the originator in patients with well-controlled disease; these trials would not be eligible for this review. Additionally, biosimilars are often approved for a condition without direct evidence in that condition.

Biologic doses were restricted to the approved doses, whereas conventional synthetic DMARD doses were not restricted. This is explained in the description of the interventions above. We included combinations of conventional synthetic DMARDs, with each combination analysed separately, but excluded trials that evaluated combinations of biologic/targeted synthetic DMARDs.

A trial had to compare at least one eligible intervention against placebo/no treatment or against another eligible intervention, and have planned to administer the intervention(s) for at least 12 weeks. We excluded trials that compared different treatment strategies (e.g. trials using multiple therapies in a defined sequence), unless there was an initial period of at least 12 weeks when treatment changes were not allowed. In these trials, this initial period (prior to the switching) was used as the end of the trial date.

Defining the nodes of each network

The interventions listed above, which include all currently used DMARDs in rheumatoid arthritis, formed our comparison set. We considered each DMARD or DMARD combination as separate nodes. A treatment node could also be defined as a set of treatments that people could choose from, rather than one specific drug, provided all interventions were eligible for the review. For example, in a trial that evaluated a switch to a different TNFi to a switch to a non-TNF biologic therapy, separate nodes were defined (Gottenberg 2016 [31]). Placebo/no treatment arms formed an additional node that provided indirect evidence between our interventions of interest.

It is possible that additional indirect evidence was available through an intervention we have not considered in our list. These may be interventions that are no longer commonly used due to efficacy or safety considerations (e.g. auranofin), or interventions that have not yet been approved. We excluded these trials, as these interventions may not be proven to have disease-modifying properties, and the assumption of exchangeability of effect-modifiers made in an NMA may therefore not hold [32].

Living systematic review considerations

While these interventions are currently treatments of interest, additional interventions or new doses of existing interventions may be approved for use in the future. When a new treatment is approved for use by a regulatory agency (e.g. US Food and Drug Administration, Health Canada, Therapeutic Goods Administration (Australia), or European Medicines Agency), we will add the treatment to the review, at its approved dose. We may also modify the networks if we identify new DMARD combinations, or if different doses of biologic/targeted synthetic DMARDs are approved in the future.

Outcome measures

Critical outcomes

We assessed seven critical outcomes: three critical outcomes and four other critical outcomes. We chose the three critical outcomes to reflect important and distinct treatment benefits (control of disease activity, prevention of joint damage), as well as treatment harms. We chose the American College of Rheumatology 50% response (ACR50) (defined below) over other measures of disease improvement/control given its wide use in clinical trials, as well as its importance over other less stringent measures of disease improvement (e.g. ACR20 response).

Critical outcomes

1. ACR50 response (50% improvement in both tender and swollen joint counts and 50% improvement in three of the following five variables: patient global assessment, physician global assessments, pain scores, physical function (Health Assessment Questionnaire (HAQ) score), and acute-phase reactants (erythrocyte sedimentation rate (ESR) or C-reactive protein (CRP)) [33]. These seven outcomes form the ACR core set of outcome measures for rheumatoid arthritis clinical trials [34]. In addition to the composite response criteria, we also assessed pain and physical function (see below), as these individual components of the core set are of high importance to people with rheumatoid arthritis.
2. Radiographic progression as a continuous variable, measured by Larsen, Sharp or modified Larsen/Sharp scores, including Scott-modified Larsen, Genant-modified Sharp, and van der Heijde-modified Sharp [35].
3. Withdrawals due to adverse events, including death.

Other critical outcomes

1. Disease Activity Score (DAS) 28 remission, defined as DAS < 1.6 or DAS28 < 2.6 [36].
2. Pain intensity (mean or mean change, measured by visual analogue scale (VAS), numeric or categorical rating scale).
3. Functional limitation or disability, as measured by the Health Assessment Questionnaire-Disability Index (HAQ-DI) [37], modified HAQ (mHAQ) [38], 36-item Short Form Health Survey (SF-36) physical function component, or SF-12 physical function component. If a study reported multiple measures, we extracted them all and analysed them separately.
4. Serious adverse events, as defined by the study, including deaths.

Important outcomes

1. ACR20 response [33].
2. ACR70 response [33].
3. DAS [39], or DAS28 [40]. DAS28 in its original form includes four variables: ESR, patient global assessment of disease activity, tender and swollen joint counts (DAS28-4-ESR). It has been modified to include just three variables (excluding global assessment) and with the CRP instead of ESR. We extracted the original DAS28-4-ESR if it was reported in multiple formats.
4. Quality of life, as measured by EQ-5D, SF-36 (mental component score, MCS), or SF-12 (MCS). If a study reported multiple measures, we extracted them all and analysed them separately.

5. Fatigue, as defined by the study, measured using a validated scale.
6. Serious infections, as defined by the study.

We also included and described the primary findings of trials that did not report any of our specified outcomes.

Living systematic review considerations

Outcome measurement in rheumatoid arthritis may evolve over time. As other outcome measures become increasingly reported in trials or desired by decision-makers, we may modify the outcomes evaluated in this review.

Search methods for identification of studies

Electronic searches

We conducted the following database searches:

1. Cochrane Central Register of Controlled Trials via Ovid (CENTRAL; Issue 10, 2025);
2. MEDLINE via Ovid (1946 to 28 November 2025);
3. Embase via Ovid (1980 to 28 November 2025);

We used the Cochrane Highly Sensitive Search Strategy for identifying RCTs and controlled clinical trials (as described in Chapter 4 of the *Cochrane Handbook for Systematic Reviews of Interventions*), as well as MeSH and free text terms for the concept 'rheumatoid arthritis' [41]. See [Supplementary material 1](#) for the search strategies. We supplemented this with a search of clinical trial registries ClinicalTrials.gov (clinicaltrials.gov) and the World Health Organization International Clinical Trials Registry Platform (WHO ICTRP) (trialsearch.who.int), using the search term 'rheumatoid arthritis'.

We intentionally designed our search not to include terms for the 'intervention'. This allowed us to identify trials of all current and future approved DMARDs, necessary for our living review and to establish a registry of all randomised trials in rheumatoid arthritis, regardless of the intervention. Our hope is that this can be used in the future to facilitate efficient reviews of interventions other than the ones considered under our eligibility criteria. While our broad search expanded the volume of records, we used project enhancements (e.g. Cochrane Crowd) [42] to facilitate the screening process, as discussed below.

We searched all databases from their inception to 28 November 2025, without restrictions on time or language of publication.

Living systematic review considerations

We will conduct search updates monthly in all of the above databases. The search results will first be filtered through Cochrane Crowd to filter out non-RCTs before being screened by the review team for potential inclusion [42]. Search methods and strategies will be reviewed approximately yearly, to ensure they reflect any terminology changes in the topic area, or in the databases.

Searching other resources

As part of the base review, we screened existing Cochrane reviews to ensure we had identified all relevant clinical trials. Going forward, we will follow the MECIR standard to search the reference lists of included studies and relevant systematic reviews, exploring

the use of artificial intelligence (AI) tools to facilitate this step. We will continue to follow Cochrane policy for managing potentially problematic studies by checking for post-publication amendments to the included studies [41]. As this is a living systematic review, our search strategy is regularly updated to capture newly published studies and post-publication amendments (e.g. errata, corrigenda, expressions of concern, or retractions). These will be incorporated in subsequent updates of the review.

Data collection and analysis

Selection of studies

Automated identification of probable RCTs

We first screened the records using an approach that combined machine learning and crowdsourcing to identify probable RCTs. We have previously described and evaluated our approach [12]. Search results were uploaded to an RCT classifier [43], available in the Cochrane Register of Studies (CRS-Web). The classifier uses machine-learning algorithms to assign a probability of each citation being a true RCT. We assumed citations with a < 1% probability of being an RCT to be non-RCTs and excluded them.

We then uploaded the remaining records to the citizen science platform Cochrane Crowd, launched in 2016. Cochrane Crowd uses a crowdsourcing approach to conduct small, discrete tasks which help identify and describe health research in different areas. Reviewers are categorised as novices, experts (individuals who screened > 1000 abstracts and had at least 95% recall and specificity with respect to a reference standard), and resolvers (individuals who are used to settle any disagreements or uncertainties in judgements). If only novices are involved, three consecutive consistent judgements are required to move a study to the next stage. If at least one screener is an expert, then only two consecutive agreements are required. Any designation as unsure sends an abstract to a resolver, as do any disagreements.

Any citations that were deemed to be potential RCTs by Cochrane Crowd were returned to the review authors for screening. This hybrid machine-learning/crowdsourcing approach has been validated in a study which demonstrated a specificity of 99.8% and an overall recall of 98% when identifying RCTs [43].

Further title and abstract screening and PICO annotation

In the second stage, we reviewed the records identified as RCTs for eligibility and, at the same time, classified them to the appropriate PICO question(s). We uploaded abstracts to a custom Cochrane PICO annotator tool that we developed for this review.

The annotator asked reviewers to answer the following three questions for each abstract.

1. Is the study an RCT? (yes/no/unclear)
2. Is the population adults with rheumatoid arthritis? (yes/no/unclear)
3. Is the intervention a DMARD? (yes/no/unclear)

The screening was completed by a large group of reviewers working independently. Reviewers were typically trainees, who were invited through invitation from authors, or relevant organisations (e.g. trainees in undergraduate, graduate, or post-doctoral programmes; clinical trainees through rheumatology associations; early career researchers).

All participants received detailed teaching materials explaining the tasks, and were asked to first review a test set of 10 to 20 records, which were reviewed to ensure accuracy. Teaching sessions were arranged as necessary. Prior to participating, all invited reviewers completed an International Committee of Medical Journal Editors (ICJME) conflicts of interest form [44]. Only reviewers without conflicts of interest were invited to participate. Exceptions could be made on a case-by-case basis, ensuring consistency with Cochrane's conflicts of interest policies. Each record was reviewed by two to four different reviewers. Any record deemed eligible or unclear by one reviewer proceeded to full-text review.

Full-text review

The filtered records were screened by one expert reviewer (GH, SW, RB, JP) to confirm eligibility and resolve any disagreements between the different reviewers at the title/abstract screen. The reviewer retrieved the full text when necessary, and annotated each record with the clinical trial registration number and study acronym (if available), which facilitated grouping of the individual records into unique studies.

This process was facilitated using a custom online application we developed, which has been previously described [12]. Any records flagged for exclusion by the single reviewer were reviewed by a second reviewer for confirmation, and discussed as a group to resolve discrepancies if necessary. The full-text publication of all eligible records was then retrieved and uploaded into Covidence for data extraction.

Living systematic review considerations

We will apply the above-described selection process (including title and abstract screening, custom PICO annotation, and full-text screening) to the monthly search results. We will update the 'What's new' section of the review (or the Updating Classification System, when it is implemented) to indicate the status of the search results. Every three months, the team will review the list of inclusions with completed PICO annotation and decide whether the review requires an update. Updates should occur when new evidence is likely to impact review findings, as per Cochrane recommendations [16]. Specifically, we expect to update our analyses and republish the review when at least one of the following conditions apply.

1. A new DMARD is approved. We will conduct quarterly searches of the US Food and Drug Administration and European Medicines Agency websites to identify newly approved drugs.
2. The findings of the newly published trial are likely to impact the pooled estimated treatment effects.
3. Study findings are likely to increase the certainty of evidence (GRADE assessment) for a treatment effect(s), particularly if the certainty of evidence is likely to go from very low/low to moderate/high.

In the context of an NMA, we are particularly interested in trials that are likely to impact estimated treatment effects between active treatments. While this can occur with indirect evidence, it is much more likely to occur with a trial that provides a head-to-head (direct) comparison. As such, any such new trial will be flagged and reviewed carefully to determine if analyses need to be updated.

Data extraction and management

Data extraction

Two review authors extracted relevant data from included studies in duplicate and independently. A third review author resolved any disagreements. For each trial, we extracted outcome data (described below) and relevant trial characteristics, including details on the study method, inclusion and exclusion criteria, baseline characteristics of the population (including potential effect-modifiers that will be explored through meta-regression, as described below), funding for trials, and notable declarations of interest of trial authors. For all trials, we also sought data from clinical trial registries (clinicaltrials.gov; trialsearch.who.int). If there were discrepancies between the data as reported in the article and the clinical trial registry, we explored and resolved discrepancies, and contacted trial authors for their input as needed. Where the data were presented in two different formats (e.g. imputed versus not imputed), we favoured the data from the peer-reviewed publication. If data were only presented graphically, we used software to digitally extract the information (e.g. [45]). All graphical data were extracted by two review authors working independently, and averaged or corrected if an obvious discrepancy was apparent.

Dichotomous outcomes

We extracted dichotomous efficacy outcomes as the number of participants with an event and the total number of participants in each arm, on an intention-to-treat (ITT) basis, as the number of participants randomised to the arm who received at least one dose of medication. If this was not reported, we extracted the total number of participants randomised to the arm. We extracted outcome data for five dichotomous outcomes: ACR20 response, ACR50 response, ACR70 response, DAS remission, and DAS28 remission.

Continuous outcomes

For each of the continuous outcomes, we extracted both the mean of the final values and the mean of the change in values from baseline. If the mean value was not available, we extracted the median. We collected the measures of variability in the following order depending on availability: standard deviations (SDs), standard errors (SEs), 95% confidence intervals (CIs), and exact P value from a t-test. Data transformations were made at the analysis stage to convert these into SDs, according to Cochrane guidance [46]. We extracted outcome data for seven continuous outcomes: DAS, DAS28, pain intensity, HAQ, fatigue, quality of life, and radiographic progression. Where both unadjusted and adjusted values were reported for the same outcome, we extracted the unadjusted values. Where trialists reported data based on the ITT sample and another sample (e.g. per-protocol, as-treated), we extracted the ITT-analysed data.

Rates (withdrawals and adverse event outcomes)

We extracted adverse event outcomes as the number of events and treatment exposure (person-years) in each arm. This allowed adverse events to be assigned to the treatment received at the time of the event, and is important for trials that involve a switch in therapy or an 'early escape' arm (see [Unit of analysis issues](#) section). If the total number of events was not reported, we used the total number of participants with at least one event, which should be a close approximation for uncommon events. If a study

reported both the total number of events and the total number of participants with at least one event, we extracted both values.

Time point of outcome assessment

We extracted all outcomes at trial end, defined as the end of the randomised treatment period, and at several time points (with time windows): 3 months (≥ 12 weeks, < 20 weeks); 6 months (≥ 20 weeks, < 39 weeks); and 12 months (≥ 39 weeks, < 78 weeks). Trial duration was calculated from the time of randomisation, which, in trials with 'run-in' periods, is not necessarily the study start.

Data management

We used Covidence to extract and house data for our review ([Supplementary material 5](#)) [13]. Covidence is a web-based software platform that links with Cochrane to simplify the systematic review process from importing citations to exporting data. Covidence allowed us to have a shared database accessible to all review authors, and to easily organise efforts with workflow management. Covidence allows both risk of bias and data extraction to be completed on one platform, which facilitated comparisons between review authors.

Living systematic review considerations

As new trials are identified, data will be extracted in real-time into Covidence, using the process described above.

Risk of bias assessment in included studies

We assessed the risk of bias in included trials using Cochrane's RoB 1 tool [47]. We assessed studies as having a 'low risk', 'high risk', or 'unclear risk' of bias across the seven specified domains: sequence generation, allocation concealment, blinding of participants and personnel, blinding of outcome assessment, incomplete outcome data, selective outcome reporting, and other sources of bias, such as baseline imbalance. We assessed the risk of bias separately for different outcome types for domains where the risk of bias could differ. Specifically, we separately assessed the two domains related to blinding for radiographic outcomes, as blinding of the outcome assessor occurs separately for radiographs, and the impact of an unblinded study on the risk of bias for the domain 'blinding of participants and personnel' was rated as 'unclear', not 'high', given that radiographs are objectively scored.

We also separately assessed the domain 'incomplete outcome data' for radiographs, as the amount of missing data may be substantially higher than for other clinical outcomes, and for the outcome 'withdrawals due to adverse events', as the amount of missing data due to participant withdrawal will not affect the risk of bias for this outcome, provided the reasons for withdrawal are adequately reported.

After rating the risk of bias for each of the seven domains, we made an overall risk of bias rating for each outcome type where the risk of bias could differ: clinical outcomes (all outcomes excluding radiographs and withdrawals), radiographic outcomes, and withdrawals. The overall rating included a judgement across all domains for that outcome, although all domains were not necessarily weighted equally. The domains that were weighted most heavily for the overall risk rating were those relating to blinding of participants and personnel, and incomplete outcome data.

The risk of bias assessment was completed in duplicate and independently by pairs of review authors and entered into Covidence. Review authors used a teaching guide with context-specific clarifications and examples to assist them with their judgements ([Supplementary material 6](#)).

Measures of treatment effect

We evaluated treatment effects on dichotomous outcomes using odds ratios (ORs). For continuous outcomes that were measured on the same scale, we evaluated treatment effects as mean differences (MDs), with final values or change in values from baseline. If final values were unavailable, but changes from baseline and baseline values were present, we back-calculated the final values, using the baseline SD for analysis. For continuous outcomes measured on different scales (radiographic progression, fatigue), we evaluated treatment effects using standardised mean differences (SMDs), and then back-transformed them to the usual scale for ease of interpretability. We summarised toxicity outcomes as risk ratios (RRs) to allow for differences in exposure between arms in early escape and cross-over trials.

Unit of analysis issues

For cross-over trials or trials comparing different DMARD sequences, we extracted data at the time of the initial switch in treatment for efficacy outcomes, provided this was at least 12 weeks. For adverse event outcomes in cross-over trials, we extracted data from the longest follow-up available, provided the adverse event was assigned to the treatment received at the time of the adverse event (on-treatment data).

In 'early escape' trials, a switch in treatment is allowed or required for participants who fail to achieve a certain response at a given time (usually 12 to 16 weeks). Participants who receive early escape therapy are typically treated as treatment failures at all future time points. For early escape trials, we used end-of-trial (carried forward) data for efficacy outcomes. For adverse event outcomes, we used on-treatment data from the longest follow-up available, similar to the approach for cross-over trials.

Dealing with missing data

For all trials, we sought data from clinical trial registries ([clinicaltrials.gov](#); [trialsearch.who.int](#)). For missing variability data (continuous outcomes), we calculated SDs from SEs, 95% CIs, exact P values from t-tests, or interquartile ranges (IQR) using formulas published in the *Cochrane Handbook for Systematic Reviews of Interventions* [48]. If no variability data were available, we used the baseline SD as the final SD. For adverse event outcomes, if the exposure (patient-years) was not reported, we calculated it, accounting for participant withdrawals. If the timing of withdrawal was not reported, we assumed the withdrawal occurred at a constant rate over time. For example, in a study arm of 200 participants, if we knew that 120 people had completed the trial, 50 people withdrew by 3 months, and another 30 people withdrew between 3 and 6 months, the exposure would be calculated as follows: total exposure (person-months) = 120*6 months + 50*1.5 (people who withdrew between 0 and 3 months assumed to contribute 1.5 months on average) + 30*4.5 (people who withdrew between 3 and 6 months).

Reporting bias assessment

To assess outcome reporting bias, we checked trial protocols against published reports. For studies published after 1 July 2005, we searched the WHO ICTRP ([trialsearch.who.int](#)) for the a priori trial protocol.

If there were > 10 studies for any pair-wise comparison, we generated a funnel plot to detect small-study effects that may be suggestive of publication bias.

Synthesis methods

Direct evidence (pair-wise meta-analyses)

To assess the direct evidence, we identified studies contributing to each estimate. When only one study was available, we carried out a single-study analysis to obtain the effect estimates [48]. For analyses involving multiple studies, the pooled estimate was derived using a Bayesian random-effects meta-analysis model, to be consistent with the Bayesian approach used for the NMA analysis. For each pooled estimate, forest plots were generated using the forest plot package in R statistical software, with the percentage weight derived using the meta package [49].

Network meta-analysis

We conducted NMAs for all critical and important outcomes using end-of-trial data. For the primary analysis of each outcome, we excluded trials at high risk of bias for that outcome (this was subjected to a sensitivity analysis for the three critical outcomes, as discussed below). Random-effects arms-based Bayesian NMAs were fitted, using models that account for the correlation in multiple-arm trials [50]. In the arms-based model, the treatment effect of each trial is modelled around a common mean effect. This mean effect is then broken down into 'basic parameters' unique for each treatment, with the basic parameter for the reference treatment set to zero. The basic parameters can then be compared to estimate the treatment effect between any pair of treatments.

For our main analysis, we used a semi-informative prior probability distribution for the between-study variance. We selected semi-informative priors based on published values for different outcome types (continuous, dichotomous, rate) and intervention comparison types (pharmacologic versus placebo/control, pharmacological versus pharmacological, non-pharmacologic). Specifically, we used the semi-objective and pharmacological versus pharmacological priors [51, 52] for all outcome types for this review ([Supplementary material 7](#)). Semi-informative priors can improve model stability and produce estimates that better reflect real-world treatment effects [52]. The prior distribution of the random effect was varied in sensitivity analyses, as discussed below. We used uninformative prior distributions for all other model parameters. Markov chain Monte Carlo sampling was used to obtain samples from posterior distributions, with 10,000 burn-in iterations followed by 10,000 monitoring iterations. Convergence was assessed by analysing the differences between the Markov chains. Specifically, we ran three chains, inspected the sampling history plots, and calculated Gelman-Rubin-Brooks (GBR) statistics [53]. We assessed model fit by using residual deviance and Deviance Information Criterion (DIC). We performed all data analyses using R statistical software [54] with rjags running Just Another Gibbs Sampler (JAGS) [55].

For each pair-wise comparison, we summarised the treatment effects as the median and 95% credible interval (CrI). For critical outcomes, we converted treatment effects into absolute risks by using the median of the placebo (reference) arm as the baseline risk. This process involved identifying studies with a placebo arm, pooling the results, and using the median estimate as the assumed baseline risk [56]. Using outcome-specific formulae, such as OR or RR, we then converted relative effects into absolute effects.

Living systematic review considerations

As part of this project, we will be working to automate the analyses, such that with each update, the analyses can be conducted rapidly with relevant changes highlighted. We will consider new statistical approaches as they are developed over time.

Investigation of heterogeneity and subgroup analysis

Heterogeneity

We conducted separate analyses for each network defined above. Within each of the networks, we assessed the clinical heterogeneity of each direct comparison by considering the characteristics of the included studies, including disease duration, measures of disease severity, and co-interventions (corticosteroids and concurrent DMARDs, including dosing of MTX). Borrowing an approach from the CINeMA framework for NMAs, each trial was assigned a rating for its indirectness as either 'low', 'some concerns', or 'high' [57, 58]. This was done independently by two review authors as a separate step following data extraction. When making these judgements, each review author reviewed summary information on the trial characteristics and had access to the full-text publication. Review authors rated the indirectness in relation to the typical patient that would be seen in contemporary clinical practice, and rated down when there was concern regarding differences in PICO characteristics in the trials that could modify the effect of the intervention. Discrepancies were discussed and a consensus judgement made. We used this information when making judgements about the certainty of the evidence.

We evaluated the statistical heterogeneity for each direct comparison by comparing the point estimates, CrIs, and I^2 values.

For the NMA analyses, we assumed that the random effects were shared across all interventions in the network. We evaluated the coherence of the direct and indirect evidence for each pair-wise comparison by calculating a P value, using a Bayesian approach [26]. A P value of < 0.05 indicates that there is less than 5% overlap between the CrIs of the indirect and direct estimates. We also calculated the between-study variance in the random-effects model to provide an overall measure of statistical heterogeneity across all comparisons within the NMA.

Subgroup analysis

We planned to conduct meta-regression to explore the association between trial-level variables and treatment effects (effect modification). We limited our meta-regression to ACR50 response and the following variables.

1. Baseline response rate
2. Disease duration
3. Duration of trial
4. Year of publication of trial

5. Baseline disease activity (low versus moderate/high)

For the meta-regression analyses, we modelled a common effect-modifier relative to the reference treatment (placebo). Treatment effects for other comparisons (e.g. head-to-head trials) were not adjusted, unless there were 10 or more trials for that comparison to specify an additional effect-modifier. For the baseline response rate, we used the modelled response rate as opposed to the actual response rate in each trial to limit the effect of regression to the mean [52, 59]. If any of the variables were found to have statistically significant effect modification, adjusted pair-wise treatment effects were also reported and compared to the unadjusted analysis.

Equity-related assessment

We did not undertake an equity-related assessment for this review.

Sensitivity analysis

We conducted the following sensitivity analyses on the critical outcomes of ACR50 response and withdrawals due to adverse events.

1. Alternate specification for the Bayesian prior to the between-study variance, comparing semi-informative priors (main analysis) to uninformative priors and fixed-effect models.
2. Including trials at high risk of bias.
3. Conducting separate networks with data from different time points (3, 6, and 12 months).

Living systematic review considerations

We will review the scope and methods of this review approximately yearly (or more frequently if appropriate) in light of potential changes in the topic area, or the evidence being included in the review (e.g. additional populations, outcomes, or new review methods available). Given the scope of our review and the continual emergence of new therapeutic options and trials, we believe our review will require maintenance over the foreseeable future. However, we will make decisions about whether to stop updating the living NMA when appropriate (e.g. if the review question is no longer a priority for decision-making, or no new evidence is likely).

Certainty of the evidence assessment

We used the GRADE approach (minimally contextualised framework) to rate the certainty of evidence for each of the seven critical outcomes (ACR50, withdrawals due to adverse events, radiographic progression, DAS28, VAS, HAQ, and serious adverse events) in the summary of findings table [46, 60]. The chosen thresholds for determining the minimally important difference (MID) are presented in Table 1. As per the GRADE method, we assessed the certainty of evidence for each of the five GRADE considerations (study limitations, inconsistency of effect, imprecision, indirectness, and publication bias) using a semi-automated approach that leveraged a novel tool developed for this project [61]. We first assessed the certainty of evidence for the direct evidence, using GRADE guidance. For the indirectness domain, our judgements were informed by the indirectness ratings for the individual studies, using the same GRADE guidance for how individual trial ratings for risk of bias are used to inform the summary judgement for the GRADE domain study limitations.

Certainty ratings for the indirect evidence considered the quality of the direct comparisons that formed the indirect evidence, and

the likelihood of 'intransitivity' (i.e. if there was heterogeneity in the trials that formed the different comparisons within the indirect evidence). We assessed intransitivity by comparing the indirectness ratings of the trials that contributed to the indirect evidence, focusing on the most dominant branches of indirect evidence, as recommended by GRADE [32, 62]. This process was expedited using a tool developed specifically for this project, which generated automated ratings for indirect and network estimates [63]. We then provided a certainty of evidence assessment for the overall NMA estimate following GRADE guidance, considering the certainty of the direct and indirect evidence, their contribution to the overall NMA estimate, and the presence or absence of statistical incoherence, estimated through the node-splitting analyses [26]. The parlance used to describe the certainty of evidence using the minimally contextualised GRADE approach is outlined in Table 2.

Summary of findings table

In the summary of findings table, we presented the results of the seven critical outcomes at trial end for each treatment relative to the reference (placebo). The results include the relative effect, the absolute risks, and the certainty of evidence ratings. We presented the results according to GRADE guidance for NMAs [64]. Additionally, we reported the results of all pair-wise comparisons for the seven critical outcomes, highlighting any estimates that were statistically significant, and the corresponding certainty of evidence.

Living systematic review considerations

How to best present summary of findings tables for an NMA is an emerging area of research within GRADE. As new presentation formats are developed, we will incorporate them into our review.

Consumer involvement

We did not involve consumers in the conduct of this review, although the review informs living Clinical Practice Guidelines, which have been developed with consumer involvement [10, 11].

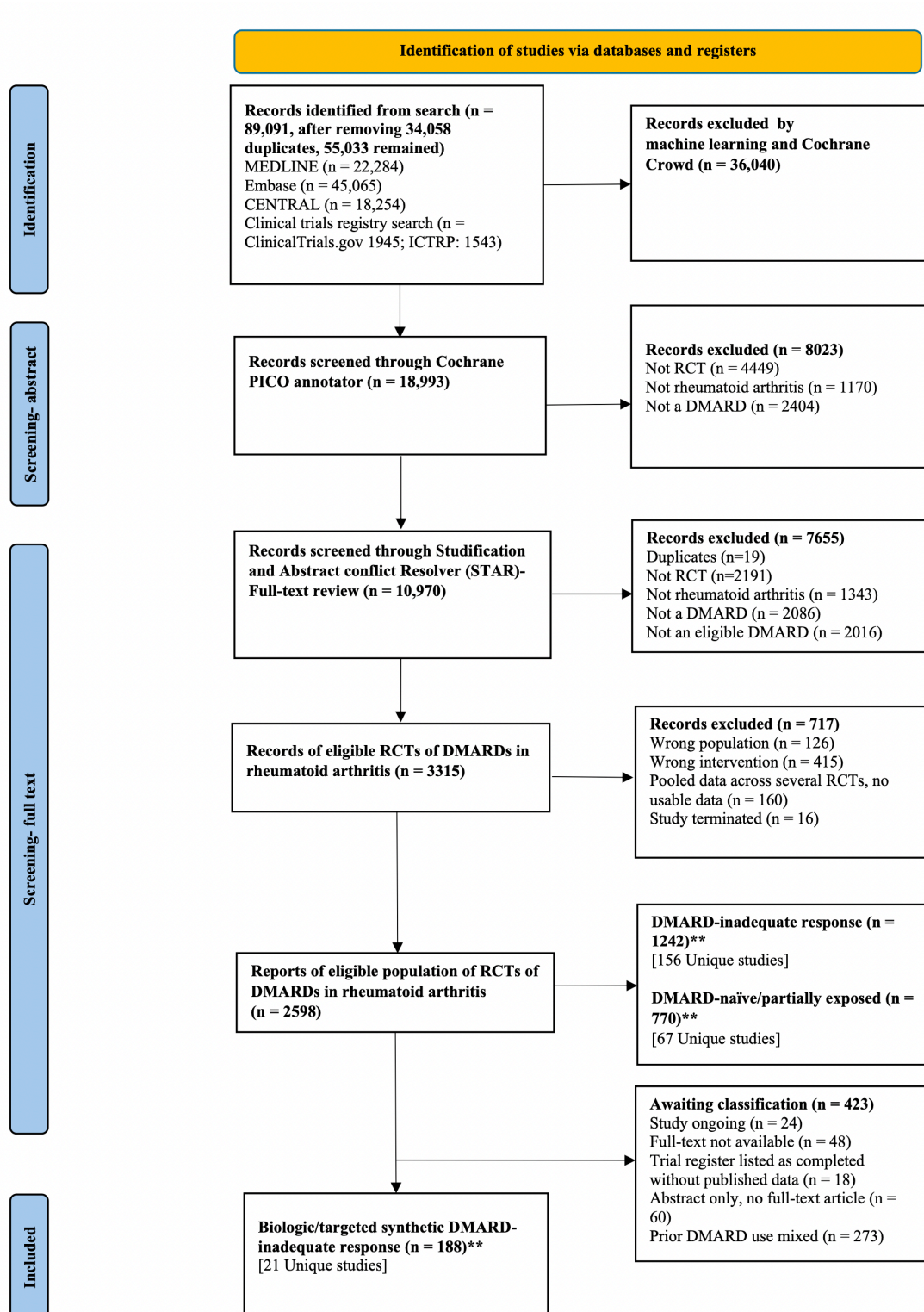
RESULTS

Description of studies

Results of the search

Our search identified 55,033 unique records. Following removal of 36,040 records that were not RCTs, 18,993 records underwent title and abstract screening using the Cochrane PICO annotator tool, and 8023 of these records were excluded. The remaining 10,970 records were assessed by rheumatologists with content and methods expertise (GH, SW, RB) using the Studification and Abstract Conflict Resolver (STAR), a tool developed for the current review. After the exclusion of 7655 additional records using the STAR tool, 3315 records underwent full-text review. Of these, 717 records were further excluded (Figure 1), leaving 2598 eligible records across the three reviews. We categorised 423 trials that were potentially eligible for any of the three reviews as awaiting classification; 10 of these trials are specific to this review and are either actively recruiting participants, terminated, complete but abstract only, or unclear if still ongoing (Supplementary material 3; Supplementary material 4; Supplementary material 8).

Figure 1. PRISMA diagram of the various screening stages and included studies (Final included records have overlapping records from one or more populations of interest).**



For this review, 188 records were eligible, representing 21 unique studies. The remaining records were eligible for one of the other two reviews (DMARD-naïve, n = 770; DMARD-inadequate response,

n = 1242). The latest search date was 28 November 2025. Of the 21 unique studies, 19 had a patient population that had previously failed a TNFi. The remaining two studies had a patient population

that had previously failed a biologic DMARD and were moved to awaiting classification until additional studies can be identified and an NMA can be performed. A PRISMA diagram is presented in [Figure 1](#).

Included studies

The 19 included studies are presented in [Table 3](#), with further details in [Supplementary material 2](#). All studies were available in English. Eleven trials included a placebo control, and eight included active or inactive comparator arms.

Study design and setting

All included studies were parallel-design RCTs ([Table 3](#)) (Blanco 2017 [65]; Brown 2018 [66]; Burmester 2013 [67]; Cohen 2006 [68]; Elmedany 2019 [69]; Emery 2008 [70]; Emery 2019 [71]; Fleischmann 2017 [72]; Furst 2007 [73]; Genovese 2005 [74]; Genovese 2016 [75]; Genovese 2018 [76]; Gottenberg 2016; Huffstutter 2017 [77]; Manders 2015 [78]; Schiff 2014 [79]; Smolen 2009 [80]). Fourteen studies had two eligible arms (Blanco 2017; Burmester 2013; Cohen 2006; Elmedany 2019; Emery 2008; Fleischmann 2017; Furst 2007; Genovese 2005; Genovese 2014 [81]; Genovese 2018; Gottenberg 2016; Huffstutter 2017; Humby 2021 [82]; Schiff 2014), and five studies had three or more eligible intervention arms (Brown 2018; Emery 2019; Genovese 2016; Manders 2015; Smolen 2009).

Most trials were international (Blanco 2017; Burmester 2013; Cohen 2006; Emery 2008; Emery 2019; Fleischmann 2017; Furst 2007; Genovese 2005; Genovese 2016; Genovese 2018; Huffstutter 2017; Schiff 2014; Smolen 2009), while five studies were conducted in one country only, including France (Gottenberg 2016), Saudi Arabia (Elmedany 2019), the Netherlands (Manders 2015), the UK (Brown 2018), and the USA (Furst 2007). All were conducted in the outpatient setting.

Nine trials received at least some funding from manufacturers (Burmester 2013; Cohen 2006; Emery 2008; Emery 2019; Fleischmann 2017; Genovese 2005; Genovese 2014; Genovese 2018; Huffstutter 2017), while 10 trials did not report manufacturer funding (Blanco 2017; Brown 2018; Elmedany 2019; Furst 2007; Genovese 2016; Gottenberg 2016; Humby 2021; Manders 2015; Schiff 2014; Smolen 2009).

Participants

The trials were performed in a well-established rheumatoid arthritis population, with the median baseline disease duration ranging from 6.4 to 14 years. The median age of participants ranged from 49 to 58 years, and all trials had a higher proportion of female participants. Participants had moderate to high baseline disease activity (DAS28 ranged from 4.87 to 6.79) and moderate baseline physical disability (HAQ-DI scores ranged from 0.96 to 1.8). All trials included participants that had failed one or more TNFi.

Interventions

Eleven trials were placebo-controlled. In these trials, participants stopped their current TNFi and were randomised to placebo or an active comparator in combination with background conventional synthetic DMARDs. Of these studies, four evaluated a JAK inhibitor, including baricitinib (Genovese 2016), tofacitinib (Burmester 2013), or upadacitinib (Genovese 2018); two evaluated the T-cell co-stimulation inhibitor abatacept (Blanco 2017; Genovese 2005);

one evaluated the B-cell depletor rituximab (Cohen 2006); three evaluated IL-6 inhibitors, tocilizumab (Emery 2008; Genovese 2014), and sarilumab (Fleischmann 2017); and two evaluated a switch to a TNFi not previously tried (Schiff 2014; Smolen 2009). For Schiff 2014, participants switched to certolizumab pegol after failing either etanercept, infliximab, adalimumab, or golimumab. For Smolen 2009, participants switched to golimumab after failing either etanercept, infliximab, or adalimumab.

Of the other eight trials, two had an inactive comparator, but participants continued their failed anti-TNF therapy, as opposed to discontinuing it. This was treated as a separate node in the network. In both trials, participants could also continue conventional synthetic DMARDs. One of these trials compared the continuation of a failed TNFi (etanercept) to a new TNFi (infliximab; Furst 2007). The other trial compared an alternate route of administration of golimumab, where failed subcutaneous golimumab was continued in one arm, and intravenous golimumab was administered in the other arm (Huffstutter 2017).

Six trials had active comparator arms. One trial compared a TNFi not previously tried to a non-TNF biologic (either abatacept, rituximab, or tocilizumab) (Gottenberg 2016). One trial compared tocilizumab and intravenous abatacept (Elmedany 2019). Two studies were three-arm trials that compared abatacept, rituximab, and a TNFi not previously tried, although abatacept was administered subcutaneously in one trial (Brown 2018) and intravenously in the other trial (Manders 2015). One trial compared sarilumab and tocilizumab (Emery 2019), and the remaining trial compared rituximab and tocilizumab (Humby 2021).

Outcomes

The critical outcome measures reported in each study are presented in an ORBIT matrix in [Table 4](#). The ORBIT matrix for the important outcomes is presented in [Table 5](#).

Critical outcomes

Twelve studies reported ACR50; 10 studies reported withdrawals due to adverse events; and two studies reported radiographic progression.

Other critical outcomes

Eleven studies reported DAS28 remission; three studies reported pain as measured by VAS; 15 studies reported physical functioning as measured by HAQ; and 13 studies reported serious adverse events.

Important outcomes

Fourteen studies reported ACR20; 10 studies reported ACR70; seven studies reported serious infections; two studies reported fatigue; and three studies reported quality of life.

Excluded studies

We excluded 717 records across our three reviews at the full-text stage due to ineligible interventions or population, unusable data that were pooled across several trials, or early termination. A list of excluded trials and the reasons for their exclusion are available in Open Science Framework.

Risk of bias in included studies

The risk of bias in the included trials varied across studies (Figure 2; Figure 3)

Figure 2. Percentages of studies rated as low, high, or unclear risk of bias for each domain. Domains without full bars indicate that some studies did not report radiographic outcomes.

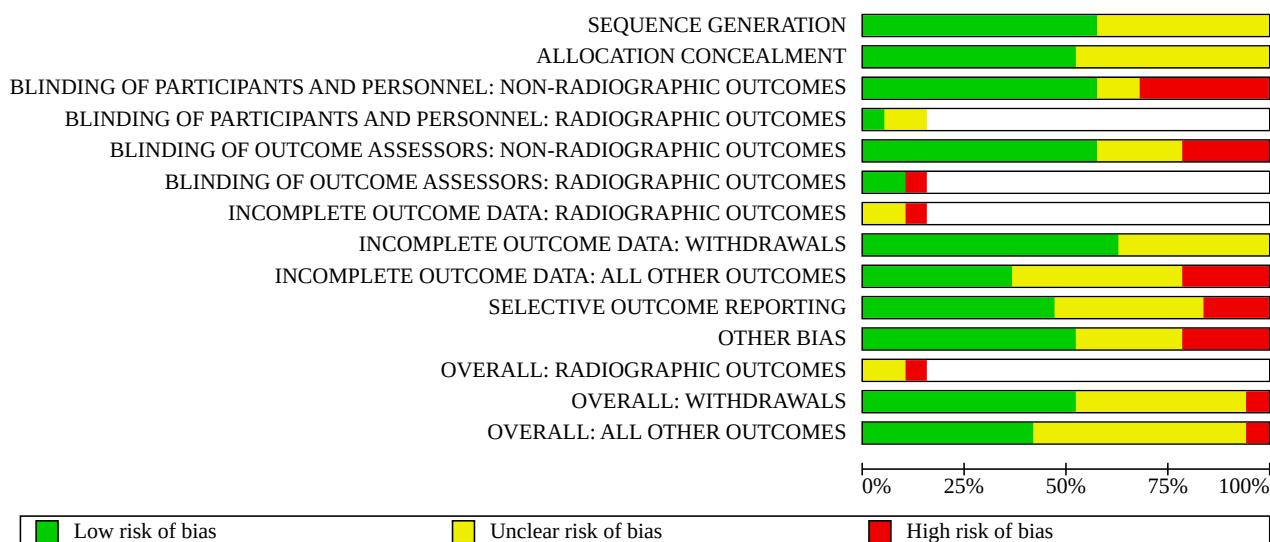


Figure 3. Summary of risk of bias judgements for each included study. Blank cells indicate not applicable (no radiographic outcomes reported).

	SEQUENCE GENERATION	ALLOCATION CONCEALMENT	BLINDING OF PARTICIPANTS AND PERSONNEL: NON-RADIOGRAPHIC OUTCOMES	BLINDING OF PARTICIPANTS AND PERSONNEL: RADIOGRAPHIC OUTCOMES	BLINDING OF OUTCOME ASSESSORS: NON-RADIOGRAPHIC OUTCOMES	BLINDING OF OUTCOME ASSESSORS: RADIOGRAPHIC OUTCOMES	INCOMPLETE OUTCOME DATA: RADIOGRAPHIC OUTCOMES	INCOMPLETE OUTCOME DATA: WITHDRAWALS	INCOMPLETE OUTCOME DATA: ALL OTHER OUTCOMES	SELECTIVE OUTCOME REPORTING	OTHER BIAS	OVERALL: RADIOGRAPHIC OUTCOMES	OVERALL: WITHDRAWALS	OVERALL: ALL OTHER OUTCOMES
Blanco 2017	?	?	+		+			+	+	+	+		+	+
Brown 2018	+	+	-		-			+	?	?	+		?	?
Burmester 2013	+	+	+		+			+	+	+	-		+	+
Cohen 2006	?	?	+	+	+	+	-	?	-	?	-	?	?	?
Elmedany 2019	+	?	-		-			+	?	?	+		?	?
Emery 2008	?	?	+		+			+	?	+	?		?	?
Emery 2019	?	?	+		+			?	?	?	-		+	+
Fleischmann 2017	+	?	+		+			+	?	?	?		+	+
Furst 2007	?	?	-	?	?	+	?	?	-	-	+	?	-	-
Genovese 2005	?	+	+		+			+	?	+	?		+	+
Genovese 2014	+	+	+		+			?	+	?	?		+	+
Genovese 2016	+	+	+		+			+	+	+	+		+	+
Genovese 2018	+	+	+		+			+	+	+	-		+	+
Gottenberg 2016	+	+	-	?	-	-	?	+	+	-	+	-	?	?
Huffstutter 2017	?	?	?		?			+	-	+	?		+	?

Figure 3. (Continued)

Huffstutter 2017	?	?	?		?			+	-	+	?		+	?
Humby 2021	+	+	-		?			+	-	+	+		?	?
Manders 2015	+	+	-		-			?	?	?	+		?	?
Schiff 2014	?	?	?		?			?	+	-	+		+	?
Smolen 2009	+	+	+		+			?	?	+	+		?	?

Sequence generation and allocation concealment (selection bias)

Nine studies adequately generated a random sequence and concealed the sequence until allocation and were thus assessed as at low risk of selection bias (Brown 2018; Burmester 2013; Genovese 2005; Genovese 2014; Genovese 2016; Genovese 2018; Gottenberg 2016; Manders 2015; Smolen 2009). Ten studies did not report their method of randomisation, and we judged their risk of selection bias as unclear (Blanco 2017; Cohen 2006; Elmedany 2019; Emery 2008; Emery 2019; Fleischmann 2017; Furst 2007; Huffstutter 2017; Humby 2021; Schiff 2014).

Blinding of participants and personnel (performance bias)

Clinical outcomes and withdrawals

Eleven studies adequately blinded participants and investigators to the treatment given through use of a placebo that was consistent with the active drug in route of administration and appearance (Blanco 2017; Burmester 2013; Cohen 2006; Emery 2008; Emery 2019; Fleischmann 2017; Genovese 2005; Genovese 2014; Genovese 2016; Genovese 2018; Smolen 2009). We thus considered these studies as at low risk of performance bias. One study was double-blinded and did not report additional details, and so was rated as at unclear risk of performance and detection bias (Schiff 2014). We considered seven trials to have high risk of performance bias due to their open-label design (Brown 2018; Elmedany 2019; Furst 2007; Gottenberg 2016; Huffstutter 2017; Humby 2021; Manders 2015).

Radiographic outcomes

Three trials had prespecified radiographic outcomes, but only two reported radiographic outcomes. One trial that reported radiographic outcomes described sufficient procedures for blinding of participants and personnel and was rated as at low risk of performance bias (Cohen 2006). The other trial that reported radiographic outcomes was open-label and was rated as at unclear risk of performance bias (Furst 2007). The trial that prespecified radiographic outcomes but did not report them was open-label, indicating an unclear risk of performance bias (Gottenberg 2016). The remaining trials were left blank for this risk of bias domain.

Blinding of outcome assessors (detection bias)

Clinical outcomes and withdrawals

We assessed four trials as at unclear risk of detection bias relating to the blinding of the outcome assessors for clinical outcomes and withdrawals because the authors did not provide details about the blinding procedures beyond stating that it was "double-blind" (Furst 2007; Huffstutter 2017; Humby 2021; Schiff 2014). We

assessed four trials as at high risk of detection bias due to lack of blinding in the open-label design of their trial (Brown 2018; Elmedany 2019; Gottenberg 2016; Manders 2015). The remaining 11 trials provided adequate details on the procedure used to blind outcome assessors and were therefore rated as at low risk of detection bias.

Radiographic outcomes

Three trials had prespecified radiographic outcomes, but only two reported radiographic outcomes. Both trials that reported radiographic outcomes described sufficient procedures for blinding outcome assessors, indicating a low risk of detection bias (Cohen 2006; Furst 2007). The trial that prespecified radiographic outcomes but did not report them was open-label, suggesting that outcome assessors may have been aware of treatment allocation, resulting in high risk of detection bias (Gottenberg 2016). The remaining trials were left blank for this risk of bias domain.

Incomplete outcome data (attrition bias)

Clinical outcomes

We assessed seven trials as at low risk of bias for incomplete outcome data for clinical outcomes (Blanco 2017; Burmester 2013; Genovese 2014; Genovese 2016; Genovese 2018; Gottenberg 2016; Schiff 2014). We rated eight trials as at unclear risk of attrition bias due to moderate dropout rates or some imbalance between arms (Brown 2018; Elmedany 2019; Emery 2008; Emery 2019; Fleischmann 2017; Genovese 2005; Manders 2015; Smolen 2009). We assessed four trials as at high risk of attrition bias due to having both an early escape design and high dropout rates or significant imbalance between arms (Cohen 2006; Furst 2007; Huffstutter 2017; Humby 2021).

Radiographic outcomes

Three studies prespecified radiographic outcome data (Cohen 2006; Furst 2007; Gottenberg 2016), but only two of these studies reported radiographic data. We assessed Cohen 2006 as at high risk of bias for incomplete radiographic outcome data due to high dropout rates and an early escape design. We assessed Furst 2007 as at unclear risk of bias due to its early escape design, with radiographic data only reported to week 14. We rated Gottenberg 2016 as at unclear risk of bias because radiographic progression data were not published, although this outcome was listed in the protocol. The remaining trials were left blank for this risk of bias domain.

Withdrawals

We assessed 12 trials as at low risk of bias for withdrawals (Blanco 2017; Brown 2018; Burmester 2013; Elmedany 2019; Emery 2008; Fleischmann 2017; Genovese 2005; Genovese 2016; Genovese 2018; Gottenberg 2016; Huffstutter 2017; Humby 2021). We rated seven trials as at unclear risk of bias for withdrawals due to inadequate reporting or an early escape design (Cohen 2006; Emery 2019; Furst 2007; Genovese 2014; Manders 2015; Schiff 2014; Smolen 2009).

Selective outcome reporting (reporting bias)

We assessed 10 trials as at low risk of reporting bias as they had a publicly available protocol (Blanco 2017; Burmester 2013; Emery 2008; Genovese 2005; Genovese 2016; Genovese 2018; Gottenberg 2016; Huffstutter 2017; Humby 2021; Smolen 2009), while seven trials did not have an available protocol and were rated as at unclear risk of reporting bias (Brown 2018; Cohen 2006; Elmedany 2019; Emery 2019; Fleischmann 2017; Genovese 2014; Manders 2015). We assessed two trials as at high risk of reporting bias: Furst 2007 had missing prespecified radiographic outcome data, and Schiff 2014 had minor discrepancies between the clinical trial registry data and the published data, and did not publish several secondary outcomes.

Other bias

Four trials were industry sponsored, and flagged as high risk for potential sponsorship bias, although this did not impact the overall risk of bias rating (Burmester 2013; Cohen 2006; Emery 2019; Genovese 2018). Five trials had some funding from industry, but were investigator-initiated and so were rated as at unclear risk of other bias (Emery 2008; Fleischmann 2017; Genovese 2005; Genovese 2014; Huffstutter 2017). We assessed the remaining trials as at low risk of bias for this domain.

Overall risk of bias

Clinical outcomes

We assessed eight studies as at low overall risk of bias for clinical outcomes (Blanco 2017; Burmester 2013; Emery 2019; Fleischmann

2017; Genovese 2005; Genovese 2014; Genovese 2016; Genovese 2018). We assessed 10 studies as at unclear risk of bias due to an open-label or early escape design (Brown 2018; Cohen 2006; Elmedany 2019; Emery 2008; Gottenberg 2016; Huffstutter 2017; Humby 2021; Manders 2015; Schiff 2014; Smolen 2009). We rated one study as at high risk of bias due to having both an open-label and early escape design (Furst 2007).

Radiographic outcomes

We assessed the overall risk of bias for radiographic outcomes as unclear for three trials because of lack of double-blinding (Furst 2007), moderate withdrawal rates (Cohen 2006), and lack of reporting (Gottenberg 2016).

Withdrawals

We assessed 10 studies as at low overall risk of bias for withdrawals (Blanco 2017; Burmester 2013; Emery 2019; Fleischmann 2017; Genovese 2005; Genovese 2014; Genovese 2016; Genovese 2018; Huffstutter 2017; Schiff 2014). We assessed eight studies as at unclear risk of bias, three due to an early escape design (Cohen 2006; Emery 2008; Smolen 2009) and five due to an open-label design (Brown 2018; Elmedany 2019; Gottenberg 2016; Humby 2021; Manders 2015). We rated one study as at high risk of bias due to having both an open-label and early escape design, as well as having an imbalance in the dropout rate across arms (Furst 2007).

Synthesis of results

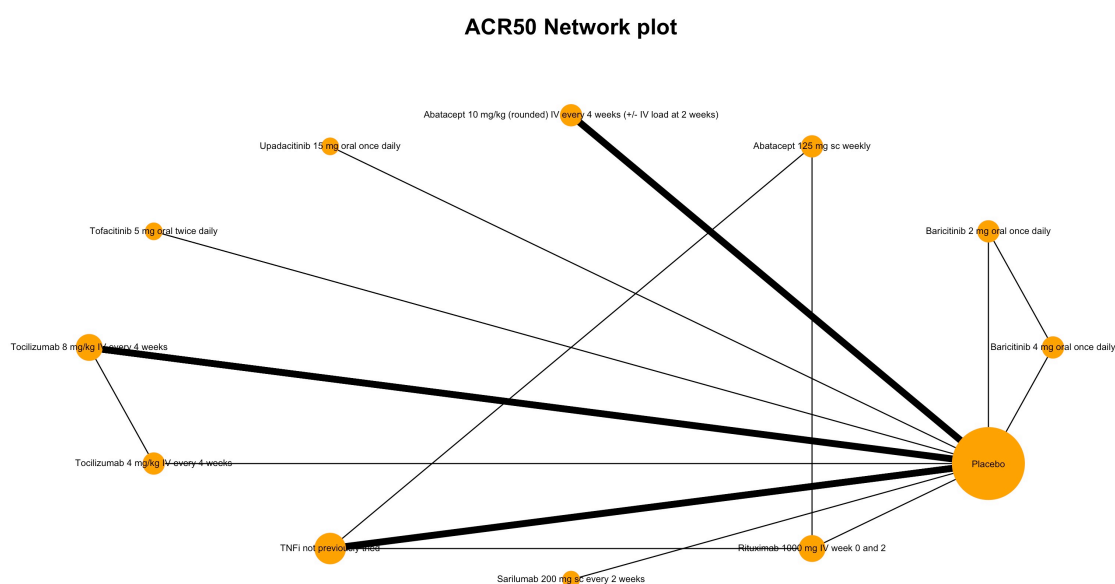
Critical outcomes

ACR50

Effects relative to placebo

We included 12 trials in the analysis of ACR50 among TNFi inadequate responders. The findings presented describe each intervention relative to placebo ([Summary of findings 1](#); [Figure 4](#)).

Figure 4. Network plot of ACR50. Each circle represents an intervention node, and connecting lines represent direct comparisons. Larger nodes indicate a greater number of participants receiving the intervention, and thicker lines indicate a greater number of studies informing the comparison.



TNF inhibitor

High-certainty evidence indicates that switching to a TNFi not previously tried resulted in more people achieving an ACR50 response compared to placebo. The number of people achieving ACR50 was 77 with placebo and 334 with a TNFi not previously tried (OR 6.04, 95% CrI 2.49 to 16.3).

IL-6 inhibitor

High-certainty evidence indicates that switching to IL-6 inhibitors resulted in more people achieving an ACR50 response compared to placebo. The number of people achieving an ACR 50 response was 77 with placebo and 206 with sarilumab (OR 3.11, 95% CrI 1.25 to 7.76), 313 with tocilizumab 4 mg/kg intravenous (IV) (OR 5.31, 95% CrI 2.09 to 12.09), and 455 with tocilizumab 8 mg/kg IV (OR 10.03, 95% CrI 3.65 to 31.27).

T-cell co-stimulation inhibitor

High-certainty evidence indicates that intravenous abatacept resulted in more people achieving an ACR50 response compared to placebo. The number of people achieving ACR50 was 77 with placebo and 275 with intravenous abatacept (OR 4.57, 95% CrI 2.21 to 10.18). Moderate-certainty evidence indicates that subcutaneous abatacept was probably more effective than placebo in achieving ACR50, with 264 people achieving ACR50 (OR 4.31, 95% CrI 0.97 to 18.28), although the credible interval crossed the null effect. We downgraded the certainty of evidence for the comparison subcutaneous abatacept versus placebo due to serious imprecision.

B-cell depletor

High-certainty evidence indicates that switching to rituximab resulted in more people achieving ACR50 compared to placebo. The number of people achieving ACR50 was 77 for placebo and 314 for rituximab (OR 5.50, 95% CrI 2.31 to 13.12).

JAK inhibitor

High-certainty evidence indicates that switching to upadacitinib, tofacitinib, or baricitinib 4 mg daily resulted in more people achieving ACR50 response compared to placebo. The number of people achieving ACR50 was 77 with placebo and 246 with upadacitinib 15 mg (OR 3.93, 95% CrI 1.53 to 10.32), 254 with tofacitinib (OR 4.09, 95% CrI 1.43 to 12.13), and 188 with baricitinib 4 mg (OR 2.79, 95% CrI 1.10 to 7.22). Moderate-certainty evidence indicates that baricitinib 2 mg was probably more effective than placebo in achieving ACR50, but the credible interval crossed the null effect. We downgraded the certainty of evidence for the comparison baricitinib versus placebo due to serious imprecision. The number of people achieving ACR50 was 77 with placebo and 143 with baricitinib 2 mg (OR 2.00, 95% CrI 0.77 to 5.23).

Pair-wise comparisons

A complete table of pair-wise comparisons for ACR50 is presented in Open Science Framework. High-certainty evidence indicates that tocilizumab 8 mg/kg IV was more effective than baricitinib 2 mg. The number of people achieving ACR50 was 314 with baricitinib 2 mg and 695 with tocilizumab 8 mg/kg IV (OR 5.01, 95% CrI 1.23 to 22.08). High-certainty evidence indicates that tocilizumab 8 mg/kg IV was more effective than tocilizumab 4 mg/kg IV. The number

of people achieving ACR50 was 206 with tocilizumab 4 mg/kg IV compared to 348 with tocilizumab 8 mg/kg IV (OR 2.06, 95% CrI 1.18 to 3.41). No other significant pair-wise comparisons were found.

Radiographic progression

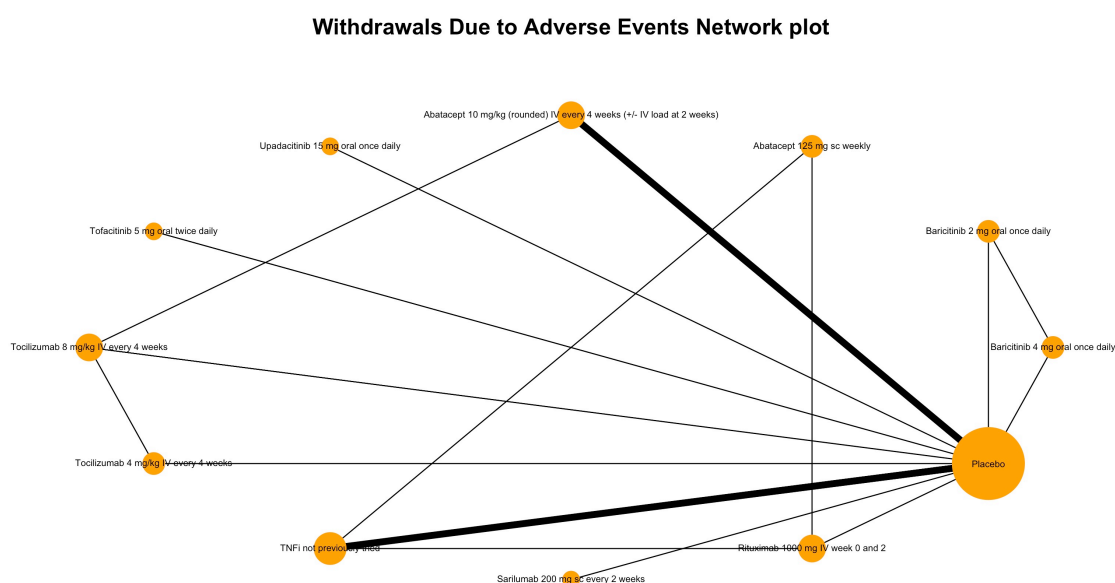
There was insufficient radiographic data available for the included trials to perform an NMA. Three studies prespecified radiographic outcome data (Cohen 2006; Furst 2007; Gottenberg 2016), but only two of these studies reported radiographic data. Cohen 2006 found significantly less radiographic progression in the rituximab

arm compared to the placebo arm. Furst 2007 found comparable changes in radiographic progression between trial arms, which were a continuation of failed etanercept and a TNFi switch to infliximab. Gottenberg 2016 prespecified radiographic data in their protocol, but did not publish any data due to missing data for more than half of participants.

Withdrawals due to adverse events

We included 10 studies in this analysis (Summary of findings 2; Figure 5).

Figure 5. Network plot of withdrawals due to adverse events. Each circle represents an intervention node, and connecting lines represent direct comparisons. Larger nodes indicate a greater number of participants receiving the intervention, and thicker lines indicate a greater number of studies informing the comparison.



Effects relative to placebo

TNF inhibitor

Moderate-certainty evidence indicates that switching to a TNFi not previously tried was probably less harmful than placebo, with lower withdrawal rates, but the credible interval crossed the null effect. The rate of withdrawals due to adverse events per 1000 person-years was 107 with placebo and 34 with TNFi (RR 0.32, 95% CrI 0.07 to 1.1). We downgraded the certainty of evidence due to serious imprecision.

IL-6 inhibitor

Moderate-certainty evidence indicates that sarilumab was probably more harmful than placebo, but the credible interval crossed the null value. Low-certainty evidence suggests that tocilizumab may be more harmful than placebo, with higher withdrawal rates, but the credible interval crossed the null effect. The rate of withdrawals due to adverse events per 1000 person-years was 107 with placebo, 190 with sarilumab (RR 1.98, 95%

CrI 0.57 to 7.19), 138 with tocilizumab 4 mg/kg (RR 1.39, 95% CrI 0.39 to 5.28), and 147 with tocilizumab 8 mg/kg (RR 1.49, 95% CrI 0.51 to 4.81). We downgraded the certainty of evidence twice for very serious imprecision for both tocilizumab doses, and once for serious imprecision for sarilumab.

T-cell co-stimulation inhibitor

Low-certainty evidence suggests that intravenous abatacept may be less harmful than placebo, while subcutaneous abatacept may be more harmful than placebo; however, the credible interval crossed the null effect for both comparisons. The rate of withdrawals due to adverse events per 1000 person-years was 107 with placebo, 93 with intravenous abatacept (RR 0.92, 95% CrI 0.34 to 2.79), and 282 with subcutaneous abatacept (RR 3.11, 95% CrI 0.05 to 249.6). We downgraded the certainty of evidence twice for very serious imprecision.

B-cell depletor

Low-certainty evidence suggests that rituximab may be more harmful than placebo, with lower withdrawal rates, but the credible interval crossed the null effect. The rate of withdrawals due to adverse events per 1000 person-years was 107 with placebo and 224 with rituximab (RR 2.38, 95% CrI 0.44 to 23.31). We downgraded the certainty of evidence twice for very serious imprecision.

JAK inhibitor

Low-certainty evidence suggests that upadacitinib and baricitinib 2 mg may be less harmful than placebo, with lower withdrawal rates, but the credible interval crossed the null effect. Low-certainty evidence suggests that tofacitinib and baricitinib 4 mg may be more harmful than placebo, but the credible interval crossed the null effect. The rate of withdrawals due to adverse events per 1000 person-years was 107 with placebo and 43 with upadacitinib 15 mg (RR 0.41, 95% CrI 0.08 to 1.83), 136 with tofacitinib (RR 1.37, 95% CrI 0.35 to 5.58), 94 with baricitinib 2 mg (RR 0.93, 95% CrI 0.22 to 4.01), and 141 with baricitinib 4 mg (RR 1.43, 95% CrI 0.38 to 5.81). We downgraded the certainty of evidence for all four comparisons twice for very serious imprecision.

Pair-wise comparisons

A complete table of pair-wise comparisons for withdrawals due to adverse events is presented in Open Science Framework. There were no statistically significant pair-wise comparisons for withdrawals due to adverse events.

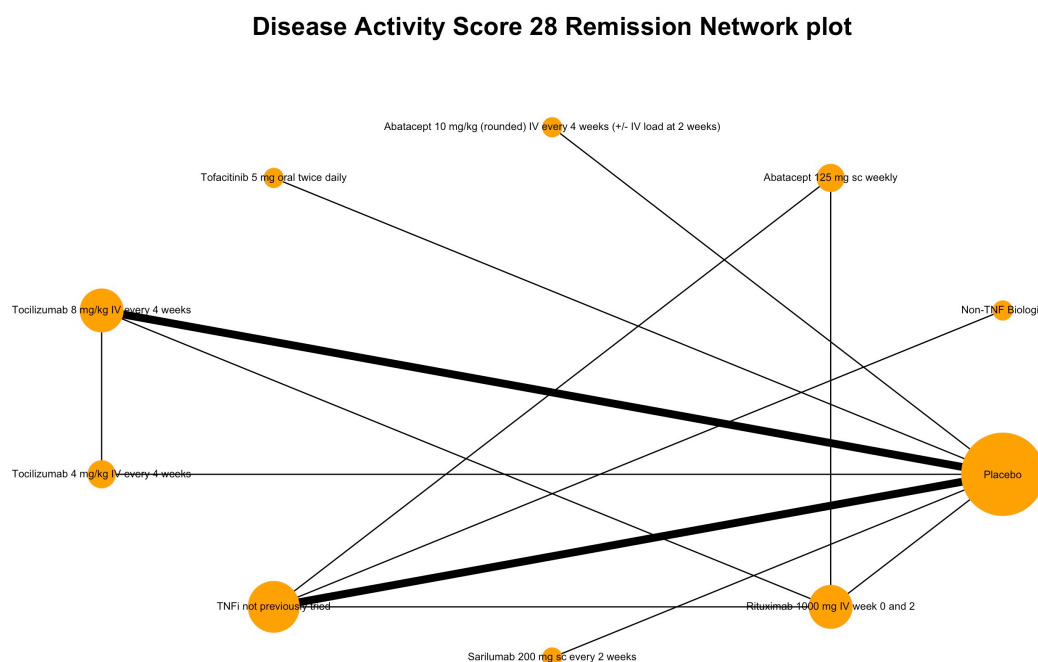
Other critical outcomes

DAS28 remission

Effects relative to placebo

We included 11 studies in this analysis ([Summary of findings 3](#); [Figure 6](#)). There was high-certainty evidence that several interventions were superior to placebo, resulting in more people achieving DAS28 remission (DAS28 < 2.6). The number of DAS28 remission events per 1000 person-years was 15 with placebo and 112 with a TNFi not previously tried (RR 8.14, 95% CrI 2.65 to 34.54), 230 for a non-TNF biologic (RR 19.33, 95% CrI 4.19 to 124.31), 76 for sarilumab (RR 5.33, 95% CrI 1.77 to 16.69), 260 for tocilizumab 8 mg/kg (RR 22.8, 95% CrI 8.29 to 78.5), 253 for intravenous abatacept (RR 21.99, 95% CrI 2.99 to 927.69), and 78 for rituximab (RR 5.51, 95% CrI 1.76 to 21.35). There was moderate-certainty evidence that tocilizumab 4 mg/kg was probably more effective than placebo, with 66 DAS28 remission events per 1000 patient-years (RR 4.60, 95% CrI 1.21 to 13.79). There was moderate-certainty evidence that the remaining interventions, subcutaneous abatacept and tofacitinib, were probably more effective than placebo in achieving DAS28 remission, but the credible intervals crossed the null effect. The number of DAS28 remission events per 1000 person-years was 15 with placebo and 50 with subcutaneous abatacept (RR 3.39, 95% CrI 0.31 to 29.04) and 71 with tofacitinib (RR 4.95, 95% CrI 0.87 to 44.54). We downgraded the certainty of evidence for serious imprecision.

Figure 6. Network plot of DAS28 remission. Each circle represents an intervention node, and connecting lines represent direct comparisons. Larger nodes indicate a greater number of participants receiving the intervention, and thicker lines indicate a greater number of studies informing the comparison.



Pair-wise comparisons

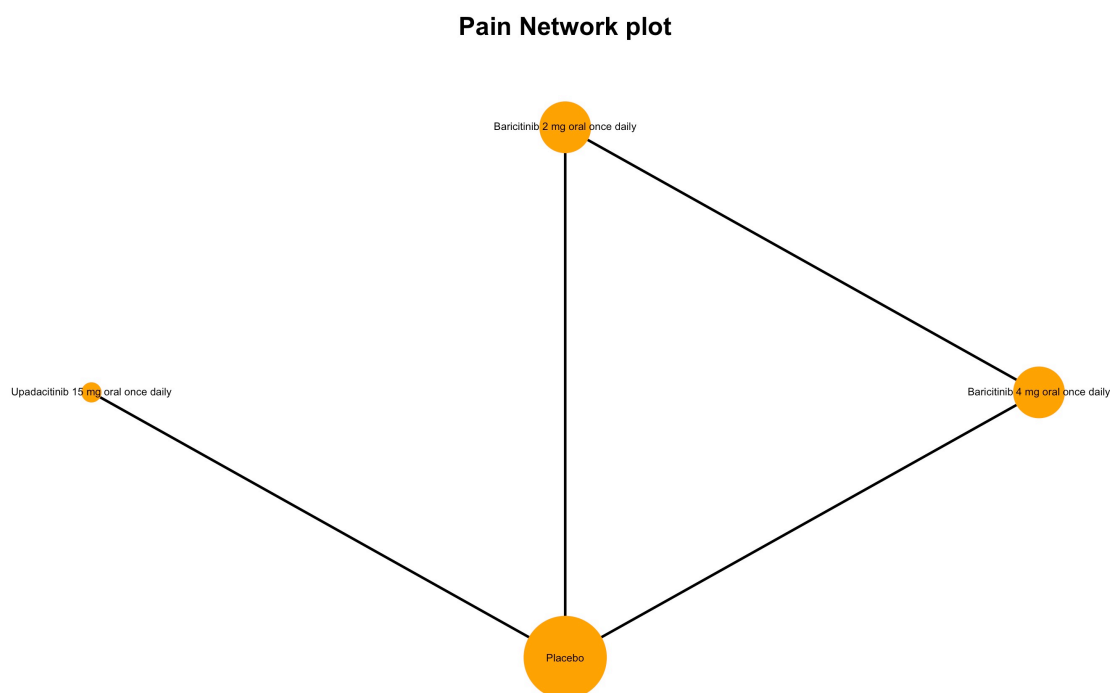
A complete table of pair-wise comparisons for DAS28 remission is presented in Open Science Framework. There was high-certainty evidence that tocilizumab 8 mg/kg IV was more effective than tocilizumab 4 mg/kg IV and rituximab. The number of people achieving DAS28 remission per 1000 person-years was 76 for tocilizumab 4 mg/kg IV and 307 for tocilizumab 8 mg/kg IV (OR 5.39, 95% CrI 12.67 to 10.04). The number of people achieving DAS28 remission per 1000 person-years was 50 for rituximab and 180 for tocilizumab 8 mg/kg IV (OR 4.2, 95% CrI 1.34 to 12.19).

Pain using the VAS

Effects relative to placebo

We included three studies in the analysis of pain scores, measured by VAS (Summary of findings 4; Figure 7). All comparisons for pain scores were based on direct evidence. There was high-certainty evidence that upadacitinib and baricitinib 2 mg were more effective than placebo in improving pain. The mean pain scores, measured by the VAS (0-to-100-point scale, lower scores indicate less pain), were 61.5 with placebo, 45.25 with upadacitinib (MD -16.25, 95% CrI -20.86 to -11.77), and 54 with baricitinib 2 mg (MD -7.5, 95% CrI -11.7 to -3.4). The evidence for baricitinib 4 mg was of very low certainty, indicating that we are very uncertain about the effect. We downgraded the certainty of evidence three times due to extremely serious imprecision.

Figure 7. Network plot of pain. Each circle represents an intervention node, and connecting lines represent direct comparisons. Larger nodes indicate a greater number of participants receiving the intervention, and thicker lines indicate a greater number of studies informing the comparison.



Pair-wise comparisons

A complete table of pair-wise comparisons for pain is presented in Open Science Framework. There was high-certainty indirect evidence that baricitinib 2 mg and baricitinib 4 mg were both more effective than upadacitinib 15 mg in reducing pain scores. The mean pain scores were 45.25 with upadacitinib 15 mg, 57.05 with baricitinib 2 mg (MD 8.8, 95% CrI 2.66 to 14.83), and 57.55 with baricitinib 4 mg (MD 12.3, 95% CrI 5.94 to 18.68).

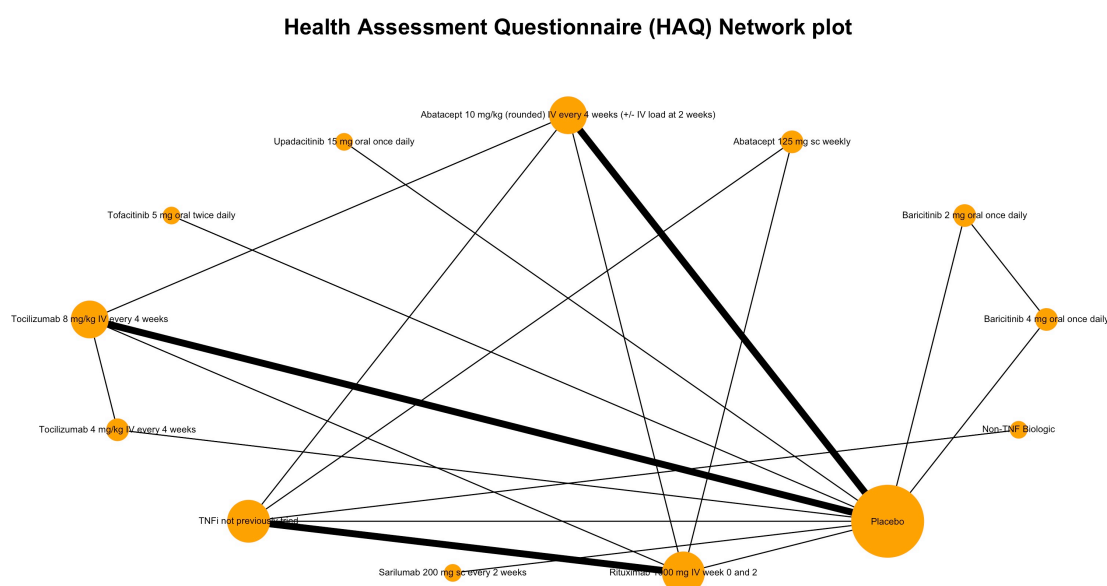
Physical functioning using the HAQ

Effects relative to placebo

We included 15 studies in this analysis (Summary of findings 5; Figure 8). There was high-certainty evidence that tocilizumab 4 mg, tocilizumab 8 mg, intravenous abatacept, and rituximab were more effective than placebo at improving physical functioning. The mean physical functioning score, measured by the HAQ (0-to-3 scale, lower scores indicate better function), was 1.56 with placebo and 1.3 with tocilizumab 4 mg (MD -0.26, 95% CrI -0.39 to -0.13), 1.31 with tocilizumab 8 mg (MD -0.25, 95% CrI -0.44 to -0.05), 1.23 with intravenous abatacept (MD -0.33, 95% CrI -0.5 to -0.1), and 1.29 with rituximab (MD -0.27, 95% CrI -0.48

to -0.04). There was moderate-certainty evidence that sarilumab, tofacitinib, baricitinib 2 mg, and baricitinib 4 mg were probably more effective than placebo in improving physical functioning, but the credible intervals crossed the null effect. The mean HAQ score was 1.56 with placebo and 1.32 with sarilumab (MD -0.24, 95% CrI -0.55 to 0.07), 1.32 with tofacitinib (MD -0.24, 95% CrI -0.57 to 0.1), 1.28 with baricitinib 2 mg (MD -0.28, 95% CrI -0.58 to 0.03), and 1.26 with baricitinib 4 mg (MD -0.3, 95% CrI -0.61 to 0.01). We downgraded the certainty of evidence to moderate due to serious imprecision. There was moderate-certainty evidence that a TNFi not previously tried probably did not differ from placebo in improving physical functioning (MD -0.19, 95% CrI -0.5 to 0.07). We downgraded the certainty of evidence to moderate due to serious imprecision. There was low-certainty evidence that subcutaneous abatacept may be more effective than placebo, but the credible interval crossed the null effect (MD -0.3, 95% CrI -0.97 to 0.35). Low-certainty evidence suggests that compared with placebo, non-TNF biologics and upadacitinib may not improve physical functioning, with MDs of -0.16 (95% CrI -0.62 to 0.22) and -0.12 (95% CrI -0.43 to 0.19), respectively. We downgraded the certainty of evidence twice due to very serious imprecision.

Figure 8. Network plot of Health Assessment Questionnaire. Each circle represents an intervention node, and connecting lines represent direct comparisons. Larger nodes indicate a greater number of participants receiving the intervention, and thicker lines indicate a greater number of studies informing the comparison.



Pair-wise comparisons

A complete table of pair-wise comparisons for the HAQ is presented in Open Science Framework. There were no significant pair-wise comparisons for the HAQ.

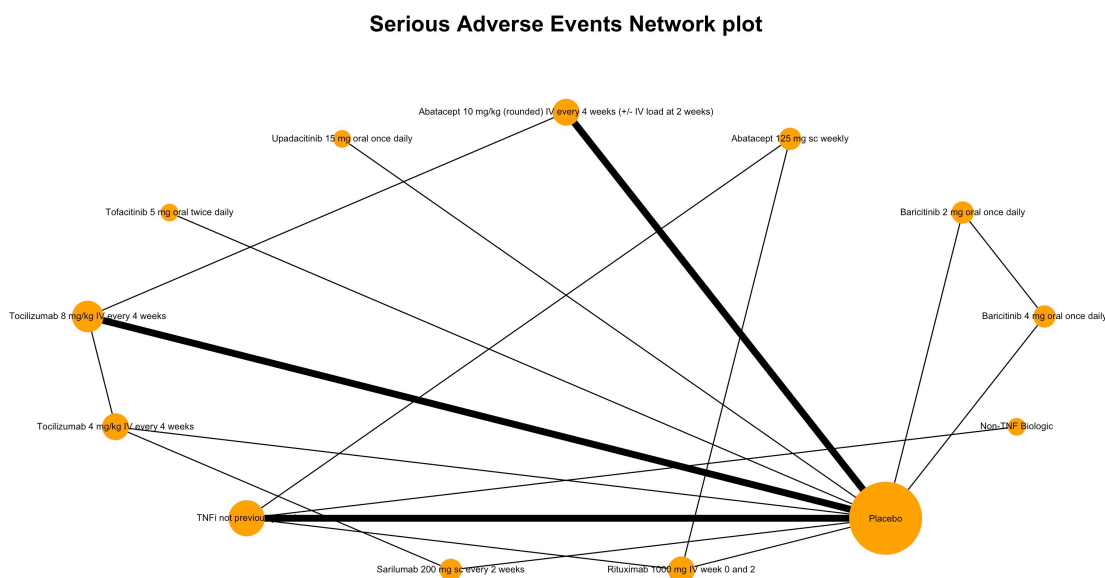
Serious adverse events

Effects relative to placebo

We included 12 studies in this analysis (Summary of findings 6; Figure 9). There was moderate-certainty evidence that intravenous abatacept and rituximab were probably less harmful than placebo, with lower rates of serious adverse events, but the credible interval crossed the null effect. The rate of serious adverse events per 1000 patient years was 187 with placebo and 108 with intravenous

abatacept (RR 0.61, 95% CrI 0.28 to 1.26) and 106 with rituximab (RR 0.6, 95% CrI 0.36 to 1.72). We downgraded the certainty of evidence due to serious imprecision. Low-certainty evidence suggests that tofacitinib and baricitinib 2 mg may be less harmful than placebo, with lower rates of serious adverse events, but the credible interval crossed the null effect. The rate of serious adverse events per 1000 patient years was 187 with placebo and 53 with tofacitinib (RR 0.29, 95% CrI 0.03 to 1.6) and 84 with baricitinib 4 mg (RR 0.49, 95% CrI 0.13 to 1.63). We downgraded the certainty of evidence twice for very serious imprecision. We are very uncertain about the safety of the remaining interventions (TNFi not previously tried, sarilumab, tocilizumab 4 mg, tocilizumab 8 mg, subcutaneous abatacept, and baricitinib 4 mg) compared to placebo, as the certainty of evidence was downgraded to very low due to extremely serious imprecision.

Figure 9. Network plot of serious adverse events. Each circle represents an intervention node, and connecting lines represent direct comparisons. Larger nodes indicate a greater number of participants receiving the intervention, and thicker lines indicate a greater number of studies informing the comparison.



Pair-wise comparisons

A complete table of pair-wise comparisons for serious adverse events is presented in Open Science Framework. There were no statistically significant pair-wise comparisons for serious adverse events.

Important outcomes

Five important outcomes are presented in Table 6.

ACR20/ACR70

We included 11 studies in the analysis of ACR20 and ACR70. Several interventions demonstrated improved rates of ACR20 response compared to placebo, including TNFi not previously tried (OR 4.45, 95% CrI 2.24 to 12.17), sarilumab (OR 3.08, 95% CrI 1.18 to 8.12), tocilizumab 4 mg/kg (OR 3.84, 95% CrI 1.45 to 10.5), tocilizumab 8 mg/kg (OR 8.67, 95% CrI 3.89 to 19.78), intravenous abatacept (OR 3.82, 95% CrI 1.86 to 7.82), rituximab (OR 4.12, 95% CrI 1.85 to 9.46), and upadacitinib (OR 4.65, 95% CrI 1.75 to 12.51).

Some of the interventions which were statistically superior to placebo in ACR20 response were also superior in ACR70 response, including TNFi not previously tried (OR 5.4, 95% CrI 1.6 to 20.39), tocilizumab 8 mg/kg (OR 12.97, 95% CrI 2.44 to 113.84), intravenous abatacept (OR 4.12, 95% CrI 1.53 to 12.87), and rituximab (OR 6.27, 95% CrI 1.84 to 22.77). Interventions that were statistically superior to placebo in ACR70 but not ACR20 included tofacitinib (OR 12.02, 95% CrI 2.25 to 112.43), baricitinib 2 mg (OR 4.49, 95% CrI 1.19 to 19.14), and baricitinib 4 mg (OR 6.04, 95% CrI 1.6 to 25.17).

DAS28

We included two studies in the analysis of DAS28. Three interventions were more effective than placebo at reducing DAS28 scores, including TNFi not previously tried (MD -1.99, 95% CrI -2.82 to -0.62), subcutaneous abatacept (MD -1.33, 95% CrI -2.46 to -0.18), and rituximab (MD -1.37, 95% CrI -2.08 to -0.62).

Quality of life

We included three studies in this analysis, though none of the interventions was convincingly different from placebo in improving quality of life.

Fatigue

We included two studies in this analysis. Rituximab was more effective than placebo at reducing fatigue (MD -8.43, 95% CrI -10.77 to -6.13).

Serious infections

We included seven studies in this analysis. The effects of several interventions were not estimable due to the rarity of events. The remaining interventions did not differ significantly from placebo in the rate of serious infections.

Subgroup analysis

We conducted meta-regression to assess four variables that may have impacted the NMA for ACR50 response; these variables were baseline response rate, disease duration, duration of trial, and year of publication. We did not conduct the planned meta-regression for a fifth variable, baseline disease activity (low disease activity versus

moderate and high disease activity), as all included trials sampled a population with moderate or high disease activity.

The results of the subgroup analysis are presented in [Table 7](#). Data were sparse for each of the variables, leading to wide CIs. None were found to be statistically significant.

Sensitivity analysis

Sensitivity analyses for two critical outcomes (ACR50 and withdrawals due to adverse events) are presented in [Table 8](#) and [Table 9](#). The prespecified sensitivity analyses were 1) including studies with high risk of bias, using semi-informative priors and a fixed-effect model, and 2) creating networks from different time points (three and six months). There were minor differences between the main analysis and the sensitivity analyses.

The inclusion of studies at high risk of bias altered some point estimates for ACR50 and withdrawals due to adverse events, but the credible intervals overlapped. Superiority of subcutaneous abatacept over placebo for ACR50 was observed when studies at high risk of bias were included (OR 4.78, 95% CrI 1.15 to 19.58), but not when they were excluded (OR 4.31, 95% CrI 0.97 to 18.28).

The main analysis used semi-informative priors and a random-effects model. The sensitivity analysis using uninformative priors showed larger credible intervals than the main analysis with semi-informative priors. Using uninformative priors altered some of the findings for ACR50 response, with several interventions no longer demonstrating superiority over placebo (sarilumab, tocilizumab 4 mg/kg, upadacitinib, tofacitinib, and baricitinib 4 mg). No material change was observed for withdrawals due to adverse events, as credible intervals crossed the null in both analyses. Using a fixed-effect model, point estimates were similar to those from the random-effects model, although credible intervals were narrower for both ACR50 response and withdrawals due to adverse events. Subcutaneous abatacept (OR 4.26, 95% CrI 1.23 to 13.46) and baricitinib 2 mg (OR 2.00, 95% CrI 1.14 to 3.55) were superior to placebo for ACR50 under the fixed-effect model but not under the random-effects model used in the main analysis (abatacept: OR 4.31, 95% CrI 0.97 to 18.28; baricitinib 2 mg: OR 2.00, 95% CrI 0.77 to 5.23).

Minor changes in point estimates and corresponding credible intervals for ACR50 response were observed when analyses were conducted separately at three and six months; however, data were not available for all interventions at both time points. Tocilizumab 8 mg/kg demonstrated superiority over placebo at six months (OR 10.99, 95% CrI 3.13 to 43.42), but not at three months (OR 8.82, 95% CrI 0.95 to 320.15).

Heterogeneity and coherence

Clinical heterogeneity was assessed and incorporated into indirectness ratings. We judged all ratings to be low, indicating no major concerns related to clinical heterogeneity. Statistical heterogeneity among direct comparisons was not substantial. Assessment of incoherence using the node-splitting approach did not identify any serious disagreement between direct and indirect evidence requiring downgrading.

Additional material

Additional datasets and material relevant to this review are hosted on Open Science Framework. This includes guidance used for risk

of bias assessments, an annotated list of excluded studies, detailed summary of findings tables, and pair-wise comparisons.

Equity assessment

We did not assess equity in this version of the review.

Reporting biases

We were not able to pool more than 10 studies, therefore we did not create a funnel plot to formally explore possible small-study and publication biases.

DISCUSSION

This systematic review and NMA included 19 RCTs and 4779 participants. The included trials assessed biologic/targeted synthetic DMARDs in participants with rheumatoid arthritis who had experienced an inadequate response to a TNFi and were pooled for NMA.

Summary of main results

There were 12 treatment nodes within the 19 included trials: TNFi not previously tried, sarilumab (200 mg subcutaneously (SC) biweekly), tocilizumab (4 mg/kg), tocilizumab (8 mg/kg), abatacept (125 mg SC), abatacept (10 mg/kg IV), rituximab (1000 mg), upadacitinib (15 mg), tofacitinib (5 mg), baricitinib (2 mg), and baricitinib (4 mg). Two treatment nodes (continue failed TNFi and switch in route of administration of TNFi) were excluded from the primary analysis due to high risk of bias in the underlying trials.

There was high-certainty evidence that nine interventions were superior to placebo in improving rheumatoid arthritis disease activity, as measured by the ACR50 response criteria in participants with inadequate response to a TNFi ([Summary of findings 1](#)). Of these, the higher dose of tocilizumab had the largest magnitude of effect, although pair-wise comparisons did not show a statistically significant difference between tocilizumab and other interventions. Other efficacious drugs included a TNFi not previously tried, two IL-6 inhibitors (sarilumab and tocilizumab), intravenous abatacept, the B-cell depletor rituximab, and three JAK inhibitors (upadacitinib, tofacitinib, and baricitinib 4 mg). Moderate-certainty evidence showed that subcutaneous abatacept and baricitinib 2 mg were probably more effective than placebo, but the credible interval crossed the null effect.

Based on moderate-/high-certainty evidence, the findings for DAS28 remission were generally consistent with the ACR50 response analysis ([Summary of findings 3](#)). Based on high-certainty evidence, upadacitinib and baricitinib 2 mg were more efficacious than placebo at improving pain VAS scores, while the effect of other interventions was uncertain ([Summary of findings 4](#)). Based on moderate-/high-certainty evidence, most interventions improved physical function more than placebo, although the credible intervals often crossed the null effect ([Summary of findings 5](#)).

An important feature of this review was a lack of clarity regarding the safety of active interventions compared to placebo due to the low-certainty evidence in the analysis of withdrawals due to adverse events and serious adverse events ([Summary of findings 2](#); [Summary of findings 6](#)). Given this uncertainty, we were unable to determine the magnitude of harm associated with each intervention. However, the studies included in this

review are a small subset of all randomised and non-randomised trials of these interventions in rheumatoid arthritis and other diseases, which would provide additional data on harms associated with the interventions. There are rare risks associated with each intervention. For example, administration of TNFi comes with a risk of demyelination causing adverse neurological events, although a recent large cohort study found that the incidence of neuroinflammatory disease is low, with an estimated absolute risk of less than 1 per 1000 patients per year [83, 84]. There are also data suggesting that JAK inhibitors may increase the risk of thrombotic events and malignancy in older populations with pre-existing cardiovascular risk factors, although a recent study found no overall increase in cancer risk with exposure to JAK inhibitors [85, 86, 87].

Limitations of the evidence included in the review

Participants in the included trials were generally representative of a subset of people with rheumatoid arthritis seen in clinical practice who have not responded to TNFi. Participants were similar in terms of disease duration and severity and number of failed therapies. All had failed one or more TNFi, although participants varied in which TNFi was tried.

We employed the GRADE minimally contextualised approach to rate certainty of network evidence for six of the critical outcomes (ACR50, withdrawals due to adverse events, DAS28 remission, pain as measured by the VAS, physical functioning as measured by the HAQ, and serious adverse events). There were insufficient data to analyse radiographic progression, the seventh critical outcome, using NMA. Radiographic outcomes are infrequently assessed in trials of biologic/targeted synthetic DMARD-experienced populations; however, they are commonly assessed in DMARD-insufficient responder populations, where the outcome has been well established. Comparisons between placebo and each intervention for the critical outcomes had varying certainty of evidence ratings. The reason for downgrading in all cases was imprecision, and the level of downgrade was related to the severity of imprecision.

To summarise the certainty of evidence in this review, we will outline the ratings and reasons for downgrading each of the six critical outcomes.

1. High-certainty evidence showed that a TNFi not previously tried, sarilumab, tocilizumab (4 mg/kg and 8 mg/kg), intravenous abatacept, rituximab, upadacitinib, tofacitinib 5 mg, and baricitinib 4 mg were all more effective than placebo in improving symptoms from baseline, as measured by the ACR50. Moderate-certainty evidence showed that subcutaneous abatacept and baricitinib 2 mg were probably more effective than placebo, but the credible interval crossed the null effect. We downgraded the certainty of evidence for both comparisons to placebo due to serious imprecision.
2. There was profound uncertainty regarding the short-term harms associated with biologic/targeted synthetic DMARDs in this review, as certainty of evidence ratings for withdrawals due to adverse events and serious adverse events were impacted by significant imprecision within the network. Moderate-certainty evidence showed that a TNFi not previously tried probably resulted in fewer withdrawals due to adverse events, but the credible interval crossed the null effect. There was low-

certainty evidence for the remaining interventions, and all were downgraded for serious imprecision.

3. High-certainty evidence showed that a TNFi not previously tried, a non-TNF biologic, sarilumab, tocilizumab 8 mg/kg, intravenous abatacept, and rituximab were more effective than placebo in achieving remission, according to the DAS28. Moderate-certainty evidence showed that subcutaneous abatacept, tocilizumab 4 mg/kg, and tofacitinib were probably more effective than placebo, but the evidence was downgraded due to serious imprecision.
4. High-certainty evidence showed that upadacitinib and baricitinib 2 mg were more effective than placebo in improving pain. There was very low-certainty evidence for the comparison between placebo and baricitinib 4 mg, leading to uncertainty about the effect. We downgraded the certainty of evidence three times due to extremely serious imprecision.
5. High-certainty evidence showed that tocilizumab (4 mg/kg and 8 mg/kg), intravenous abatacept, and rituximab were more effective than placebo in improving physical functioning as measured by the HAQ. Moderate-certainty evidence showed that sarilumab, tofacitinib, and baricitinib (2 mg and 4 mg) were probably more effective than placebo, though the credible interval crossed the null effect. We downgraded the certainty of evidence once for serious imprecision. There was moderate-/low-certainty evidence that a TNFi not previously tried, a non-TNF biologic, and upadacitinib may not differ from placebo, as the point estimate did not cross the MID. We downgraded the certainty of evidence due to very serious imprecision.
6. Moderate-certainty evidence showed that intravenous abatacept and rituximab probably resulted in fewer serious adverse events than placebo, but the credible interval crossed the null effect. We downgraded the certainty of evidence once for serious imprecision. Low-certainty evidence suggests that tofacitinib and baricitinib 2 mg may be less harmful than placebo, but the credible interval crossed the null effect. We downgraded the certainty of evidence twice for very serious imprecision. There was very low-certainty evidence for the comparisons of placebo versus a TNFi not previously tried, sarilumab, tocilizumab (4 mg/kg and 8 mg/kg), subcutaneous abatacept, and baricitinib 4 mg. We downgraded the certainty of evidence three times due to extremely serious imprecision.

Gaps in the current body of direct evidence comparing interventions used after failure of a TNFi was a limiting factor in this review. Due to the scarcity of head-to-head trials, comparisons between active interventions were largely based on indirect evidence. The low certainty of evidence of these head-to-head comparisons limited our ability to make judgements on comparative efficacy between active treatments.

Limitations of the review processes

The method used for identifying trials was intensive and included electronic searches of relevant databases, a first-line screening approach that combined machine learning and crowdsourcing (e.g. Cochrane Crowd), followed by PICO annotation and then full-text review. Data extraction and risk of bias assessments were performed in Covidence, with review authors following a detailed extraction guide. All screening and data extraction procedures were performed independently and in duplicate, with conflicts resolved by an expert review author. We did not assess publication bias at the domain level of the GRADE process

because of the limited number of studies available for each direct comparison. Instead, we reviewed trial registers to ensure thorough examination in alignment with our PICO framework. Furthermore, when evaluating imprecision using the minimally contextualised framework, we did not take the optimal information size (OIS) into account.

We did not include publications that were abstract-only because they often include insufficient information to be suitable in an NMA [88]. However, excluding abstract-only trials may have introduced publication bias if the difference between published and unpublished trials was substantial enough to change treatment effect estimates or ranking. The two abstract-only trials identified in our search did not report on any of our critical or important clinical outcomes and were instead focused on changes to bone mineral density, so we expect the publication bias in this review to be minimal.

Agreements and disagreements with other studies or reviews

Previous systematic reviews that have explored the effects of biologic DMARDs for rheumatoid arthritis treatment differed from our review in population parameters, outcomes measured, and interventions. Our review had specific inclusion parameters, limited to trials with participants with rheumatoid arthritis who had an inadequate response to at least one TNFi or biologic/targeted synthetic DMARD. Previous reviews have included trials with participants who were unsuccessfully treated by conventional DMARDs [89, 90, 91]. The comprehensive Cochrane review by Singh JA and colleagues (2017), which assessed the comparative effectiveness of biologic DMARDs in rheumatoid arthritis, has some overlapping features with the current review; however, several key differences distinguish the two [91]. Our review expands the classes of medications included and adds medications within each class. Furthermore, our review is in living mode to ensure all new trials are captured to reflect the most up-to-date understanding of the body of evidence on this topic [91].

In contrast, Singh S and colleagues (2020) focused on the risk of major adverse cardiovascular events and stroke associated with biologic/targeted synthetic DMARD use [89]. Janke and colleagues did not include trials evaluating JAK inhibitors [90]. Our review followed the GRADE approach to rating certainty of evidence, while Janke and colleagues assessed certainty using the approach by the Institute for Quality and Efficiency in Health Care (IQWiG) [90].

Overall, the findings of the current review are in agreement with previous evidence indicating that biologic/targeted synthetic DMARDs are superior to placebo in improving clinical outcomes for rheumatoid arthritis [89, 90, 91]. Consistent with our review, Singh JA and colleagues (2017) reported that biologic/targeted synthetic DMARDs demonstrated greater clinical benefit compared with placebo. Janke and colleagues identified few statistically significant pair-wise differences and pointed to the scarcity of head-to-head trials [90]. Singh S and colleagues (2020) reported low-certainty evidence regarding withdrawals and serious adverse events [89].

AUTHORS' CONCLUSIONS

Implications for practice

The main implications for practice are the clear efficacy of biologic/targeted synthetic disease-modifying antirheumatic drugs (DMARDs) compared to placebo, but the considerable uncertainty surrounding head-to-head comparisons. Clinical treatment decisions should be guided by individual patient characteristics and preferences.

Based upon both direct and indirect evidence, high-certainty evidence shows that nine of the 11 biologic/targeted synthetic DMARDs assessed in this systematic review and network meta-analysis (tumour necrosis factor (TNF) inhibitor not previously tried, sarilumab, tocilizumab 4 mg/kg and 8 mg/kg, intravenous abatacept, rituximab, upadacitinib, tofacitinib, and baricitinib 4 mg) are superior to placebo in improving American College of Rheumatology 50% (ACR50) response in individuals with rheumatoid arthritis who have responded inadequately to a TNF inhibitor. The two interventions with less certain efficacy in ACR50 response (subcutaneous abatacept and baricitinib 2 mg) have point estimates sufficiently large (i.e. above the minimally important difference) to suggest that they are probably more effective than placebo, even though the credible intervals crossed the null effect. Pair-wise comparisons showed no significant differences between interventions in ACR50 response.

Based upon both direct and indirect evidence, moderate-certainty evidence shows that switching to a TNF inhibitor not previously tried is probably less harmful than placebo as measured by the rate of withdrawals due to adverse events, but the credible interval crossed the null effect. The evidence for withdrawals due to adverse events for the remaining interventions is low quality, leading to uncertainty regarding demonstrated harms. Pair-wise comparisons showed no significant differences between interventions in the rate of withdrawals due to adverse events.

The scarcity of head-to-head trials and reliance on mainly indirect evidence for pair-wise comparisons contributes to the uncertainty regarding which biologic/targeted synthetic DMARD is best. In view of the lack of pair-wise differences in efficacy among the nine interventions with demonstrated efficacy, other factors, such as route or frequency of administration, concurrent medications, rare risks, comorbidities, and cost, may drive treatment decisions after inadequate response to TNF inhibitors. Operating within a shared decision-making framework, patients and clinicians should discuss the benefits and risks of available treatments and tailor the treatment plan to each patient's goals and preferences.

This review is linked to living rheumatology guidelines in Canada and Australia through the Canadian Rheumatology Association and Australian Rheumatology Association [92, 93]. These living guidelines provide treatment recommendations to support decision-making in rheumatology care. This review will be living and will be updated annually to capture emerging evidence.

Equity-related implications for practice

Given the number of effective treatments highlighted in this review, individuals with rheumatoid arthritis have the opportunity to make treatment choices based on their unique circumstances. Choices may be driven by equity considerations, including health

vulnerabilities, treatment-related values, access to care, and insurance coverage and associated out-of-pocket cost.

Implications for research

The paucity of head-to-head trials in advanced rheumatoid arthritis treatments contributed to the low-certainty evidence for pair-wise comparisons in this review. This underscores the importance of new randomised controlled trials with active comparator arms, particularly for biologic and targeted synthetic DMARDs, as there are few available in the current body of evidence. Trials that compare Janus kinase (JAK) inhibitors to biological DMARDs with different mechanisms of action (interleukin-6 (IL-6) inhibitors, B-cell depletors, T-cell co-stimulation inhibitors) would provide useful insight, as would trials comparing drugs from different classes of biologic DMARDs.

The network meta-analysis presented in this review is limited to group-level data. The inclusion of individual patient data could enhance the precision of treatment effect estimates and allow for an exploration of covariates that contribute to heterogeneity. The inclusion of individual patient data could highlight potential factors that influence therapeutic response to certain drugs, which could allow for a more personalised approach to treatment.

Equity-related implications for research

Randomised controlled trials typically have narrow inclusion criteria, resulting in a homogeneous sample of the population. While restriction of certain population subgroups can limit the potential for confounding effects, it may impact the generalisability of the results. Future trials could consider adopting a more pragmatic approach to eligibility criteria to increase the diversity of people enrolled.

SUPPLEMENTARY MATERIALS

Supplementary materials are available with the online version of this article: [10.1002/14651858.CD013562.pub2](https://doi.org/10.1002/14651858.CD013562.pub2).

Supplementary material 1 Search strategies

Supplementary material 2 Characteristics of included studies

Supplementary material 3 Characteristics of studies awaiting classification

Supplementary material 4 Characteristics of ongoing studies

Supplementary material 5 Data package

Supplementary material 6 Risk of bias template

Supplementary material 7 Semi-informative priors

Supplementary material 8 Study characteristics of unpublished or awaiting classification, ongoing and terminated trials

ADDITIONAL INFORMATION

Acknowledgements

Elements of the [Methods](#) section are based on the standard Cochrane Musculoskeletal protocol template.

Daksh Choudhary, Dennis Neuen, Julia New-Tolley, Maria Powell, Alannah Quinlivan, and Amro Qaddoura contributed to the development of the protocol for this systematic review.

Editorial and peer-reviewer contributions

Cochrane Musculoskeletal Health and Pain Thematic Group supported the authors in the development of this review.

The following people conducted the editorial process for this article:

- Sign-off Editor (final editorial decision): Toby Lasserson, Deputy Editor in Chief, Cochrane;
- Managing Editor (selected peer reviewers, provided editorial guidance to authors, edited the article): Alan Thomas, Cochrane Central Editorial Service;
- Editorial Assistant (conducted editorial policy checks, collated peer-reviewer comments, and supported the editorial team): Lisa Wydrzynski, Cochrane Central Editorial Service;
- Copy Editor (copy editing and production): Lisa Winer, Cochrane Central Production Service;
- Peer reviewers (provided comments and recommended an editorial decision): Luiz Sergio F de Carvalho, Hospital Sirio-Libanes (Brasília, DF, Brazil) (clinical/content review), Vadim Khaychuk (clinical/content review), Brian Duncan (patient and public review), Dr Kerry Dwan, University of York (methods review), Jo Platt, Central Editorial Information Specialist (search review).

Contributions of authors

Jocelyn Thomas: data extraction, quality assessment, writing of review

Mohammed Mujaab Kamso: data analysis, statistical analysis design, editing of protocol

Samuel L Whittle: content expertise, methodology, writing of protocol

George A Wells: content expertise, methodology, statistician

Shannon E Kelly: content expertise, methodology

Jordi Pardo Pardo: content expertise, methodology, writing of protocol, editing of review

Peter Tugwell: content expertise, methodology, editing of protocol

Renea V Johnston: content expertise, methodology, editing of review

Vanessa Glennon: content expertise, methodology, data extraction

Maede Ejaredar: data extraction, quality assessment, editing of protocol

Megan Thomas: co-ordination of pilot testing, data extraction form design, writing of protocol

Chloe Lee: pilot-tested data collection, quality assessment, and PICO annotation, editing of protocol

Jesse Elliott: content expertise, methodology

Abir Mokbel: content expertise, methodology, data extraction, quality assessment

Jodie C Avery: content expertise, methodology, data extraction, quality assessment

Liesl Grobler: content expertise, methodology, data extraction, quality assessment

Sheila Cyril: content expertise, methodology

Rob Deardon: statistical analysis design

Elie A Akl: content expertise, methodology, data analysis consulting

Lara J Maxwell: content expertise, methodology

Rachelle Buchbinder: content expertise, methodology, writing of protocol, editing of review

Glen S Hazlewood: conception and design of analysis structure, content expertise, methodology, writing of protocol, editing of review

Declarations of interest

Jocelyn Thomas: none known

Mohammed Mujaab Kamso: none known

Samuel L Whittle: consultant rheumatologist in a public hospital

George A Wells: none known

Shannon E Kelly: none known

Jordi Pardo Pardo: none known

Peter Tugwell: rheumatologist who treats people with rheumatoid arthritis (public practice)

Renea V Johnston: none known

Vanessa Glennon: none known

Maede Ejaredar: none known

Megan Thomas: none known

Chloe Lee: MD, PGY4 trainee in physical medicine and rehabilitation at the University of Calgary

Jesse Elliott: none known

Abir Mokbel: rheumatologist who treats people with rheumatoid arthritis (public and private practice)

Jodie C Avery: none known

Liesl Grobler: none known

Sheila Cyril: none known

Rob Deardon: none known

Elie A Akl: general internist, but does not treat rheumatoid arthritis

Lara J Maxwell: none known

Rachelle Buchbinder: rheumatologist, Malvern Rheumatology

Glen S Hazlewood: rheumatologist who treats people with rheumatoid arthritis (public practice)

Sources of support

Internal sources

- School of Public Health and Preventative Medicine, Monach University, Australia

In-kind support (authors: S Whittle, R Johnston, R Buchbinder)

External sources

- Australia and New Zealand Musculoskeletal (ANZMUSC) Clinical Trial Network NHMRC Centre of Research Excellence, Australia

S Whittle is supported in part by an Australia and New Zealand Musculoskeletal (ANZMUSC) Clinical Trial Network Practitioner Fellowship

- The Hospital Research Foundation, Australia

S Whittle is supported in part by The Hospital Research Foundation which provides part funding for his ANZMUSC Practitioner Fellowship

- Australian National Health and Medical Research Council (NHMRC), Australia

R Buchbinder is supported by an Australian National Health and Medical Research Council (NHMRC) Fellowship

- National Health and Medical Research Council (NHMRC), Other

NHMRC Cochrane Collaboration ROund 7 Funding Program Grant to R Johnston, R Buchbinder and S McDonald, to support the activities of Cochrane Musculoskeletal - Australia and Cochrane Australia. The funders had no role in the design, conduct or publication of this review.

- Canadian Institutes of Health Research (CIHR), Canada

CIHR Project Grant [Funding Reference Numbers (FRN) 178375 and 180324]

Registration and protocol

This study was outlined in a Cochrane protocol (CD013562; DOI: 10.1002/14651858.CD013562).

Data, code and other materials

Additional datasets and material relevant to this review are hosted on Open Science Framework (<https://osf.io/x6ct9/>). This includes guidance used for risk of bias assessments, an annotated of list excluded studies, detailed summary of findings tables, and pairwise comparisons.

Notes

This protocol is based on a template developed by Cochrane Musculoskeletal editorial base.

History

Protocol first published: Issue 3, 2020

REFERENCES

1. Cross M, Smith E, Hoy D, Carmona L, Wolfe F, Kamalaraj N, et al. The global burden of rheumatoid arthritis: estimates from the Global Burden of Disease 2010 study. *Annals of the Rheumatic Diseases* 2014;**73**(7):1316-22. [DOI: [10.1136/annrheumdis-2013-204627](https://doi.org/10.1136/annrheumdis-2013-204627)]
2. McInnes IB, Schett G. The pathogenesis of rheumatoid arthritis. *New England Journal of Medicine* 2011;**365**:2205-19. [DOI: [10.1056/NEJMr1004965](https://doi.org/10.1056/NEJMr1004965)]
3. Smolen JS, van der Heijde D, Machold KP, Aletaha D, Landewe R. Proposal for a new nomenclature of disease-modifying antirheumatic drugs. *Annals of the Rheumatic Diseases* 2014;**73**(1):3-5.
4. Yamaoka K, Tanaka Y. Targeting the Janus kinases in rheumatoid arthritis: focus on tofacitinib. *Expert Opinion on Pharmacotherapy* 2014;**15**(1):103-13.
5. Choi MY, Barnabe C, Barber CE, Bykerk V, Pope JE, Hazlewood GS. Pragmatism of randomized controlled trials of biologic treatment with methotrexate in rheumatoid arthritis: a systematic review. *Arthritis Care & Research* 2019;**71**:620-8.
6. Lumley T. Network meta-analysis for indirect treatment comparisons. *Statistics in Medicine* 2002;**21**(16):2313-24.
7. Pope JE, Hazlewood GS. Randomized trials, meta-analyses, and systematic reviews: using examples from rheumatology. *Rheumatic Diseases Clinics of North America* 2018;**44**:295-305.
8. Elliott JH, Synnot A, Turner T, Simmonds M, Akl EA, McDonald S, et al; Living Systematic Review Network. Living systematic review: 1. Introduction – the why, what, when, and how. *Journal of Clinical Epidemiology* 2017;**91**:23-30.
9. Fraenkel L, Bathon JM, England BR, St Clair EW, Arayssi T, Carandang K, et al. 2021 American College of Rheumatology guideline for the treatment of rheumatoid arthritis. *Arthritis & Rheumatology* 2021;**73**(7):924-39.
10. Australia & New Zealand Musculoskeletal Clinical Trials Network (ANZMUSC). An Australian Living Guideline for the Pharmacological Management of Inflammatory Arthritis (2020 (version 0.3); 2022 (version 1.13)). mskguidelines.org.
11. Hazlewood GS, Pardo Pardo J, Barnabe C, Schieir O, Barber CE, Proulx L, et al. Canadian Rheumatology Association living guidelines for the pharmacological management of rheumatoid arthritis with disease-modifying anti-rheumatic drugs. *Journal of Rheumatology* 2022;**49**(10):1092-9.
12. Kamso MM, Pardo Pardo J, Whittle SL, Buchbinder R, Wells G, Glennon V, et al. Crowdsourcing and automation facilitated the identification and classification of randomized controlled trials in a living review. *Journal of Clinical Epidemiology* 2023;**164**:1-8.
13. Covidence. Version accessed 16 February 2025. Melbourne, Australia: Veritas Health Innovation, 2025. Available at <https://www.covidence.org>.
14. Sharma D, Thomas J, Kamso MM, Pardo Pardo J, Hazlewood G. Open Science Framework: Living systematic review (LSR) of rheumatoid arthritis randomized trials. https://osf.io/ph892/?view_only=fa6cb34d46ce40829d692726b76418d0.
15. Hazlewood GS, Whittle SL, Kamso MM, Akl EA, Wells GA, Tugwell P, et al. Disease-modifying anti-rheumatic drugs for rheumatoid arthritis: a systematic review and network meta-analysis. *Cochrane Database of Systematic Reviews* 2020, Issue 3. Art. No: CD013562. [DOI: [10.1002/14651858.CD013562](https://doi.org/10.1002/14651858.CD013562)]
16. Cochrane. Guidance for the production and publication of Cochrane living systematic reviews: Cochrane Reviews in living mode. community.cochrane.org/sites/default/files/uploads/inline-files/Transform/201912_LSR_Revised_Guidance.pdf.
17. Ghogomu EA, Maxwell LJ, Buchbinder R, Rader T, Pardo Pardo J, Johnston RV, et al. Updated method guidelines for Cochrane musculoskeletal group systematic reviews and metaanalyses. *Journal of Rheumatology* 2014;**41**(2):194-205.
18. Higgins JPT, Lasserson T, Thomas J, Flemyng E, Churchill R. Methodological Expectations of Cochrane Intervention Reviews. Cochrane: London. Version August 2023.
19. Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. *BMJ* 2021;**372**(n71):1-9. [DOI: [10.1136/bmj.n71](https://doi.org/10.1136/bmj.n71)]
20. Hazlewood GS, Barnabe C, Tomlinson G, Marshall D, Devoe D, Bombardier C. Methotrexate monotherapy and methotrexate combination therapy with traditional and biologic disease modifying antirheumatic drugs for rheumatoid arthritis: abridged Cochrane systematic review and network meta-analysis. *BMJ* 2016;**353**:i7777.
21. Shafaat, N. Myocrisin (Sodium aurothiomalate) injection: Permanent discontinuation – end of supply in 2019. Sanofi 10 June 2019.
22. Curteis T, Wigle A, Michaels CJ, Nikolakopoulou A. Ranking of treatments in network meta-analysis: incorporating minimally important differences. *BMC Medical Research Methodology* 2025;**25**(1):67. [DOI: [10.1186/s12874-025-02499-0](https://doi.org/10.1186/s12874-025-02499-0)]
23. Mavridis D, Porcher R, Nikolakopoulou A, Salanti G, Ravaud P. Extensions of the probabilistic ranking metrics of competing treatments in network meta-analysis to reflect clinically important relative differences on many outcomes. *Biometrical Journal* 2019;**62**(2):375-85. [DOI: [10.1002/bimj.201900026](https://doi.org/10.1002/bimj.201900026)]
24. Trinquart L, Attiche N, Bafeta A, Porcher R, Ravaud P. Uncertainty in treatment rankings: reanalysis of network meta-analyses of randomized trials. *Annals of Internal Medicine* 2016;**164**(10):666-73. [DOI: [10.7326/m15-2521](https://doi.org/10.7326/m15-2521)]
25. Ren S, Oakley JE, Stevens JW. Incorporating genuine prior information about between-study heterogeneity in random

effects pairwise and network meta-analyses. *Medical Decision Making* 2018;**38**(4):531-42. [DOI: [10.1177/0272989x18759488](https://doi.org/10.1177/0272989x18759488)]

26. Schünemann H, Brożek J, Guyatt G, Oxman A, editor(s). *Handbook for grading the quality of evidence and the strength of recommendations using the GRADE approach* (updated October 2013). GRADE Working Group, 2013. Available from <https://gdt.guidelinedevelopment.org/app/handbook/handbook.html>.

27. Aletaha D, Neogi T, Silman AJ, Funovits J, Felson DT, Bingham CO 3rd, et al. 2010 rheumatoid arthritis classification criteria: an American College of Rheumatology/European League Against Rheumatism collaborative initiative. *Annals of the Rheumatic Diseases* 2010;**69**(9):1580-8.

28. Arnett FC, Edworthy SM, Bloch DA, McShane DJ, Fries JF, Cooper NS, et al. The American Rheumatism Association 1987 revised criteria for the classification of rheumatoid arthritis. *Arthritis and Rheumatism* 1998;**31**(3):315-24.

29. Ropes MW, Bennett GA, Cobb S, Jacox R, Jessar RA. 1958 Revision of diagnostic criteria for rheumatoid arthritis. *Bulletin on the Rheumatic Diseases* 1958;**9**(4):175-6.

30. Hazlewood GS, Barnabe C, Tomlinson G, Marshall D, Devoe DJ, Bombardier C. Methotrexate monotherapy and methotrexate combination therapy with traditional and biologic disease modifying anti-rheumatic drugs for rheumatoid arthritis: A network meta-analysis. *Cochrane Database of Systematic Reviews* 2016, Issue 8. Art. No: CD010227. [DOI: [10.1002/14651858.CD010227.pub2](https://doi.org/10.1002/14651858.CD010227.pub2)]

31. Gottenberg JE, Brocq O, Perdriger A, Lassoued S, Berthelot JM, Wendling D, et al. Non-TNF-targeted biologic vs a second anti-TNF drug to treat rheumatoid arthritis in patients with insufficient response to a first anti-TNF drug: a randomized clinical trial. *JAMA* 2016;**316**(11):1172-80. [DOI: [10.1001/jama.2016.13512](https://doi.org/10.1001/jama.2016.13512)]

32. Jansen JP, Naci H. Is network meta-analysis as valid as standard pairwise meta-analysis? It all depends on the distribution of effect modifiers. *BMC Medicine* 2013;**11**:1-8.

33. Felson DT, Anderson JJ, Boers M, Bombardier C, Furst D, Goldsmith C, et al. American College of Rheumatology. Preliminary definition of improvement in rheumatoid arthritis. *Arthritis and Rheumatism* 1995;**38**(6):727-35.

34. Felson DT, Anderson JJ, Boers M, Bombardier C, Chernoff M, Fried B, et al. The American College of Rheumatology preliminary core set of disease activity measures for rheumatoid arthritis clinical trials. *Arthritis and Rheumatism* 1993;**36**(6):729-40.

35. Ory PA. Interpreting radiographic data in rheumatoid arthritis. *Annals of the Rheumatic Diseases* 2003;**62**(7):597-604.

36. Fransen J, van Riel PL. DAS remission cut points. *Clinical and Experimental Rheumatology* 2006;**24**(6 Suppl 43):S-29-32.

37. Fries JF, Spitz PW, Young DY. The dimensions of health outcomes: the Health Assessment Questionnaire, disability and pain scales. *Journal of Rheumatology* 1982;**9**(5):789-93.

38. Pincus T, Yazici Y, Bergman M. Development of a multi-dimensional health assessment questionnaire (MDHAQ) for the infrastructure of standard clinical care. *Clinical and Experimental Rheumatology* 2005;**23**(5 Suppl 39):S19-28.

39. van der Heijde DM, Van't Hof MA, van Riel PL, van Leeuwen MA, van Rijswijk MH, van de Putte LB. Validity of single variables and composite indices for measuring disease activity in rheumatoid arthritis. *Annals of the Rheumatic Diseases* 1992;**51**(2):177-81.

40. Prevoo ML, van 't Hof MA, Kuper HH, van Leeuwen MA, van de Putte LB, van Riel PL. Modified disease activity scores that include twenty-eight-joint counts. Development and validation in a prospective longitudinal study of patients with rheumatoid arthritis. *Arthritis and Rheumatism* 1995;**38**(1):44-8.

41. Lefebvre C, Glanville J, Briscoe S, Featherstone R, Littlewood A, Metzendorf M-I, et al. Chapter 4: Searching for and selecting studies (last updated March 2025). In: Higgins JP, Thomas J, Chandler J, Cumpston M, Li T, Page MJ, et al, editor(s). *Cochrane Handbook for Systematic Reviews of Interventions* Version 6.5.1 Cochrane, 2025. Available from <https://www.cochrane.org/handbook>.

42. Cochrane Crowd. <http://crowd.cochrane.org> (accessed 1 December 2019).

43. Wallace BC, Noel-Storr A, Marshall IJ, Cohen AM, Smalheiser NR, Thomas J. Identifying reports of randomized controlled trials (RCTs) via a hybrid machine learning and crowdsourcing approach. *Journal of the American Medical Informatics Association* 2017;**24**(6):1165-8.

44. International Committee of Medical Journal Editors. <http://www.icmje.org/> (accessed prior to 5 June 2026).

45. WebPlotDigitizer. Ankit Rohatgi, 2019. Available at <https://automeris.io/WebPlotDigitizer/>.

46. Schünemann HJ, Oxman AD, Higgins JP, Vist GE, Glasziou P, Akl E, et al, editor(s). Chapter 11: Completing 'Summary of findings' tables and grading the confidence in or quality of the evidence. In: Higgins JP, Churchill R, Chandler J, Cumpston MS, editor(s). *Cochrane Handbook for Systematic Reviews of Interventions* version 5.2.0 (updated June 2017). Cochrane, 2017. Available from <https://www.cochrane.org/authors/handbooks-and-manuals/handbook/archive/v5.2.0>.

47. Higgins JP, Altman DG, Gøtzsche PC, Jüni P, Moher D, Oxman AD, et al; Cochrane Statistical Methods Group. The Cochrane Collaboration's tool for assessing risk of bias in randomised trials. *BMJ* 2011;**343**:d5928.

48. Higgins JP, Green S, editor(s). *Cochrane Handbook for Systematic Reviews of Interventions* Version 5.1.0 (updated March 2011). The Cochrane Collaboration, 2011. Available from <https://www.cochrane.org/authors/handbooks-and-manuals/handbook/archive/v5.1.0>.

49. forestplot: Advanced Forest Plot Using 'grid' Graphics. R Package Version 3.1.7. Gordon M, Lumley T. Available at <https://github.com/gforge/forestplot>, 2025.

50. Dias S, Sutton AJ, Ades AE, Welton NJ. Evidence synthesis for decision making 2: a generalized linear modeling framework for pairwise and network meta-analysis of randomized controlled trials. *Medical Decision Making* 2013;**33**(5):607-17.
51. Turner RM, Davey J, Clarke MJ, Thompson SG, Higgins JP. Predicting the extent of heterogeneity in meta-analysis, using empirical data from the Cochrane Database of Systematic Reviews. *International Journal of Epidemiology* 2012;**41**(3):818-27.
52. Rhodes KM, Turner RM, Higgins JP. Predictive distributions were developed for the extent of heterogeneity in meta-analyses of continuous outcome data. *Journal of Clinical Epidemiology* 2015;**68**(1):52-60.
53. Brooks S, Gelman A. General methods for monitoring convergence of iterative simulations. *Journal of Computational and Graphical Statistics* 1998;**7**(4):434-55.
54. R: A language and environment for statistical computing. Version 3.4.2. Vienna, Austria: R Foundation for Statistical Computing, 2017. Available at <https://www.r-project.org>.
55. Just Another Gibbs Sampler. Version 4.3.0. SourceForge, 18 July 2017. Available at <https://sourceforge.net/projects/mcmc-jags>.
56. Newcombe G, Bender R. Implementing GRADE: calculating the risk difference from the baseline risk and the relative risk. *Evidence-Based Medicine* 2014;**19**(1):6-8. [DOI: [10.1136/eb-2013-101340](https://doi.org/10.1136/eb-2013-101340)]
57. Nikolakopoulou A, Higgins JP, Papakonstantinou T, Chaimani A, Del Giovane C, Egger M, et al. CINeMA: an approach for assessing confidence in the results of a network meta-analysis. *PLOS Medicine* 2020;**17**(4):e1003082.
58. Salanti G, Del Giovane C, Chaimani A, Caldwell DM, Higgins JPT. Evaluating the quality of evidence from a network meta-analysis. *PLOS One* 2014;**9**(7):e99682.
59. Sharp SJ, Thompson SG, Altman DG. The relation between treatment benefit and underlying risk in meta-analysis. *BMJ* 1996;**313**(7059):735-8.
60. GRADEpro GDT. Version accessed 20 April 2025. Hamilton (ON): McMaster University (developed by Evidence Prime), 2025. Available at <https://www.gradepro.org>.
61. Kamso MM, Whittle SL, Pardo JP, Buchbinder R, Wells G, Deardon R, et al. Paper 1: a semi-automated approach facilitated the assessment of the certainty of evidence in a network meta-analysis: part 1 – Direct comparisons. *Journal of Clinical Epidemiology* 2026;**191**:112109. [DOI: [10.1016/j.jclinepi.2025.112109](https://doi.org/10.1016/j.jclinepi.2025.112109)]
62. Brignardello-Petersen R, Tomlinson G, Florez I, Rind David M, Chu D, Morgan R, et al. Grading of recommendations assessment, development, and evaluation concept article 5: addressing intransitivity in a network meta-analysis. *Journal of Clinical Epidemiology* 2023;**160**:151-9.
63. Kamso MM, Whittle SL, Pardo JP, Buchbinder R, Wells G, Deardon R, et al. Paper 2: a semi-automated approach facilitated the assessment of the certainty of evidence for in a network meta-analysis: part 2 – indirect and mixed comparisons. *Journal of Clinical Epidemiology* 2026;**191**:112110. [DOI: [10.1016/j.jclinepi.2025.112110](https://doi.org/10.1016/j.jclinepi.2025.112110)]
64. Yepes-Nuñez JJ, Li SA, Guyatt G, Jack SM, Brozek JL, Beyene J, et al. Development of the summary of findings table for network meta-analysis. *Journal of Clinical Epidemiology* 2019;**115**:1-13. [DOI: [10.1016/j.jclinepi.2019.04.018](https://doi.org/10.1016/j.jclinepi.2019.04.018)]
65. Blanco Francisco J, Möricke R, Dokoupilova E, Codding C, Neal J, Andersson M, et al. Secukinumab in active rheumatoid arthritis: a phase III randomized, double-blind, active comparator- and placebo-controlled study. *Arthritis & Rheumatology* 2017;**69**(6):1144-53. [DOI: [10.1002/art.40070](https://doi.org/10.1002/art.40070)]
66. Brown S, Everett Colin C, Naraghi K, Davies C, Dawkins B, Hulme C, et al. Alternative tumour necrosis factor inhibitors (TNFi) or abatacept or rituximab following failure of initial TNFi in rheumatoid arthritis: the SWITCH RCT. *Health Technology Assessment (Winchester, England)* 2018;**22**(34):1-280. [DOI: [10.3310/hta22340](https://doi.org/10.3310/hta22340)]
67. Burmester GR, Blanco R, Charles-Schoeman C, Wollenhaupt J, Zerbini C, Benda B, et al; Oral Step investigators. Tofacitinib (CP-690,550) in combination with methotrexate in patients with active rheumatoid arthritis with an inadequate response to tumour necrosis factor inhibitors: a randomised phase 3 trial. *Lancet* 2013;**381**(9865):451-60. [DOI: [10.1016/S0140-6736\(12\)61424-X](https://doi.org/10.1016/S0140-6736(12)61424-X)]
68. Cohen SB, Emery P, Greenwald MW, Dougados M, Furie RA, Genovese MC, et al; Reflex Trial Group. Rituximab for rheumatoid arthritis refractory to anti-tumour necrosis factor therapy: Results of a multicenter, randomized, double-blind, placebo-controlled, phase III trial evaluating primary efficacy and safety at twenty-four weeks. *Arthritis and Rheumatism* 2006;**54**(9):2793-806. [DOI: [10.1002/art.22025](https://doi.org/10.1002/art.22025)]
69. Elmedany Samah H, Mohamed Aly E, Galil Sahar Mahfouz A. Efficacy and safety profile of intravenous tocilizumab versus intravenous abatacept in treating female Saudi Arabian patients with active moderate-to-severe rheumatoid arthritis. *Clinical Rheumatology* 2019;**38**(8):2109-17.
70. Emery P, Keystone E, Tony HP, Cantagrel A, van Vollenhoven R, Sanchez A, et al. IL-6 receptor inhibition with tocilizumab improves treatment outcomes in patients with rheumatoid arthritis refractory to anti-tumour necrosis factor biologicals: results from a 24-week multicentre randomised placebo-controlled trial. *Annals of the Rheumatic Diseases* 2008;**67**(11):1516-23. [DOI: [10.1136/ard.2008.092932](https://doi.org/10.1136/ard.2008.092932)]
71. Emery P, Rondon J, Parrino J, Lin Y, Pena-Rossi C, van Hoogstraten H, et al. Safety and tolerability of subcutaneous sarilumab and intravenous tocilizumab in patients with rheumatoid arthritis. *Rheumatology (Oxford, England)* 2019;**58**(5):849-58. [DOI: [10.1093/rheumatology/key361](https://doi.org/10.1093/rheumatology/key361)]
72. Fleischmann R, van Adelsberg J, Lin Y, Castelar-Pinho G, Geraldo da R, Brzezicki J, Hrycaj P, et al. Sarilumab and

nonbiologic disease-modifying antirheumatic drugs in patients with active rheumatoid arthritis and inadequate response or intolerance to tumor necrosis factor inhibitors. *Arthritis & Rheumatology* 2017;**69**(2):277-90. [DOI: [10.1002/art.39944](https://doi.org/10.1002/art.39944)]

73. Furst DE, Gaylis N, Bray V, Olech E, Yocum D, Ritter J, et al. Open-label, pilot protocol of patients with rheumatoid arthritis who switch to infliximab after an incomplete response to etanercept: the OPPOSITE study. *Annals of the Rheumatic Diseases* 2007;**66**(7):893-9. [DOI: [10.1136/ard.2006.068304](https://doi.org/10.1136/ard.2006.068304)]

74. Genovese MC, Becker JC, Schiff M, Luggen M, Sherrer Y, Kremer J, et al. Abatacept for rheumatoid arthritis refractory to tumor necrosis factor alpha inhibition. *New England Journal of Medicine* 2005;**353**(11):1114-23. [DOI: [10.1056/NEJMoa050524](https://doi.org/10.1056/NEJMoa050524)]

75. Genovese Mark C, Kremer J, Zamani O, Ludivico C, Krogulec M, Xie L, et al. Baricitinib in patients with refractory rheumatoid arthritis. *New England Journal of Medicine* 2016;**374**(13):1243-52. [DOI: [10.1056/NEJMoa1507247](https://doi.org/10.1056/NEJMoa1507247)]

76. Genovese Mark C, Fleischmann R, Combe B, Hall S, Rubbert-Roth A, Zhang Y, et al. Safety and efficacy of upadacitinib in patients with active rheumatoid arthritis refractory to biologic disease-modifying anti-rheumatic drugs (SELECT-BEYOND): a double-blind, randomised controlled phase 3 trial. *Lancet* 2018;**391**(10139):2513-24. [DOI: [10.1016/S0140-6736\(18\)31116-4](https://doi.org/10.1016/S0140-6736(18)31116-4)]

77. Huffstutter JE, Kafka S, Brent LH, Matucci-Cerinic M, Tang KL, Chevrier M, et al. Clinical response to golimumab in rheumatoid arthritis patients who were receiving etanercept or adalimumab: results of a multicenter active treatment study. *Current Medical Research and Opinion* 2017;**33**(4):657-66. [DOI: [10.1080/03007995.2016.1277195](https://doi.org/10.1080/03007995.2016.1277195)]

78. Manders SH, Kievit W, Adang E, Brus HL, Moens HJ, Hartkamp A, et al. Cost-effectiveness of abatacept, rituximab, and TNFi treatment after previous failure with TNFi treatment in rheumatoid arthritis: a pragmatic multi-centre randomised trial. *Arthritis Research & Therapy* 2015;**17**:134. [DOI: [10.1186/s13075-015-0630-5](https://doi.org/10.1186/s13075-015-0630-5)]

79. Schiff MH, von Kempis J, Goldblum R, Tesser JR, Mueller RB. Rheumatoid arthritis secondary non-responders to TNF can attain an efficacious and safe response by switching to certolizumab pegol: a phase IV, randomised, multicentre, double-blind, 12-week study, followed by a 12-week open-label phase. *Annals of the Rheumatic Diseases* 2014;**73**(12):2174-7. [DOI: [10.1136/annrheumdis-2014-205325](https://doi.org/10.1136/annrheumdis-2014-205325)]

80. Smolen JS, Kay J, Doyle MK, Landewe R, Matteson EL, Wollenhaupt J, et al; Go-After study investigators. Golimumab in patients with active rheumatoid arthritis after treatment with tumour necrosis factor alpha inhibitors (GO-AFTER study): a multicentre, randomised, double-blind, placebo-controlled, phase III trial. *Lancet* 2009;**374**(9685):210-21. [DOI: [10.1016/S0140-6736\(09\)60506-7](https://doi.org/10.1016/S0140-6736(09)60506-7)]

81. Genovese MC, Fleischmann R, Furst D, Janssen N, Carter J, Dasgupta B, et al. Efficacy and safety of olokizumab in patients with rheumatoid arthritis with an inadequate response to TNF inhibitor therapy: outcomes of a randomised Phase IIb study.

Annals of the Rheumatic Diseases 2014;**73**(9):1607-15. [DOI: [10.1136/annrheumdis-2013-204760](https://doi.org/10.1136/annrheumdis-2013-204760)]

82. Humby F, Durez P, Buch MH, Lewis MJ, Rizvi H, Rivellese F, et al. Rituximab versus tocilizumab in anti-TNF inadequate responder patients with rheumatoid arthritis (R4RA): 16-week outcomes of a stratified, biopsy-driven, multicentre, open-label, phase 4 randomised controlled trial. *Lancet* 2021;**397**(10271):305-17. [DOI: [10.1016/s0140-6736\(20\)32341-2](https://doi.org/10.1016/s0140-6736(20)32341-2)]

83. Kalén E, Piehl F, Andersson M. Demyelinating events following anti-tumor necrosis factor alpha therapy: Rare but challenging to treat. *European Journal of Neurology* 2022;**29**:2047-55.

84. Kopp TI, Delcoigne B, Arkema EV, Jacobsen RK, Magyari M, Ibfelt EH, et al. Risk of neuroinflammatory events in arthritis patients treated with tumour necrosis factor alpha inhibitors: a collaborative population-based cohort study from Denmark and Sweden. *Annals of the Rheumatic Diseases* 2020;**79**(5):566-72. [DOI: [10.1136/annrheumdis-2019-216693](https://doi.org/10.1136/annrheumdis-2019-216693)]

85. Russell MD, Stovin C, Alveyn E, Adeyemi O, Chan CK, Patel V, et al. JAK inhibitors and the risk of malignancy: a meta-analysis across disease indications. *Annals of the Rheumatic Diseases* 2023;**82**(8):1059-67.

86. Ytterberg SR, Bhatt DL, Mikuls TR, Koch GG, Fleischmann R, Rivas JL, et al. Cardiovascular and cancer risk with tofacitinib in rheumatoid arthritis. *New England Journal of Medicine* 2022;**386**(4):316-26. [DOI: [10.1056/nejmoa2109927](https://doi.org/10.1056/nejmoa2109927)]

87. Ahmed S, Pfeiffer M, Jackson S, Song M, Mariette X, Seror R, et al. Cancer risk of Janus kinase inhibitors and biological disease-modifying antirheumatic drugs among older Americans with rheumatoid arthritis. *Rheumatology (Oxford, England)* 2025;**64**(7):4172-80. [DOI: [10.1093/rheumatology/keaf092](https://doi.org/10.1093/rheumatology/keaf092)]

88. Hackenbroich S, Kranke P, Meybohm P, Weibel S. Include or not to include conference abstracts in systematic reviews? Lessons learned from a large Cochrane network meta-analysis including 585 trials. *Systematic Reviews* 2022;**11**(1):178. [DOI: [10.1186/s13643-022-02048-6](https://doi.org/10.1186/s13643-022-02048-6)]

89. Singh S, Fumery M, Singh AG, Singh N, Prokop LJ, Dulai PS, et al. Comparative risk of cardiovascular events with biologic and synthetic disease-modifying antirheumatic drugs in patients with rheumatoid arthritis: a systematic review and meta-analysis. *Arthritis Care & Research* 2020;**72**(4):561-76. [DOI: [10.1002/acr.23875](https://doi.org/10.1002/acr.23875)]

90. Janke K, Biester K, Krause D, Richter B, Schürmann C, Hirsch K, et al. Comparative effectiveness of biological medicines in rheumatoid arthritis: systematic review and network meta-analysis including aggregate results from reanalysed individual patient data. *BMJ* 2020;**370**:m2288. [DOI: [10.1136/bmj.m2288](https://doi.org/10.1136/bmj.m2288)]

91. Singh JA, Hossain A, Tanjong Ghogomu E, Mudano AS, Maxwell LJ, Buchbinder R, et al. Biologics or tofacitinib for people with rheumatoid arthritis unsuccessfully treated with biologics: a systematic review and network meta-analysis. *Cochrane Database of Systematic Reviews* 2017, Issue 3. Art. No: CD012591. [DOI: [10.1002/14651858.CD012591](https://doi.org/10.1002/14651858.CD012591)]

92. Australian Rheumatology Association. An Australian Living Guideline for the Pharmacological Management of Inflammatory Arthritis (version #2). <https://app.magicapp.org/#/guideline/LqRV3n/section/jDJelJ>.

93. Canadian Rheumatology Association. Canadian Rheumatology Association living guidelines for the pharmacological management of rheumatoid arthritis with disease-modifying anti-rheumatic drugs (version #2). <https://app.magicapp.org/#/guideline/jNxxw7n>.

ADDITIONAL TABLES

Table 1. Chosen thresholds for rating the certainty of evidence for imprecision following the minimally contextualised framework

Outcomes	Small effect (MID)	Large effect threshold
ACR50	5%	20%
DAS28 remission	5%	20%
Withdrawal due to adverse events	5%	20%
Serious adverse events	5%	10%
HAQ-DI (0 to 3)	0.21	0.5
Pain (VAS 0 to 100)	10	30
Radiographic progression (SvdH)	5	10

Abbreviations: ACR50, American College of Rheumatology response criteria-50; DAS28 remission, Disease Activity Score in 28 joints remission; HAQ-DI, Health Assessment Questionnaire-Disability Index; MID, minimally important difference; SvdH, Sharpe van der Heijde; VAS, visual analogue scale

Table 2. Describing levels of certainty of the GRADE framework using the minimally contextualised approach

Certainty level	Category 0		Category 1
	Includes the null (point estimate not above MID)	Includes the null (point estimate above MID)	Excludes the null
High	Not applicable (would be downgraded for imprecision)		More effective/less harmful than placebo
Moderate	Probably not different than placebo	Probably more effective/less harmful than placebo, but the credible interval crosses the null effect	Probably more effective/less harmful than placebo
Low	May not be different than placebo	May be more effective/less harmful than placebo, but the credible interval crosses the null effect	May be more effective/less harmful than placebo
Very low	The evidence is very uncertain about the effect.		

Abbreviations. MID, minimally important difference

Table 3. Characteristics of included trials

Comparison	Number of studies	Study name(s)	Number of participants	Year published, median (range)	Trial duration, median (range) months	Base-line age, median (range) years	Base-line disease duration, median (range) years	Base-line proportion female (0,1), median (range)	DAS28 base-line, median (range)	HAQ-DI base-line, median (range)	Base-line proportion maintenance of oral steroid (0,1), median (range)
Placebo-controlled trials											
Abatacept 10 mg/kg (rounded) IV every 4 weeks (+/- IV load at 2 weeks) vs placebo	2	Genovese 2005; Blanco 2017	665	2011 (2005 to 2017)	6	53.05 (51.6 to 55.5)	10.85 (10.2 to 12.2)	0.785 (0.76 to 0.83)	6.5 [Missing n = 1]	1.8 (1.7 to 1.8)	0.675 (0.65 to 0.7) [Missing n = 1]
TNFi not previously tried vs placebo	2	Schiff 2014; Smolen 2009	343	2012 (2009 to 2014)	4 (3 to 6)	55.55 (54 to 59)	10.9 (9.6 to 14)	0.825 (0.74 to 0.85)	[Missing n = 2]	1.3 (1.1 to 1.5) [Missing n = 1]	[Missing n = 2]
Sarilumab 200 mg SC every 2 weeks vs placebo	1	Fleischmann 2017	365	2017	6	52.5	12.35	0.835	[Missing n = 1]	1.8	0.615
Upadacitinib 15 mg oral once daily vs placebo	1	Genovese 2018	333	2018	3	56.95	13.45	0.845	[Missing n = 1]	1.65	0.475
Rituximab 1000 mg IV week 0 and 2 vs placebo	1	Cohen 2006	499	2006	6	52.565	11.945	0.815	6.85	1.9	0.63
Tofacitinib 5 mg oral twice a day vs placebo	1	Burmester 2013	263	2013	6	54.9	12.15	0.825	[Missing n = 1]	1.615	0.635
Baricitinib 4 mg oral once daily vs baricitinib 2 mg oral once daily vs placebo	1	Genovese 2016	527	2016	6	56	14	0.82	[Missing n = 1]	1.74	0.54

Table 3. Characteristics of included trials (Continued)

Tocilizumab 8 mg/kg IV every 4 weeks vs placebo	1	Gen-ovese 2014	65	2014	3	55.52	9.99	0.849	5.77	[Missing n = 1]	0.52
Tocilizumab 8 mg/kg IV every 4 weeks vs tocilizumab 4 mg/kg IV every 4 weeks vs placebo	1	Emery 2008	489	2008	6	53.4	11.4	0.81	6.79	1.7	0.58
Trials with inactive comparator arms											
TNFi not previously tried vs continue failed TNFi	1	Furst 2007	27	2007	7	48.65	10.85	0.923	6.35	1.45	0.3
Switch in TNFi route from SC to IV vs continue failed TNFi	1	Huffstutter 2017	275	2017	12	56.5	6.4	0.82	[Missing n = 1]	[Missing n = 1]	[Missing n = 1]
Trials with active comparator arms											
Sarilumab 200 mg SC every 2 weeks vs tocilizumab 4 mg/kg IV every 4 weeks	1	Emery 2019	153	2019	6	51.05	10.65	0.785	5.895	1.745	[Missing n = 1]
TNFi not previously tried vs non-TNF biologic	1	Gottenberg 2016	235	2016	12	57.05	10.5	0.83	5.1	1.3	0.53
Abatacept 125 mg SC weekly vs TNFi not previously tried vs rituximab 1000 mg IV week 0 and 2	1	Brown 2018	122	2018	11	57.8	6.9	0.8	6.2	[Missing n = 1]	0.195
TNFi not previously tried vs rituximab 1000 mg IV week 0 and 2 vs abatacept 10 mg/kg IV every 4 weeks	1	Manders 2015	139	2015	12	56.16	6.56	0.74	4.87	1.39	0.1
Tocilizumab 8 mg/kg IV every 4 weeks vs abatacept 10 mg/kg IV every 4 weeks	1	Elmedany 2019	118	2019	6	51.12	6.97	1.00	5.44	0.96	0.088
Tocilizumab 8 mg/kg IV every 4 weeks vs rituximab 1000 mg IV week 0 and 2	1	Humby 2021	161	2021	4	57.65	11	0.855	5.87	1.75	[Missing n = 1]

Abbreviations: DAS28, Disease Activity Score 28; HAQ-DI, Health Assessment Questionnaire-Disability Index; IV, intravenous; SC, subcutaneous; TNFi, tumour necrosis factor inhibitor

Table 4. Outcome Reporting Bias In Trials (ORBIT) matrix for critical outcomes

Comparison	Critical outcomes			Other critical outcomes			
	ACR50	Withdrawals due to adverse events	Radiographic outcomes	DAS28	Pain measured by VAS	HAQ	Serious adverse events
Placebo-controlled trials							
Abatacept 10 mg/kg (rounded) IV every 4 weeks (+/- IV load at 2 weeks) vs placebo							
Genovese 2005	Yes	Yes	-	Yes	-	Yes	Yes
Blanco 2017	Yes	Yes	-	-	-	Yes	Yes
TNFi not previously tried vs placebo							
Schiff 2014	Yes	Yes	-	Yes	-	-	Yes
Smolen 2009	Yes	Yes	-	Yes	Yes	Yes	Yes
Upadacitinib 15 mg oral once daily vs placebo							
Genovese 2018	Yes	Yes	-	-	Yes	Yes	Yes
Sarilumab 200 mg SC every 2 weeks vs placebo							
Fleischmann 2017	Yes		-	Yes	-	Yes	Yes
Rituximab 1000 mg IV week 0 and 2 vs placebo							
Cohen 2006	Yes	Yes	Yes	Yes	-	Yes	Yes
Tofacitinib 5 mg oral twice a day vs placebo							
Burmester 2013	Yes	Yes	-	Yes	-	Yes	Yes
Baricitinib 4 mg oral once daily vs baricitinib 2 mg oral once daily vs placebo							

Table 4. Outcome Reporting Bias In Trials (ORBIT) matrix for critical outcomes (Continued)

Genovese 2016	Yes	Yes	-	-	Yes	Yes	Yes
Tocilizumab 8 mg/kg IV every 4 weeks vs placebo							
Genovese 2014	Yes	-	-	Yes	-	Yes	Yes
Tocilizumab 8 mg/kg IV every 4 weeks vs tocilizumab 4 mg/kg IV every 4 weeks vs placebo							
Emery 2008	Yes	Yes	-	Yes	-	Yes	Yes
Trials with active comparator arms							
TNFi not previously tried vs continue failed TNFi							
Furst 2007	-	-	Yes	-	-	-	-
Continue failed TNFi vs switch in TNFi route from SC to IV							
Huffstutter 2017	-	-	-	-	-	-	-
TNFi not previously tried vs non-TNF biologic							
Gottenberg 2016	-	-	-	Yes	-	Yes	-
Sarilumab 200 mg SC every 2 weeks vs tocilizumab 4 mg/kg IV every 4 weeks							
Emery 2019	-	-	-	-	-	-	Yes
Abatacept 125 mg SC weekly vs TNFi not previously tried vs rituximab 1000 mg IV week 0 and 2							
Brown 2018	Yes	-	-	Yes	-	Yes	-
TNFi not previously tried vs rituximab 1000 mg IV week 0 and 2 vs abatacept 10 mg/kg IV every 4 weeks							
Manders 2015	-	-	-	-	-	Yes	-
Tocilizumab 8 mg/kg IV every 4 weeks vs abatacept 10 mg/kg IV every 4 weeks							
Elmedany 2019	-	Yes	-	-	-	Yes	Yes
Tocilizumab 8 mg/kg IV every 4 weeks vs rituximab 1000 mg IV week 0 and 2							



Table 4. Outcome Reporting Bias In Trials (ORBIT) matrix for critical outcomes *(Continued)*

Humby 2021	-	-	-	Yes	-	Yes	-
------------	---	---	---	-----	---	-----	---

Abbreviations: ACR50, American College of Rheumatology 50%; DAS28, Disease Activity Score 28; HAQ, Health Assessment Questionnaire; IV, intravenous; SC, subcutaneous; TNFi, tumour necrosis factor inhibitor; VAS, visual analogue scale

Table 5. Outcome Reporting Bias In Trials (ORBIT) matrix for important outcomes

Comparison	Important outcomes				
	ACR20	ACR70	Serious infections	Fatigue	Quality of life
Placebo-controlled trials					
Abatacept 10 mg/kg (rounded) IV every 4 weeks (+/- IV load at 2 weeks) vs placebo					
Genovese 2005	Yes	Yes	Yes	-	-
Blanco 2017	Yes	Yes	-	-	-
TNFi not previously tried vs placebo					
Schiff 2014	Yes	-	-	-	-
Smolen 2009	Yes	Yes	Yes	Yes	-
Upadacitinib 15 mg oral once daily vs placebo					
Genovese 2018	Yes	Yes	Yes	-	-
Sarilumab 200 mg SC every 2 weeks vs placebo					
Fleischmann 2017	Yes	Yes	Yes	-	-
Rituximab 1000 mg IV week 0 and 2 vs placebo					
Cohen 2006	Yes	Yes		Yes	-
Tofacitinib 5 mg oral twice a day vs placebo					
Burmester 2013	Yes	Yes	Yes	-	-
Baricitinib 4 mg oral once daily vs baricitinib 2 mg oral once daily vs placebo					
Genovese 2016	Yes	Yes	Yes	-	-
Tocilizumab 8 mg/kg IV every 4 weeks vs placebo					
Genovese 2014	Yes	-	-	-	-
Tocilizumab 8 mg/kg IV every 4 weeks vs tocilizumab 4 mg/kg IV every 4 weeks vs placebo					
Emery 2008	Yes	Yes	-	-	-
Trials with active comparator arms					
TNFi not previously tried vs continue failed TNFi					
Furst 2007	Yes	-	-	-	-
Continue failed TNFi vs switch in TNFi route from SC to IV					

Table 5. Outcome Reporting Bias In Trials (ORBIT) matrix for important outcomes (Continued)

Huffstutter 2017	Yes	-	-	-	-
TNFi not previously tried vs non-TNF biologic					
Gottenberg 2016	-	-	-	-	-
Sarilumab 200 mg SC every 2 weeks vs tocilizumab 4 mg/kg IV every 4 weeks					
Emery 2019	-	-	-	-	-
Abatacept 125 mg SC weekly vs TNFi not previously tried vs rituximab 1000 mg IV week 0 and 2					
Brown 2018	Yes	Yes	-	-	Yes
TNFi not previously tried vs rituximab 1000 mg IV week 0 and 2 vs abatacept 10 mg/kg IV every 4 weeks					
Manders 2015	-	-	-	-	Yes
Tocilizumab 8 mg/kg IV every 4 weeks vs abatacept 10 mg/kg IV every 4 weeks					
Elmedany 2019	-	-	Yes	-	-
Tocilizumab 8 mg/kg IV every 4 weeks vs rituximab 1000 mg IV week 0 and 2					
Humby 2021	-	-	-	-	Yes

Abbreviations: ACR20, American College of Rheumatology 20%; ACR70, American College of Rheumatology 70%; IV, intravenous; SC, subcutaneous; TNFi, tumour necrosis factor inhibitor

Table 6. Important outcomes

	ACR20 OR (95% CrI)	ACR70 OR (95% CrI)	DAS28 MD (95% CrI)	Quality of life MD (95% CrI)	Fatigue MD (95% CrI)	Serious infections RR (95% CrI)
Patient population: individuals with rheumatoid arthritis who have previously failed a TNFi						
Intervention: TNFi (TNFi not previously tried, continuation of failed TNFi, switch in route of TNFi from SC to IV), non-TNFi biologic, IL-6 inhibitor (tocilizumab 4 mg/kg IV every 4 weeks, tocilizumab 8 mg/kg IV every 4 weeks), T-cell co-stimulation inhibitor (abatacept 125 mg SC weekly, abatacept 10 mg/kg (rounded) IV every 4 weeks (+/- IV load at 2 weeks)), B-cell depletor (rituximab 1000 mg IV week 0 and 2), and JAK inhibitor (upadacitinib 15 mg oral once daily, tofacitinib 5 mg oral twice a day, baricitinib 2 mg oral once daily, baricitinib 4 mg oral once daily)						
Comparator: placebo						
Outcomes: 20% improvement in rheumatoid arthritis symptoms from baseline (ACR20); 70% improvement in rheumatoid arthritis symptoms from baseline (ACR70); serious infections; fatigue; quality of life						
Setting: outpatient						
TNF inhibitor						
TNFi not previously tried (2 RCTs; 343 participants)	4.45 (2.24, 12.17)	5.4 (1.6, 20.39)	-1.99 (-2.82, -0.62)	-	1.5 (0.29, 2.72)	0.4 (0.06, 3.53)

Table 6. Important outcomes (Continued)

Continue failed TNFi	1 (0.13, 8.08)	-	-	-	-	-
Switch in TNFi route from SC to IV	1.02 (0.1, 11.15)	-	-	-	-	-
Non-TNFi biologic						
Non-TNFi biologic (indirect evidence only)	-	-	-	-	-	0.39 (0.04, 5.29)
IL-6 inhibitor						
Sarilumab 200 mg SC every 2 weeks (1 RCT; 365 participants)	3.08 (1.18, 8.12)	2.57 (0.76, 8.94)	-	-	-	0.6 (0.07, 5.46)
Tocilizumab 4 mg/kg IV every 4 weeks (1 RCT; 319 participants)	3.84 (1.45, 10.5)	4.62 (0.77, 43.68)	-	-	-	-
Tocilizumab 8 mg/kg IV every 4 weeks (1 RCT; 328 participants)	8.67 (3.89, 19.78)	12.97 (2.44, 113.84)	-	-	-	-
T-cell co-stimulation inhibitor						
Abatacept 125 mg SC weekly (indirect evidence only)	2.79 (0.82, 11.01)	4.04 (0.48, 28.43)	-1.33 (-2.46, -0.18)	-	-	-
Abatacept 10 mg/kg (rounded) IV every 4 weeks (+/- IV load at 2 weeks) (2 RCTs; 665 participants)	3.82 (1.86, 7.82)	4.12 (1.53, 12.87)	-	-	-	0.8 (0.24, 2.94)
B-cell depletor						
Rituximab 1000 mg IV week 0 and 2 (1 RCT; 499 participants)	4.12 (1.85, 9.46)	6.27 (1.84, 22.77)	-1.37 (-2.08, -0.62)	-	-8.43 (-10.77, -6.13)	1.68 (0.33, 10.14)
JAK inhibitor						
Upadacitinib 15 mg oral once daily (1 RCT; 333 participants)	4.65 (1.75, 12.51)	1.91 (0.54, 7.06)	-	-	-	NE
Tofacitinib 5 mg oral twice a day (1 RCT; 263 participants)	2.21 (0.81, 6.11)	12.02 (2.25, 112.43)	-	-	-	NE
Baricitinib 2 mg oral once daily (1 RCT; 350 participants)	2.18 (0.82, 5.79)	4.49 (1.19, 19.14)	-	-	-	0.77 (0.14, 4.21)
Baricitinib 4 mg oral once daily (1 RCT; 353 participants)	2.32 (0.87, 5.79)	6.04 (1.6, 25.17)	-	-	-	1.11 (0.24, 5.36)

Abbreviations: ACR20, American College of Rheumatology 20%; ACR70, American College of Rheumatology 70%; CrI, credible interval; DAS28, Disease Activity Score-28; IL-6, interleukin-6; JAK, Janus kinase; MD, mean difference; MID, minimally important difference; NE, not estimable; OR, odds ratio; RCT, randomised controlled trial; RR, risk ratio; TNFi, tumour necrosis factor inhibitor

Table 7. Subgroup analysis using meta-regression for ACR50 response

Covariates	Log odds ratio	Odds ratio	Interpretation
	β (95% CrI)	β (95% CrI)	
Baseline response rate	-0.012 (-0.519, 0.13)	0.99 (0.95, 1.01)	Decrease in OR of 0.99 times (0.95, 1.01) for every 10% increase in the baseline response rate
Disease duration	0.196 (-0.086, 0.505)	1.22 (0.92, 1.66)	Increase in OR of 1.22 times (0.92 to 1.66) for every 6 years of disease duration
Duration of trial	0.131 (-0.252, 0.523)	1.14 (0.78, 1.69)	Increase in OR of 1.14 times (0.78 to 1.69) for every 6 months of trial duration
Year of trial publication	0.465 (-1.522, 0.696)	0.63 (0.22, 2.01)	Decrease in OR of 0.63 times (0.22 to 2.01) for each year later of publication (range of years 2005 to 2018)

Abbreviations: ACR50, American College of Rheumatology 50%; CrI, credible interval; OR, odds ratio

Table 8. Sensitivity analyses for ACR50

	Relative effect OR (95% CrI)					
	Main analysis	Including studies at high risk of bias	Uninformative prior	Fixed-effect model	3 months	6 months
TNF inhibitors						
TNFi not previously tried (2 RCTs; 343 participants)	6.04 (2.49, 16.3)	7.07 (3.25, 17.18)	6.24 (1.98, 29.17)	5.82 (2.89, 12.5)	4.97 (1.61, 23.89)	-
Continue failed TNFi (indirect evidence only)	-	3.4 (0.35, 20.76)	-	-	NE	-
Switch in route of TNFi (SC to IV) (indirect evidence only)	-	1.96 (0.15, 17.92)	-	-	-	-
IL-6 inhibitors						
Sarilumab 200 mg SC every 2 weeks (1 RCT; 365 participants)	3.11 (1.25, 7.76)	3.11 (1.28, 7.61)	3.13 (0.62, 15.66)	3.11 (1.93, 5.07)	-	3.11 (1.06, 9.09)
Tocilizumab 4 mg/kg IV every 4 weeks (1 RCT; 319 participants)	5.04 (1.65, 17.05)	4.36 (1.59, 13.15)	5.02 (0.99, 27.38)	5.02 (2.2, 13.24)	-	5.44 (1.5, 21.95)
Tocilizumab 8 mg/kg IV every 4 weeks	10.03 (3.65, 31.27)	8.32 (3.46, 21.28)	9.95 (2.46, 43.81)	10.06 (4.68, 25.44)	8.82 (0.95, 320.15)	10.99 (3.13, 43.42)

Table 8. Sensitivity analyses for ACR50 (Continued)
(1 RCT; 328 participants)

T-cell co-stimulation inhibitor						
Abatacept 125 mg SC weekly (indirect evidence only)	4.31 (0.97, 18.28)	4.78 (1.15, 19.58)	4.42 (0.65, 36.76)	4.26 (1.23, 13.46)	-	-
Abatacept 10 mg/kg (rounded) IV every 4 weeks (+/- IV load at 2 weeks) (2 RCTs; 665 participants)	4.57 (2.21, 10.18)	4.55 (2.23, 10.02)	4.61 (1.49, 16.35)	4.44 (2.67, 7.7)	-	4.55 (2.01, 11.39)
B-cell depletor						
Rituximab 1000 mg IV week 0 and 2 (1 RCT; 499 participants)	5.5 (2.31, 13.12)	5.79 (2.54, 13.67)	5.58 (1.44, 22.59)	5.63 (3.23, 10.42)	-	6.55 (2.08, 21.98)
JAK inhibitors						
Upadacitinib 15 mg oral once daily (1 RCT; 333 participants)	3.93 (1.53, 10.32)	3.92 (1.54, 10.22)	3.94 (0.77, 20.28)	3.92 (2.24, 7.04)	3.93 (1.07, 14.38)	-
Tofacitinib 5 mg oral twice a day (1 RCT; 263 participants)	4.09 (1.43, 12.13)	4.09 (1.47, 11.89)	4.08 (0.75, 22.13)	4.08 (1.99, 8.93)	4.06 (1.05, 15.92)	-
Baricitinib 2 mg oral once daily (1 RCT; 350 participants)	2 (0.77, 5.23)	2 (0.79, 5.17)	1.98 (0.38, 10.18)	2 (1.14, 3.55)	-	2 (0.65, 6.13)
Baricitinib 4 mg oral once daily (1 RCT; 353 participants)	2.8 (1.12, 7.16)	2.79 (1.1, 7.22)	2.79 (0.54, 14.1)	2.8 (1.63, 4.91)	-	2.79 (0.93, 8.5)

Abbreviations: ACR50, American College of Rheumatology 50%; CrI, credible interval; IL-6, interleukin-6; IV, intravenous; JAK, Janus kinase; NE, no evidence; OR, odds ratio; SC, subcutaneous; TNFi, tumour necrosis factor inhibitor

Table 9. Sensitivity analyses for withdrawal due to adverse events

	Relative effect OR (95% CrI)			
	Main analysis	Including studies at high risk of bias	Uninformative prior	Fixed-effect model
TNF inhibitors				
TNFi not previously tried (2 RCTs; 343 participants)	0.32 (0.06, 1.1)	0.33 (0.07, 1.16)	0.27 (0.02, 1.44)	0.35 (0.11, 0.99)
Continue failed TNFi (indirect evidence only)	-	0.15 (0, 2.69)	-	-

Table 9. Sensitivity analyses for withdrawal due to adverse events (Continued)

Switch in route of TNFi (SC to IV) (indirect evidence only)	-	0.26 (0, 7.06)	-	-
IL-6 inhibitors				
Sarilumab 200 mg SC every 2 weeks (1 RCT; 365 participants)	1.99 (0.56, 7.41)	2 (0.59, 7.01)	2 (0.21, 19.24)	1.99 (0.9, 4.74)
Tocilizumab 4 mg/kg IV every 4 weeks (1 RCT; 319 participants)	1.41 (0.38, 5.48)	1.4 (0.39, 5.17)	1.43 (0.17, 13.3)	1.37 (0.54, 3.53)
Tocilizumab 8 mg/kg IV every 4 weeks (1 RCT; 328 participants)	1.5 (0.52, 4.95)	1.5 (0.52, 4.95)	1.57 (0.17, 10.98)	1.42 (0.65, 3.26)
T-cell co-stimulation inhibitor				
Abatacept 125 mg SC weekly (indirect evidence only)	2.8 (0.05, 240.01)	2.57 (0.05, 166.37)	3.18 (0.04, 384.02)	2.76 (0.06, 150.7)
Abatacept 10 mg/kg (rounded) IV every 4 weeks (+/- IV load at 2 weeks) (2 RCTs; 665 participants)	0.92 (0.34, 2.87)	0.91 (0.35, 2.71)	0.98 (0.23, 5.69)	0.86 (0.4, 1.95)
B-cell depletor				
Rituximab 1000 mg IV week 0 and 2 (1 RCT; 499 participants)	2.3 (0.42, 22.7)	2.3 (0.43, 21.28)	2.44 (0.26, 39.97)	2.21 (0.54, 15.91)
JAK inhibitors				
Upadacitinib 15 mg oral once daily (1 RCT; 333 participants)	0.4 (0.08, 1.84)	0.4 (0.08, 1.77)	0.41 (0.03, 4.36)	0.41 (0.11, 1.28)
Tofacitinib 5 mg oral twice a day (1 RCT; 263 participants)	1.39 (0.35, 5.8)	1.38 (0.36, 5.56)	1.38 (0.14, 14.13)	1.37 (0.52, 3.79)
Baricitinib 2 mg oral once daily (1 RCT; 350 participants)	0.94 (0.21, 4.11)	0.94 (0.23, 3.88)	0.93 (0.09, 9.72)	0.94 (0.32, 2.77)
Baricitinib 4 mg oral once daily (1 RCT; 353 participants)	1.43 (0.37, 5.95)	1.44 (0.39, 5.61)	1.43 (0.14, 14.62)	1.42 (0.65, 3.93)

Abbreviations: CrI, credible interval; IL-6, interleukin-6; IV, intravenous; JAK, Janus kinase; OR, odds ratio; SC, subcutaneous; TNFi, tumour necrosis factor inhibitor